Домашняя работа (опциональная) по дисциплине «МОВС: Компьютерное зрение»

Разбор нейросетевого модуля трекинга

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😊 Первый раз в жизни делаю посадку на юпитер 😊

ANSI коды для цветов текста

In [1]:
```
RED = "\033[31m"
GREEN = "\033[32m"
YELLOW = "\033[33m"
BLUE = "\033[34m"
RESET = "\033[0m"  # сброс цветов до стандартных
```

Шаблоны markdown

```
# Это комментарий в коде
def greet(name):
print(f"Hello, {name}!")
```

```
code
```

Установка библиотек (Необязательно)

In [2]:
```
# !pip install ultralytics pycocotools numpy scipy tqdm pillow scikit-learn albumen
```

Подготовка к написанию кода

Импорт библиотек

In [399…

```
print(f"{YELLOW}"+60*"-"+f"{RESET}")
print(f"Библиотеки: \n")

# Дополнительные библиотеки
import platform # Узнать версию паймона ;)
import os
import logging
import time
import sys
import random
import re
import shutil
from tqdm import tqdm
from functools import wraps
from typing import Any, Tuple, Union, Optional, List, Type, Callable, Dict
import collections
from collections import defaultdict
from collections import Counter
import zipfile
import rarfile
from dataclasses import dataclass
import tempfile

# Основные библиотеки
import IPython.display as ipd # Добавляет виджеты для ячеек юпитера
from IPython.display import HTML
from IPython import get_ipython
import ipykernel
print(f"python: {BLUE}{platform.python_version()}{RESET} ")
import matplotlib # Для рисунков
import matplotlib.pyplot as plt
print(f"matplotlib: {BLUE}{matplotlib.__version__}{RESET}")
import numpy as np # Для работы с массивами
print(f"numpy: {BLUE}{np.__version__}{RESET}")
import pandas as pd # Работа с таблицами
print(f"pandas: {BLUE}{pd.__version__}{RESET}")
import sklearn # Много полезного для ML
from sklearn.metrics import precision_score, recall_score, f1_score, accuracy_score
from sklearn.manifold import TSNE
print(f"sklearn: {BLUE}{sklearn.__version__}{RESET}")
import scipy
from scipy.spatial.distance import cdist, euclidean
from scipy.optimize import linear_sum_assignment
print(f"scipy: {BLUE}{scipy.__version__}{RESET}")
import pydantic # Для валидации данных
from pydantic import (BaseModel, Field, StrictStr, condecimal, StrictInt, StrictBoo
                      FilePath, DirectoryPath, ValidationError, root_validator, Con
print(f"pydantic: {BLUE}{pydantic.__version__}{RESET}")
```

```
import fastapi
from fastapi import HTTPException, status
print(f"fastapi: {BLUE}{fastapi.__version__}{RESET}")
import torch
import torch.nn as nn
import torch.optim as optim
import torch.nn.functional as F
from torch.optim.lr_scheduler import ReduceLROnPlateau
from torch.utils.data import DataLoader, Subset, WeightedRandomSampler, Dataset
print(f"torch: {BLUE}{torch.__version__}{RESET}")
import torchvision
from torchvision import datasets
import torchvision.transforms as transforms
from torchvision.datasets import ImageFolder
from torchvision.transforms.functional import to_pil_image
import torchvision.models as models
print(f"torchvision: {BLUE}{torchvision.__version__}{RESET}")
import requests
print(f"requests: {BLUE}{requests.__version__}{RESET}")
import PIL
from PIL import Image, ImageOps
print(f"Pillow: {BLUE}{PIL.__version__}{RESET}")
import cv2
print(f"opencv: {BLUE}{cv2.__version__}{RESET}")
import ultralytics
from ultralytics import YOLO
print(f"ultralytics: {BLUE}{ultralytics.__version__}{RESET}")
import albumentations as A
print(f"albumentations: {BLUE}{A.__version__}{RESET}")
import pycocotools
import pycocotools.mask as maskUtils
from pytorch_metric_learning import losses


from rich.theme import Theme
from rich.logging import RichHandler
from rich.console import Console
from rich.pretty import install as pretty_install
from rich.traceback import install as traceback_install


print(f"{YELLOW}"+60*"-"+f"{RESET}")
```

```
------------------------------------------------------------
Библиотеки:

python: 3.9.0
matplotlib: 3.7.2
numpy: 1.23.0
pandas: 2.2.2
sklearn: 1.3.2
scipy: 1.13.1
pydantic: 2.10.6
fastapi: 0.115.7
torch: 1.12.1+cu116
torchvision: 0.13.1+cu116
requests: 2.32.3
Pillow: 10.4.0
opencv: 4.9.0
ultralytics: 8.3.78
albumentations: 2.0.4
------------------------------------------------------------
```

Дополнительные настройки

In [4]:
```python
import warnings
warnings.filterwarnings("ignore")
import logging
# Отключение логирования для cmdstanpy
logging.getLogger('cmdstanpy').setLevel(logging.ERROR)
# Полезна при разработке собственных библиотек, юпитер будет переимпортировать моду
%load_ext autoreload
%autoreload 1
```

Кастомный setup_logging

In [5]:
```python
log = None

def setup_logging(clean=False, debug=False):
    global log

    if log is not None:
        return log

    try:
        if clean and os.path.isfile('setup.log'):
            os.remove('setup.log')
        time.sleep(0.1)  # prevent race condition
    except:
        pass

    if sys.version_info >= (3, 9):
        logging.basicConfig(level=logging.DEBUG, format='%(asctime)s | %(levelname)
                            filename='setup.log', filemode='a', encoding='utf-8', f
    else:
        logging.basicConfig(level=logging.DEBUG, format='%(asctime)s | %(levelname)
```

```
                            filename='setup.log', filemode='a', force=True)

    console = Console(log_time=True, log_time_format='%H:%M:%S-%f', theme=Theme({
        "traceback.border": "black",
        "traceback.border.syntax_error": "black",
        "inspect.value.border": "black",
    }))
    pretty_install(console=console)
    traceback_install(console=console, extra_lines=1, width=console.width, word_wra
                      suppress=[])
    rh = RichHandler(show_time=True, omit_repeated_times=False, show_level=True, sh
                     rich_tracebacks=True, log_time_format='%H:%M:%S-%f',
                     level=logging.DEBUG if debug else logging.INFO, console=consol
    rh.set_name(logging.DEBUG if debug else logging.INFO)
    log = logging.getLogger("sd")
    log.addHandler(rh)

    return log
```

In [6]:
```
log = setup_logging()
```

Валидация входящих данных для каждого класса

validate_with_pydantic

In [7]:
```
def validate_with_pydantic(model_cls):
    """
    Декоратор для валидации данных с использованием Pydantic-модели.
    """

    def decorator(func):
        @wraps(func)
        def wrapper(*args, **kwargs):
            # Проверяем данные в аргументах функции
            try:
                data = kwargs.get("entry", args[0] if args else None)
                if not data:
                    raise HTTPException(status_code=status.HTTP_400_BAD_REQUEST,
                                        detail="No data provided for validation.")
                # Валидация данных
                if isinstance(data, BaseModel):
                    data = data.dict(by_alias=True)
                validated_data = model_cls(**data)
                # Передаем валидированные данные дальше
                kwargs["entry"] = validated_data
                return func(*args, **kwargs)
            except ValidationError as ve:
                log.exception("Validation failed", exc_info=ve)
                raise HTTPException(status_code=status.HTTP_400_BAD_REQUEST,
                                    detail="Invalid data for Pydantic model.") from

        return wrapper
```

```
    return decorator
```

auto_generate_docstring

In [8]:
```
def auto_generate_docstring(cls: Type[BaseModel]) -> Type[BaseModel]:
    """
    Декоратор для автоматического добавления docstring в классы Pydantic.
    """

    def generate_docstring(model: Type[BaseModel]) -> str:
        """
        Генерация строки документации из описания полей модели Pydantic.
        """
        docstring = []
        for field_name, field_info in model.__fields__.items():
            field_details = f"Field '{field_name}':\n"
            if field_info.description:  # Получение описания
                field_details += f"  Description: {field_info.description}\n"
            if field_info.examples:  # Получение примеров
                field_details += f"  Examples: {field_info.examples}\n"
            docstring.append(field_details)
        return "\n".join(docstring)

    # Добавляем описание к существующему docstring
    cls.__doc__ = (cls.__doc__ or "") + "\n\n" + generate_docstring(cls)
    return cls
```

EntryGraduateModel

In [338…
```
@auto_generate_docstring
class EntryGraduateModel(BaseModel):
    """
    Класс для валидации входных данных EntryGraduateModel
    """
    model_config = ConfigDict(arbitrary_types_allowed=True)
    Prefix: Optional[StrictStr] = Field("",
                                        alias="prefix",
                                        examples=["exp1"],
                                        descripton="Префикс к названию модели")
    Models: list[str] = Field(...,
                              alias="models",
                              examples=[["efficientnet_b0", "efficientnet_b1"]],
                              description="Список моделей для обучения")
    NameOptimizers: list[str] = Field(...,
                                      alias="name_optimizers",
                                      examples=[["Adam"]],
                                      description="Лист с названиями оптимизаторов
    EmbSize: Optional[StrictInt] = Field(512,
                                         alias="emb_size",
                                         examples=[512],
                                         description="Размер ембеддинга сиамской се
    UseImageNetWeights: Optional[bool] = Field(False,
```

```
                                          alias="is_use_imagenet_weights",
                                          examples=[True],
                                          description="Использовать предобучен
    SizeImg: Optional[tuple] = Field((64, 64),
                                     alias="size_img",
                                     examples=[(224, 224)],
                                     description="Размер изображения для обучения")
    EmbSize: StrictInt = Field(1024,
                               alias="emb_size",
                               examples=[1024],
                               description="Размер эмбеддинга модели")
    PathData: Optional[StrictStr] = Field("./data",
                                          alias="path_to_data",
                                          examples=["./data"],
                                          description="Путь к директории с данными.
    PathWeights: Optional[StrictStr] = Field("./weights",
                                             alias="path_to_weights",
                                             examples=["./weights"],
                                             description="Если сохраняем веса, то к
    PathMetricsTrain: Optional[StrictStr] = Field("./metrics_train",
                                                  alias="path_to_metrics_train",
                                                  examples=["./metrics_train"],
                                                  description="Куда сохраняем трени
    PathMetricsTest: Optional[StrictStr] = Field("./metrics_test",
                                                 alias="path_to_metrics_test",
                                                 examples=["./metrics_test"],
                                                 description="Куда сохраняем тестов
    PathSavePlots: Optional[StrictStr] = Field("./plot",
                                               alias="path_to_save_plots",
                                               examples=["./plot"],
                                               description="Куда сохраняем графики
    UseDevice: Optional[StrictStr] = Field(None,
                                           alias="use_device",
                                           examples=["cuda"],
                                           description="Какое устройство использова
    StartLerningRate: Optional[condecimal(ge=0.00000001, le=0.01, decimal_places=8)



    BatchSize: Optional[StrictInt] = Field(1,
                                           alias="batch_size",
                                           examples=[10],
                                           description="Размер пакета при обучении"
    NumWorkers: Optional[StrictInt] = Field(0,
                                            alias="num_workers",
                                            examples=[0],
                                            description="Кол. используемых потоков
    PinMemory: Optional[StrictBool] = Field(False,
                                            alias="pin_memory",
                                            examples=[False],
                                            description="Если True ускоряет загрузк
    NumEpochs: Optional[StrictInt] = Field(3,
                                           alias="num_epochs",
                                           examples=[20],
                                           description="Количество эпох обучения дл
    Seed: Optional[StrictInt] = Field(17,
```

```
                                    alias="seed",
                                    examples=[17],
                                    description="Сажает зерно")
```

EntryInferenceModel

In [353…

```
@auto_generate_docstring
class EntryInferenceModel(BaseModel):
    """
    Класс для валидации входных данных EntryInferenceModel
    """
    model_config = ConfigDict(arbitrary_types_allowed=True)
    Prefix: Optional[StrictStr] = Field("",
                                        alias="prefix",
                                        examples=["exp1"],
                                        descripton="Префикс к названию модели")
    NameModel: StrictStr = Field(...,
                                 alias="name_model",
                                 examples=["efficientnet_b0_2024_10_02)"],
                                 description="Название файла .pt")
    ImageList: list[StrictStr] = Field(...,
                                       alias="image_path_list",
                                       examples=[["./images/image.jpg"]],
                                       description="Список путей к изображениям")
    SizeImg: Optional[tuple] = Field((64, 64),
                                     alias="size_img",
                                     examples=[(224, 224)],
                                     description="Размер изображения для инференса"
    EmbSize: StrictInt = Field(1024,
                               alias="emb_size",
                               examples=[1024],
                               description="Размер эмбеддинга модели")
    PathWeights: Optional[StrictStr] = Field("./weights",
                                             alias="path_to_weights",
                                             examples=["./weights"],
                                             description="Путь к папке, где хранятс
    UseDevice: Optional[StrictStr] = Field(None,
                                           alias="use_device",
                                           examples=["cuda"],
                                           description="Какое устройство использова
    Labels: Optional[bool] = Field(False,
                                   alias="labels",
                                   examples=["True"],
                                   description="Есть ли в названиях файлов метки? Д
```

Загрузка и предобработка данных

Загрузка и распаковка набор данных PKU-Reid-Dataset

In [11]:

```python
class PKUReidDatasetDownloader:
    def __init__(self, download_url, target_dir='data'):
        self.download_url = download_url
        self.target_dir = target_dir
        self.dataset_zip = os.path.join(self.target_dir, 'PKU-Reid-Dataset.zip')
        os.makedirs(self.target_dir, exist_ok=True)

    def download_dataset(self):
        """Скачивание архива с датасетом"""
        response = requests.get(self.download_url, stream=True)
        total_size = int(response.headers.get('content-length', 0))

        with open(self.dataset_zip, 'wb') as file, tqdm(
            desc="Downloading PKU-Reid Dataset",
            total=total_size,
            unit='B',
            unit_scale=True,
            unit_divisor=1024
        ) as bar:
            for data in response.iter_content(chunk_size=1024):
                file.write(data)
                bar.update(len(data))

    def extract_dataset(self):
        """Распаковка архива"""
        with zipfile.ZipFile(self.dataset_zip, 'r') as zip_ref:
            zip_ref.extractall(self.target_dir)

    def organize_dataset(self):
        """Организация данных в структуру train/val/test"""
        source_folder = os.path.join(self.target_dir, 'PKUv1a_128x48')
        if not os.path.exists(source_folder):
            raise FileNotFoundError(f"Dataset folder {source_folder} not found!")

        # Создаем целевую структуру директорий
        splits = {
            'train': (1, 80),
            'val': (81, 97),
            'test': (98, 114)
        }

        for split, (start, end) in splits.items():
            split_dir = os.path.join(self.target_dir, split)
            os.makedirs(split_dir, exist_ok=True)

            # Фильтрация и перемещение файлов
            files = os.listdir(source_folder)
            for file in files:
                person_id = int(file.split('_')[0])
                if start <= person_id <= end:
                    src = os.path.join(source_folder, file)
                    dst = os.path.join(split_dir, file)
                    shutil.move(src, dst)

        # Удаление временных файлов
```

```
        shutil.rmtree(source_folder)
        os.remove(self.dataset_zip)

    def visualize_person(self, person_id, split=None):
        """Визуализация всех изображений конкретного человека"""
        splits = ['train', 'val', 'test'] if split is None else [split]
        found_images = []

        for split_name in splits:
            split_dir = os.path.join(self.target_dir, split_name)
            if not os.path.exists(split_dir):
                continue

            # Поиск файлов по ID
            files = [f for f in os.listdir(split_dir)
                    if int(f.split('_')[0]) == person_id]

            # Сортировка по камере и номеру кадра
            files.sort(key=lambda x: (int(x.split('_')[1]),
                                      int(x.split('_')[2].split('.')[0])))

            found_images.extend([os.path.join(split_dir, f) for f in files])

        # Отрисовка изображений
        if not found_images:
            print(f"No images found for person {person_id}")
            return

        plt.figure(figsize=(20, 3))
        for i, img_path in enumerate(found_images):
            plt.subplot(1, len(found_images), i+1)
            img = Image.open(img_path).convert('RGB')
            plt.imshow(img)
            plt.axis('off')
            plt.title(os.path.basename(img_path))
        plt.tight_layout()
        plt.show()

    def download_and_prepare(self):
        """Полный цикл подготовки данных"""
        self.download_dataset()
        self.extract_dataset()
        self.organize_dataset()
```

In [12]:
```
dataset_url = "https://drive.usercontent.google.com/u/0/uc?id=0B176otcbAxRITkRWYUwy
downloader = PKUReidDatasetDownloader(download_url=dataset_url)
downloader.download_and_prepare()
# Визуализация человека с ID=42 из обучающей выборки
downloader.visualize_person(42, split='train')
```

```
Downloading PKU-Reid Dataset: 100%|████████████████████████████████████████████|
4.06M/4.06M [00:00<00:00, 17.2MB/s]
```

Формирование датасета для обучения

Класс SiameseDataset

In [276…

```
class SiameseDataset(Dataset):
    def __init__(self, root_dir: str, transform: Optional[callable] = None) -> None
        self.root_dir = root_dir
        self.transform = transform
        self.image_paths = []
        self.labels = []
        self.label_to_indices = defaultdict(list)
        self._prepare_dataset()

    def _prepare_dataset(self):
        for subdir, _, files in os.walk(self.root_dir):
            for file in files:
                if file.endswith('.png'):
                    label = int(file.split('.png')[0].split('_')[0])
                    full_path = os.path.join(subdir, file)
                    self.image_paths.append(full_path)
                    self.labels.append(label)

        self.labels = np.array(self.labels)

        # Создаем словарь для быстрого доступа к индексам по меткам
        for idx, label in enumerate(self.labels):
            self.label_to_indices[label].append(idx)

    def __len__(self) -> int:
        return len(self.image_paths)

    def __getitem__(self, index: int) -> Tuple[torch.Tensor, torch.Tensor, torch.Te
        # Получаем первое изображение
        anchor_path = self.image_paths[index]
        anchor_label = self.labels[index]

        # Генерируем пару
        if random.random() < 0.5:
            # Положительная пара (одинаковый класс)
            same_class_indices = [i for i in self.label_to_indices[anchor_label] if
            if not same_class_indices:
                # Если нет других изображений в классе, создаем отрицательную пару
                positive = False
            else:
                pair_index = random.choice(same_class_indices)
                positive = True
        else:
```

```
            # Отрицательная пара (разные классы)
            positive = False

        if not positive:
            # Выбираем случайный другой класс
            other_labels = [l for l in self.label_to_indices.keys() if l != anchor_
            if not other_labels:
                # Если только один класс в датасете
                raise RuntimeError("Dataset contains only one class")
            chosen_label = random.choice(other_labels)
            pair_index = random.choice(self.label_to_indices[chosen_label])

        # Загружаем и преобразуем изображения
        anchor_img = Image.open(anchor_path).convert('RGB')
        pair_img = Image.open(self.image_paths[pair_index]).convert('RGB')

        if self.transform:
            anchor_img = self.transform(anchor_img)
            pair_img = self.transform(pair_img)

        return (
            anchor_img,
            pair_img,
            torch.tensor(1.0 if positive else 0.0, dtype=torch.float32)
        )
```

Класс SimpleDataset

In [277…

```
class SimpleDataset(Dataset):
    def __init__(self, root_dir: str, transform: Optional[callable] = None) -> None
        """
        Инициализация датасета для валидации и тестирования.

        :param root_dir: корневая директория с изображениями
        :param transform: преобразования для изображений
        """
        self.root_dir = root_dir
        self.transform = transform
        self.image_paths = []
        self.labels = []
        self._prepare_dataset()

    def _prepare_dataset(self) -> None:
        """Сбор данных и меток из файловой структуры."""
        for root, _, files in os.walk(self.root_dir):
            for file in files:
                if file.endswith('.png'):
                    file_path = os.path.join(root, file)
                    label = int(file.split('.png')[0].split('_')[0])
                    self.image_paths.append(file_path)
                    self.labels.append(label)

    def __len__(self) -> int:
        """Возвращает общее количество изображений."""
        return len(self.image_paths)
```

```
    def __getitem__(self, index: int) -> Tuple[torch.Tensor, torch.Tensor]:
        """Возвращает тензор изображения и соответствующую метку."""
        img_path = self.image_paths[index]
        label = self.labels[index]

        img = Image.open(img_path).convert('RGB')

        if self.transform:
            img = self.transform(img)

        return (
            img,  # Тензор изображения [C, H, W]
            torch.tensor(label, dtype=torch.long)  # Метка класса
        )
```

Класс ResizeWithPadding

In [278…

```
# Класс для приведения изображения к заданному размеру
class ResizeWithPadding:
    def __init__(self, size):
        self.size = size

    def __call__(self, img):
        # Приведение изображения к нужному размеру с сохранением соотношения сторон
        img.thumbnail((self.size, self.size), Image.Resampling.LANCZOS)

        # Добавление паддингов
        delta_width = self.size - img.size[0]
        delta_height = self.size - img.size[1]
        padding = (delta_width // 2, delta_height // 2, delta_width - (delta_width

        new_img = ImageOps.expand(img, padding, fill=(0, 0, 0))

        return new_img
```

Визуализация данных для обучения

In [287…

```
def visualize_augmentations(dataset, person_ids, n_samples=8):
    fig, axs = plt.subplots(len(person_ids), n_samples, figsize=(15, 5))

    for i, person_id in enumerate(person_ids):
        # Фильтруем изображения по ID
        indices = [idx for idx, label in enumerate(dataset.labels) if label == pers
        selected_indices = np.random.choice(indices, n_samples, replace=False)

        for j, idx in enumerate(selected_indices):
            img_tensor, label = dataset[idx]

            # Конвертируем тензор в PIL Image
            img = to_pil_image(img_tensor)

            axs[i,j].imshow(img)
```

```
            axs[i,j].axis('off')
            if j == 0:
                axs[i,j].set_title(f'ID: {person_id}')

    plt.tight_layout()
    plt.show()
```

In [296…

```
def visualize_siamese_pairs(dataset, num_pairs=5):
    """
    Визуализация пар изображений из SiameseDataset.

    :param dataset: экземпляр SiameseDataset
    :param num_pairs: количество пар для отображения
    """
    fig, axs = plt.subplots(num_pairs, 2, figsize=(10, 2 * num_pairs))

    for i in range(num_pairs):
        # Получаем случайный индекс из датасета
        idx = np.random.randint(0, len(dataset))

        # Загружаем пару изображений и метку
        anchor_img, pair_img, label = dataset[idx]

        # Конвертируем тензоры в PIL Image
        anchor_img = to_pil_image(anchor_img)
        pair_img = to_pil_image(pair_img)

        # Отображаем изображения
        axs[i, 0].imshow(anchor_img)
        axs[i, 0].axis('off')
        axs[i, 0].set_title("Anchor Image")

        axs[i, 1].imshow(pair_img)
        axs[i, 1].axis('off')
        axs[i, 1].set_title(f"Pair Image (Label: {'Positive' if label == 1 else 'Ne

    plt.tight_layout()
    plt.show()
```

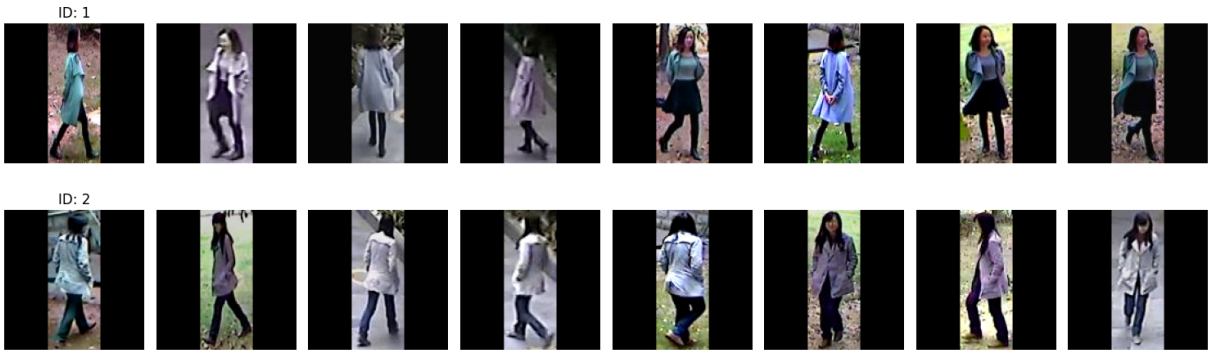
In [300…

```
# Преобразования для обучающего набора данных
train_transforms = transforms.Compose([
    ResizeWithPadding(128),
    transforms.RandomHorizontalFlip(p=0.5),
    transforms.ColorJitter(
        brightness=0.2,
        contrast=0.2,
        saturation=0.2,
        hue=0.1
    ),
    transforms.ToTensor()
])
```

In [301…

```
# Создаем временный SimpleDataset с тренировочными трансформациями
vis_dataset = SimpleDataset(root_dir='data/train', transform=train_transforms)
```

```
# Визуализируем первые два ID
visualize_augmentations(vis_dataset, person_ids=[1, 2])
```

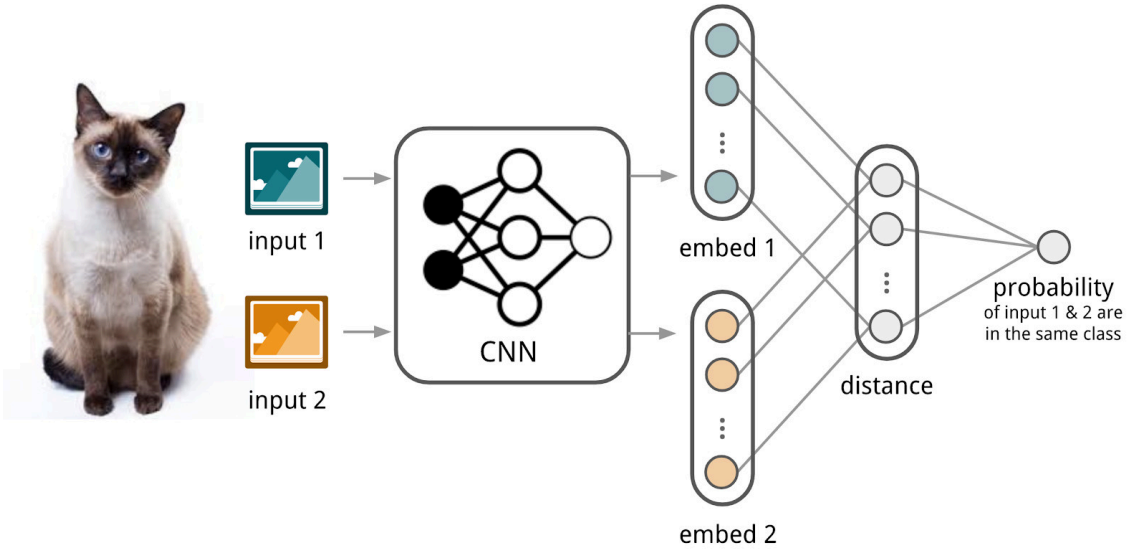


In [302…

```
# Создаем временный SiameseDataset с тренировочными трансформациями
vis_dataset = SiameseDataset(root_dir='data/train', transform=train_transforms)
# Визуализируем 5 случайных пар
visualize_siamese_pairs(vis_dataset, num_pairs=3)
```

Сиамская нейронная сеть

Класс SiameseNetwork



In [303…

```python
class SiameseNetwork(nn.Module):
    def __init__(
        self,
        name_model: str,
        emb_size: int = 512,
        imagenet: bool = True
    ):
        super().__init__()

        # Инициализируем параметры
        self.name_model = name_model
        self.emb_size = emb_size
        self.imagenet = imagenet

        # Получение списка доступных моделей
        self.model_list = sorted(name for name in models.__dict__
                                 if name.islower() and not name.startswith("__")
                                 and name != "get_weight"
                                 and callable(models.__dict__[name]))
        # Инициализируем архитектуру модели
        self.get_model()

        in_features = self.emb_size

        # Первая ветка полносвязных слоев
        self.fc = nn.Sequential(
            nn.Linear(in_features, self.emb_size * 2),
            nn.Tanh(),
            nn.Linear(self.emb_size * 2, self.emb_size * 2),
            nn.Tanh(),
```

```
            nn.Linear(self.emb_size * 2, self.emb_size * 2),
            nn.Tanh(),
            nn.Linear(self.emb_size * 2, self.emb_size),
        )

    def get_model(self):
        # Проверка доступности модели в torchvision
        if self.name_model in self.model_list:
            # Проверка доступности весов ImageNet
            if self.imagenet:
                try:
                    if self.name_model in ["inception_v3", "googlenet"]:
                        # Попытка загрузки модели с весами ImageNet
                        self.model = models.__dict__[self.name_model](init_weights=
                    else:
                        if self.name_model in ["regnet_y_128gf", "vit_h_14"]:
                            self.model = models.__dict__[self.name_model](weights="
                        else:
                            self.model = models.__dict__[self.name_model](weights="
                except KeyError:
                    # Если веса ImageNet для данной модели недоступны, загрузка без
                    print(
                        f"Предобученные веса ImageNet для модели {self.name_model}
                        f" Загрузка модели без предобученных весов.")
                    if self.name_model in ["inception_v3", "googlenet"]:
                        # Попытка загрузки модели с весами ImageNet
                        self.model = models.__dict__[self.name_model](init_weights=
                    else:
                        self.model = models.__dict__[self.name_model](weights=None)
            else:
                # Если не указано использование весов ImageNet, загрузить модель бе
                if self.name_model in ["inception_v3", "googlenet"]:
                    self.model = models.__dict__[self.name_model](init_weights=Fals
                else:
                    self.model = models.__dict__[self.name_model](weights=None)

            try:
                name_without_numbers = re.sub(r'\d+', '', self.name_model)
                # Проверка доступности слоя classifier
                if hasattr(self.model, 'classifier'):
                    if name_without_numbers == "densenet":
                        num_features = self.model.classifier.in_features
                        self.model.classifier = nn.Linear(num_features, self.emb_si
                    elif name_without_numbers == "squeezenet_":
                        num_features = 512
                        self.model.classifier[-1] = nn.Linear(num_features, self.em
                    else:
                        num_features = self.model.classifier[-1].in_features
                        self.model.classifier[-1] = nn.Linear(num_features, self.em
                elif hasattr(self.model, 'last_linear'):
                    num_features = self.model.last_linear.in_features
                    self.model.last_linear = nn.Linear(num_features, self.emb_size)
                elif hasattr(self.model, 'fc'):
                    num_features = self.model.fc.in_features
                    self.model.fc = nn.Linear(num_features, self.emb_size)
                elif hasattr(self.model, 'head'):
```

```
                    num_features = self.model.head.in_features
                    self.model.head = nn.Linear(num_features, self.emb_size)
                elif hasattr(self.model, "heads"):
                    num_features = self.model.heads[0].in_features
                    self.model.heads[0] = nn.Linear(num_features, self.emb_size)
                else:
                    print(f"Слой classifier не найден в модели. name_model:{self.na
            except Exception as ex:
                print(f"Exception: {ex}, name_model: {self.name_model}")
        else:
            print(f"Модель {self.name_model} не найдена в torchvision."
                  f"\nСписок доступных моделей: {self.model_list}")

    def forward_once(self, x: torch.Tensor) -> torch.Tensor:
        """
        Обработка одного изображения через выбранную ветку.

        :param x: входное изображение
        :param branch: номер ветки (1 или 2)
        :return: эмбеддинг изображения
        """
        features = self.model(x)
        return self.fc(features)

    def forward(self, input1: torch.Tensor, input2: torch.Tensor) -> Tuple[torch.Te
        """
        Обработка пары изображений через две ветки.

        :param input1: первое изображение
        :param input2: второе изображение
        :return: два эмбеддинга (от первой и второй ветки)
        """
        # Первое изображение обрабатывается через первую ветку
        embedding1 = self.forward_once(input1)

        # Второе изображение обрабатывается через вторую ветку
        embedding2 = self.forward_once(input2)

        return embedding1, embedding2
```

Контрастивная функция потерь

Класс ContrastiveLoss

$$L = y * ||f(I_p) - f(I_q)||_2^2 + (1 - y) * max\{0, m - ||f(I_p) - f(I_q)||_2\}^2$$

$y = 1$ if images I_p, I_q are similar (positive pairs)

$y = 0$ if images I_p, I_q are dissimilar (negative pairs)

$m -$ margin which dissimilar samples should stay apart from

$$L = y * ||f(I_p) - f(I_q)||_2^2 + (1 - y) * max(0, m - ||f(I_p) - f(I_q)||_2)^2$$

In [304…

```python
class ContrastiveLoss(nn.Module):
    def __init__(self, margin: float = 1.0):
        super().__init__()
        self.margin = margin

    def forward(self, output1: torch.Tensor, output2: torch.Tensor, label: torch.Te
        # Вычисляем Евклидово расстояние между эмбеддингами
        euclidean_distance = F.pairwise_distance(output1, output2)

        # Вычисляем потери для похожих и разных пар
        loss_similar = label * torch.pow(euclidean_distance, 2)
        loss_dissimilar = (1 - label) * torch.pow(
            torch.clamp(self.margin - euclidean_distance, min=0.0),
            2
        )

        # Усредняем потери по батчу
        loss_contrastive = torch.mean(loss_similar + loss_dissimilar)
        return loss_contrastive
```

Функция для расчета метрики mAP

In [305…

```python
def calculate_map(all_embs: List[np.ndarray], all_labels: List[int]) -> float:
    """
    Вычисление среднего значения AP (mAP) для всех эмбеддингов и их меток.

    :param all_embs: список эмбеддингов изображений.
    :param all_labels: список меток (id), соответствующих эмбеддингам.
    :return: mAP.
    """
    aps = []

    for idx in range(len(all_labels)):
        # Создаем временные копии данных
        embs = all_embs.copy()
        labels = all_labels.copy()

        # Извлекаем якорь и удаляем его из списков
        anchor_emb = embs.pop(idx)
        anchor_label = labels.pop(idx)

        # Преобразуем в питру массивы
        embs_array = np.array(embs)
        anchor_array = np.array(anchor_emb)

        # Вычисляем евклидовы расстояния
        distances = np.linalg.norm(embs_array - anchor_array, axis=1)

        # Сортируем метки по расстояниям
        sorted_indices = np.argsort(distances)
        sorted_labels = np.array(labels)[sorted_indices]
```

```
        # Вычисляем AP для текущего якоря
        correct = (sorted_labels == anchor_label)
        relevant = np.where(correct)[0]

        if len(relevant) == 0:
            ap = 0.0
        else:
            cumulative_correct = np.cumsum(correct)
            precision_at_k = cumulative_correct / (np.arange(len(correct)) + 1)
            ap = np.sum(precision_at_k[correct]) / len(relevant)

        aps.append(ap)

    return np.mean(aps)
```

Обучение моделей

GraduateModel

In [306…

```
class GraduateModel:

    def __init__(self,
                 prefix: str = "",
                 name_model: str = "efficientnet_b0",
                 path_to_data: str = "./data",
                 path_to_weights: str = "./weights",
                 path_to_metrics_train: str = "./metrics_train",
                 path_to_metrics_test: str = "./metrics_test",
                 is_use_imagenet_weights: bool = True,
                 name_optimizer: str = "Adam",
                 emb_size: int = 512,
                 num_epochs: int = 3,
                 batch_size: int = 1,
                 train_size_img: tuple = (128, 128),
                 use_device: str = None,
                 start_learning_rate: float = None,
                 pin_memory: bool = False,
                 num_workers: int = 0,
                 seed: int = 17):

        self.name_model = name_model
        self.name_model_user = f"{self.name_model}_{prefix}" if prefix != "" else s
        self.path_to_data = path_to_data
        directories_to_create = [path_to_weights, path_to_metrics_train, path_to_me
        self.create_directories_if_not_exist(*directories_to_create)
        self.path_to_weights = os.path.join(path_to_weights, f"{self.name_model_use
        self.path_to_metrics_train = os.path.join(path_to_metrics_train, f"{self.na
        self.path_to_metrics_test = os.path.join(path_to_metrics_test, f"{self.name
        self.num_epochs = num_epochs
        self.batch_size = batch_size
        self.size_img = train_size_img
```

```
        self.name_optimizer = name_optimizer
        self.emb_size = emb_size
        self.model = None
        self.train_loader = None
        self.valid_loader = None
        self.test_loader = None
        self.criterion = None
        self.scheduler = None
        self.imagenet = is_use_imagenet_weights
        self.use_device = use_device
        self.start_learning_rate = start_learning_rate
        self.pin_memory = pin_memory
        self.num_workers = num_workers
        self.seed = seed

        # Получение списка доступных моделей
        self.model_list = sorted(name for name in models.__dict__
                                 if name.islower() and not name.startswith("__")
                                 and name != "get_weight"
                                 and callable(models.__dict__[name]))

        # Перемещение модели на GPU, если CUDA доступен
        if self.use_device == None:
            self.device = torch.device("cuda" if torch.cuda.is_available() else "cp
        elif self.use_device == "cuda":
            self.device = torch.device("cuda")
        elif self.use_device == "cpu":
            if self.pin_memory == True:
                raise Exception("pin_memory musn't be True with use_device cpu")
            self.device = torch.device("cpu")
        else:
            raise Exception("use_device must be cuda or cpu or None")

    @staticmethod
    def seed_everything(seed):
        # Задаём seed для генераторов псевдослучайных чисел в Python, PyTorch и Num
        random.seed(seed)
        torch.manual_seed(seed)
        np.random.seed(seed)
        # Если доступна CUDA, задаём seed для GPU
        if torch.cuda.is_available():
            torch.cuda.manual_seed(seed)
            torch.cuda.manual_seed_all(seed)
        # Устанавливаем поведение CuDNN в детерминированный режим
        torch.backends.cudnn.deterministic = True
        # Включаем режим оптимизации CuDNN
        torch.backends.cudnn.benchmark = True
        # Создаем генератор случайных чисел PyTorch и задаём ему seed
        g = torch.Generator()
        g.manual_seed(seed)

    # Функция для установки seed в DataLoader
    @staticmethod
    def seed_worker(worker_id):
        worker_seed = torch.initial_seed() % 2**32
        np.random.seed(worker_seed)
```

```
        random.seed(worker_seed)

    def graduate(self):
        # Фиксируем зерно
        self.seed_everything(self.seed)
        # Получаем трансформер
        self.get_transform()
        # Получаем генераторы обучения, валидации и теста
        self.get_loaders()
        # Загружаем модель
        self.get_model()
        # Определяем оптимизатор, функцию потерь и планировщик
        self.get_opt_crit_sh()
        # Выводим информацию
        print(self.__str__())
        # Обучаем
        self.train_model()
        # Тестируем
        self.evaluate_model()

    def __str__(self):

        return f"""

        ----------------------------------------------------------------------

        Выбранная модель: {self.name_model}
        Пользовательское название модели: {self.name_model_user}
        Выбранный оптимизатор: {self.name_optimizer}

        ----------------------------------------------------------------------

        """

    def create_directories_if_not_exist(self, *directories):
        for directory in directories:
            if not os.path.exists(directory):
                os.makedirs(directory)

    # Функция для сохранения модели
    def save_model(self):
        # Сохраняем модель
        torch.save({
            'epoch': self.num_epochs,
            'model_state_dict': self.model.state_dict(),
            'optimizer_state_dict': self.optimizer.state_dict()},
            f"{self.path_to_weights}")

    def save_metrics_train(self,
                           train_loss_values,
                           score_values):
        metrics = {
            'train_loss': train_loss_values,
            'score': score_values
        }
        # Сохранение метрик
```

```
        torch.save(metrics, self.path_to_metrics_train)

    def save_metrics_test(self,
                          score):
        metric = {
            'score': score
        }
        # Сохранение метрик
        torch.save(metric, self.path_to_metrics_test)

    def get_transform(self):
        self.transform = {
            "train": transforms.Compose([
                ResizeWithPadding(self.size_img[0]),
                transforms.RandomHorizontalFlip(p=0.5),
                transforms.ColorJitter(
                    brightness=0.2,
                    contrast=0.2,
                    saturation=0.2,
                    hue=0.1
                ),
                transforms.ToTensor(),
                transforms.Normalize(
                    mean=[0.485, 0.456, 0.406],  # Средние значения для ImageNet
                    std=[0.229, 0.224, 0.225]    # Стандартные отклонения для Image
                )
            ]),
            "valid": transforms.Compose([
                ResizeWithPadding(self.size_img[0]),
                transforms.ToTensor(),
                transforms.Normalize(
                    mean=[0.485, 0.456, 0.406],  # Средние значения для ImageNet
                    std=[0.229, 0.224, 0.225]    # Стандартные отклонения для Image
                )
            ])
        }

    # Функция для загрузки данных
    def get_loaders(self):
        # Формирование датасетов
        self.train_dataset = SiameseDataset(root_dir='data/train', transform=self.t
        self.valid_dataset = SimpleDataset(root_dir='data/val', transform=self.tran
        self.test_dataset = SimpleDataset(root_dir='data/test', transform=self.tran
        # Формирование загрузчиков данных
        self.train_loader = DataLoader(self.train_dataset, batch_size=self.batch_si
                                       num_workers=self.num_workers, pin_memory=self.pin
        self.valid_loader = DataLoader(self.valid_dataset, batch_size=self.batch_si
                                       num_workers=self.num_workers, pin_memory=self.pin_m
        self.test_loader = DataLoader(self.test_dataset, batch_size=self.batch_size
                                      num_workers=self.num_workers, pin_memory=self.pin_

    def get_model(self):
        # Инициализация сиамской сети
        self.model = SiameseNetwork(self.name_model, self.emb_size, self.imagenet).

    def get_opt_crit_sh(self):
```

```
        self.criterion = ContrastiveLoss(margin=1.0).to(self.device)
        # Определение оптимизатора
        if self.start_learning_rate is not None:
            lr = self.start_learning_rate
        else:
            lr = 0.001

        self.optimizer = optim.__dict__[f"{self.name_optimizer}"](self.model.parame
        # Создание планировщика LR
        # ReduceLROnPlateau уменьшает скорость обучения, когда метрика перестает ум
        self.scheduler = ReduceLROnPlateau(self.optimizer, mode='min', patience=2,

    # Функция для обучения модели с валидацией
    def train_model(self):

        train_loss_values = []
        valid_loss_values = []
        mAP_values = []

        for epoch in range(self.num_epochs):
            self.model.train()
            running_loss = 0.0
            for inputs1, inputs2, labels in tqdm(self.train_loader, desc=f"Epoch {e
                                        unit="sample"):
                inputs1, inputs2, labels = inputs1.cuda(), inputs2.cuda(), labels.c
                self.optimizer.zero_grad()
                emb1, emb2 = self.model(inputs1, inputs2)
                loss = self.criterion(emb1, emb2, labels)
                loss.backward()
                self.optimizer.step()
                running_loss += loss.item() * inputs1.size(0)

            # Вычисление loss на валидационном датасете и метрик
            self.model.eval()
            valid_loss = 0.0
            best_mAP = 0.0
            all_embeds = []
            all_labels = []
            with torch.no_grad():
                for inputs, labels in tqdm(self.valid_loader, desc=f"Epoch {epoch +
                                            unit="sample"):
                    inputs, labels = inputs.cuda(), labels.cuda()
                    outputs = self.model.forward_once(inputs)
                    all_embeds += [out.cpu().numpy() for out in outputs]
                    all_labels += labels.cpu().numpy().tolist()

            epoch_loss = running_loss / len(self.train_loader.dataset)

            mAP = calculate_map(all_embeds, all_labels)  # Вычисление mAP для текущ

            # Сообщаем планировщику LR о текущей ошибке на валидационном наборе
            self.scheduler.step(mAP)

            # we want to save the model if the accuracy is the best
            if mAP > best_mAP:
                self.save_model()
```

```
                best_mAP = mAP

            # Добавление значений метрик в списки
            train_loss_values.append(epoch_loss)
            mAP_values.append(mAP)

            # Сохранение метрик
            self.save_metrics_train(train_loss_values,
                                    mAP_values)

            print(f"\nEpoch {epoch + 1}/{self.num_epochs}, Training Loss: {epoch_lo
            print(f"mAP: {mAP}")

        print("Тренировка завершена!")

    # Функция для оценки модели на тестовом датасете
    def evaluate_model(self):
        self.model.eval()
        all_embeds = []
        all_labels = []

        with torch.no_grad():
            for inputs, labels in tqdm(self.valid_loader, desc=f"(Eval)",
                                       unit="sample"):
                inputs, labels = inputs.cuda(), labels.cuda()
                outputs = self.model.forward_once(inputs)
                all_embeds += [out.cpu().numpy() for out in outputs]
                all_labels += labels.cpu().numpy().tolist()

        mAP = calculate_map(all_embeds, all_labels)  # Вычисление mAP для текущей э
        print(f"Test mAP: {mAP}")

        self.save_metrics_test(mAP)
```

GraduateModelPipeline

In [340…

```
@dataclass
class GraduateModelPipeline:
    entry: EntryGraduateModel

    def __post_init__(self):
        pass

    def graduate(self):

        prefix = self.entry.Prefix
        models = self.entry.Models
        name_optimizers = self.entry.NameOptimizers
        is_use_imagenet_weights = self.entry.UseImageNetWeights
        path_to_data = self.entry.PathData
        path_to_weights = self.entry.PathWeights
        use_device = self.entry.UseDevice
        start_learning_rate = self.entry.StartLerningRate
        train_size_img = self.entry.SizeImg
```

```
        emb_size = self.entry.EmbSize
        batch_size = self.entry.BatchSize
        num_workers = self.entry.NumWorkers
        pin_memory = self.entry.PinMemory
        num_epochs = self.entry.NumEpochs
        seed = self.entry.Seed

        path_to_metrics_train = self.entry.PathMetricsTrain
        path_to_metrics_test = self.entry.PathMetricsTest
        path_to_save_plots = self.entry.PathSavePlots

        for index, name_model in enumerate(models):

            try:
                train = GraduateModel(
                            prefix=prefix,
                            name_model=name_model,
                            path_to_data=path_to_data,
                            path_to_weights=path_to_weights,
                            path_to_metrics_train=path_to_metrics_train,
                            path_to_metrics_test=path_to_metrics_test,
                            is_use_imagenet_weights=is_use_imagenet_weights,
                            name_optimizer=name_optimizers[index],
                            num_epochs=num_epochs,
                            batch_size=batch_size,
                            train_size_img=train_size_img,
                            emb_size=emb_size,
                            use_device=use_device,
                            num_workers=num_workers,
                            pin_memory=pin_memory,
                            seed=seed)
                train.graduate()
            except Exception as ex:
                log.exception("GraduateModel\n", exc_info=ex)

        # Построение графиков
        try:
            metrics_visualizer = MetricsVisualizer(prefix=prefix,
                                                   path_to_metrics_train=path_to_me
                                                   path_to_metrics_test=path_to_met
                                                   path_to_save_plots=path_to_save_
        except Exception as ex:
            log.exception("MetricsVisualiser\n", exc_info=ex)
```

Визуализация графиков обучения

MetricsVisualizer

In [233…

```
class MetricsVisualizer:
    def __init__(self,
                 prefix: str = "",
                 path_to_metrics_train: str = "./metrics_train",
```

```
                path_to_metrics_test: str = "./metrics_test",
                path_to_save_plots: str = None):

        self.prefix = prefix
        if path_to_save_plots:
            os.makedirs(path_to_save_plots, exist_ok=True)
        self.path_to_save_plots = path_to_save_plots if path_to_save_plots is not N
        self.metrics_train_dir = os.path.join(path_to_metrics_train)
        self.metrics_test_dir = os.path.join(path_to_metrics_test)
        self.train_loss_values = {}
        self.score_values_valid = {}
        self.score_values_test = {}

        # Инициализируем сохранение графиков
        self.load_train_metrics()
        self.load_test_metrics()
        self.plot_metrics()

    def _load_metrics(self, directory, files_dict, key_name):
        for file in os.listdir(directory):
            if file.endswith(".pt") and f"_{self.prefix}" in file:
                metrics = torch.load(os.path.join(directory, file))
                model_name = file.replace('.pt', '').replace(f'_{self.prefix}', '')
                files_dict[model_name] = metrics[key_name]

    def load_train_metrics(self):
        self._load_metrics(self.metrics_train_dir, self.train_loss_values, 'train_l
        self._load_metrics(self.metrics_train_dir, self.score_values_valid, 'score'

    def load_test_metrics(self):
        self._load_metrics(self.metrics_test_dir, self.score_values_test, 'score')

    def plot_metrics(self):
        fig, axs = plt.subplots(3, 1, figsize=(14, 20))

        # Сравнение категориальной кроссэнтропии для разных моделей на тренировке
        for model, train_loss in self.train_loss_values.items():
            axs[0].plot(train_loss, label=model, linewidth=2)
        axs[0].set_title('Функция потерь на тренировке', fontsize=12)
        axs[0].set_xlabel('Эпоха')
        axs[0].set_ylabel('Значение функции потерь', fontsize=12)

        # Вычисление адаптивных лимитов для оси Y
        all_values = [value for values in self.train_loss_values.values() for value
        lower_limit = np.percentile(all_values, 0)  # нижний предел (например, 5-й
        upper_limit = np.percentile(all_values, 95)  # верхний предел (например, 95

        try:
            # Применение ограничений
            axs[0].set_ylim(lower_limit, upper_limit)
        except Exception as ex:
            pass

        for model, score in self.score_values_valid.items():
            axs[1].plot(score, label=model, linewidth=2)
        axs[1].set_title('mAP на валидации', fontsize=12)
```

```
        axs[1].set_xlabel('Эпоха')
        axs[1].set_ylabel('Значение метрики mAP', fontsize=12)

        for model, score in self.score_values_test.items():
            bar = axs[2].bar(model, score, label=model)
            axs[2].text(bar.patches[0].get_x() + bar.patches[0].get_width() / 2.,
                        bar.patches[0].get_height() + 0.01,
                        f'{score:.2f}',
                        ha='center', va='bottom', rotation=90, fontsize=8, color

        axs[2].set_title('mAP на тесте', fontsize=12)
        axs[2].set_xlabel('Название модели')
        axs[2].set_ylabel('Значение метрики mAP', fontsize=12)
        plt.setp(axs[2].get_xticklabels(), rotation=90, fontsize=6)

        # Добавление общей легенды
        handles, labels = axs[0].get_legend_handles_labels()
        fig.legend(handles, labels, loc='center right', fontsize=8, frameon=False)

        # Обрезаем графики, чтобы освободить место для легенды
        plt.subplots_adjust(left=0.1, right=0.8, top=0.9, bottom=0.1)
        path = os.path.join(self.path_to_save_plots, f"PlotsMetrics_{self.prefix}.p
        plt.savefig(path, dpi=500)
        plt.show()
```

Эксперименты

Список доступных моделей

In [308…

```
exclude_model = ["get_weight", "convnext_base", "convnext_large", "convnext_small",
                 "densenet161", "efficientnet_b5", "efficientnet_b6", "efficientnet
                 "efficientnet_v2_l", "efficientnet_v2_m", "regnet_x_8gf", "regnet_
                 "regnet_y_32gf", "regnet_y_128gf", "resnext101_32x8d", "vit_h_14",
                 "vit_l_32", "googlenet", "inception_v3", "squeezenet1_0", "squeeze

optimizer_list = ["SGD", "SGD", "SGD", "SGD", "SGD", "AdamW", "AdamW", "AdamW", "Ad
                  "AdamW", "AdamW", "AdamW", "AdamW", "AdamW", "AdamW", "AdamW", "A
                  "SGD", "SGD", "SGD", "SGD", "SGD", "SGD", "SGD", "SGD", "SGD", "S
                  "SGD", "SGD", "SGD", "SGD", "SGD", "AdamW", "AdamW", "AdamW", "Ad
                  "SGD", "SGD", "SGD", "SGD", "SGD", "SGD", "SGD", "SGD", "SGD", "S


# Получение списка доступных моделей
model_list = sorted(name for name in models.__dict__
                    if name.islower() and not name.startswith("__")
                    and name not in exclude_model
                    and callable(models.__dict__[name]))
print(model_list)
```

```
['alexnet', 'convnext_tiny', 'densenet121', 'densenet169', 'densenet201', 'efficient
net_b0', 'efficientnet_b1', 'efficientnet_b2', 'efficientnet_b3', 'efficientnet_b4',
'efficientnet_v2_s', 'mnasnet0_5', 'mnasnet0_75', 'mnasnet1_0', 'mnasnet1_3', 'mobil
enet_v2', 'mobilenet_v3_large', 'mobilenet_v3_small', 'regnet_x_16gf', 'regnet_x_1_6
gf', 'regnet_x_3_2gf', 'regnet_x_400mf', 'regnet_x_800mf', 'regnet_y_16gf', 'regnet_
y_1_6gf', 'regnet_y_3_2gf', 'regnet_y_400mf', 'regnet_y_800mf', 'regnet_y_8gf', 'res
net101', 'resnet152', 'resnet18', 'resnet34', 'resnet50', 'resnext101_64x4d', 'resne
xt50_32x4d', 'shufflenet_v2_x0_5', 'shufflenet_v2_x1_0', 'shufflenet_v2_x1_5', 'shuf
flenet_v2_x2_0', 'swin_b', 'swin_s', 'swin_t', 'vgg11', 'vgg11_bn', 'vgg13', 'vgg13_
bn', 'vgg16', 'vgg16_bn', 'vgg19', 'vgg19_bn', 'vit_b_16', 'vit_b_32', 'wide_resnet1
01_2', 'wide_resnet50_2']
```

Обучение разных бекбонов

In [310…

```python
graduate_pipeline = validate_with_pydantic(EntryGraduateModel)(GraduateModelPipelin
    entry = {
        "prefix": "Exp1",
        "models": model_list,
        "name_optimizers": optimizer_list,
        "size_img": (128, 128),
        "batch_size": 5,
        "num_epochs": 10,
        "emb_size": 512,
        "is_use_imagenet_weights": True,
        "use_device": "cuda",
        "start_learning_rate": 0.0005,
        "num_workers": 0,
        "pin_memory": True,
        "seed": 17
    }
)
```

In [311…

```python
graduate_pipeline.graduate()
```

```
----------------------------------------------------------------------

Выбранная модель: alexnet
Пользовательское название модели: alexnet_Exp1
Выбранный оптимизатор: SGD

----------------------------------------------------------------------
```

```
Epoch 1/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:11<00:00, 21.52sample/s]
Epoch 1/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:00<00:00, 122.02sample/s]
Epoch 1/10, Training Loss: 0.31296916179417167
mAP: 0.1392550996941322
Epoch 2/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:11<00:00, 23.19sample/s]
Epoch 2/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:00<00:00, 149.86sample/s]
```

```
Epoch 2/10, Training Loss: 0.2452834822179284
mAP: 0.14559630232005621
Epoch 3/10 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:10<00:00, 23.51sample/s]
Epoch 3/10 (Eval): 100%|████████████████████████████████████████████████|
55/55 [00:00<00:00, 157.53sample/s]
Epoch 3/10, Training Loss: 0.2329659157549031
mAP: 0.1591303693577224
Epoch 4/10 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:11<00:00, 22.63sample/s]
Epoch 4/10 (Eval): 100%|████████████████████████████████████████████████|
55/55 [00:00<00:00, 155.79sample/s]
Epoch 00004: reducing learning rate of group 0 to 1.0000e-04.

Epoch 4/10, Training Loss: 0.22673870463040657
mAP: 0.17422722212254446
Epoch 5/10 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:10<00:00, 23.69sample/s]
Epoch 5/10 (Eval): 100%|████████████████████████████████████████████████|
55/55 [00:00<00:00, 153.19sample/s]
Epoch 5/10, Training Loss: 0.2273974741983693
mAP: 0.17373262922965743
Epoch 6/10 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:10<00:00, 24.07sample/s]
Epoch 6/10 (Eval): 100%|████████████████████████████████████████████████|
55/55 [00:00<00:00, 160.78sample/s]
Epoch 6/10, Training Loss: 0.2250700290023815
mAP: 0.17541827158546675
Epoch 7/10 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:11<00:00, 23.17sample/s]
Epoch 7/10 (Eval): 100%|████████████████████████████████████████████████|
55/55 [00:00<00:00, 159.88sample/s]
Epoch 00007: reducing learning rate of group 0 to 1.0000e-05.

Epoch 7/10, Training Loss: 0.23040701806894504
mAP: 0.17589391418073227
Epoch 8/10 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:10<00:00, 23.95sample/s]
Epoch 8/10 (Eval): 100%|████████████████████████████████████████████████|
55/55 [00:00<00:00, 159.37sample/s]
Epoch 8/10, Training Loss: 0.22705106626381166
mAP: 0.17575465893055994
Epoch 9/10 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:10<00:00, 23.95sample/s]
Epoch 9/10 (Eval): 100%|████████████████████████████████████████████████|
55/55 [00:00<00:00, 155.85sample/s]
Epoch 9/10, Training Loss: 0.22628712162259035
mAP: 0.17610705855131628
Epoch 10/10 (Train): 100%|███████████████████████████████████████████████| 2
56/256 [00:10<00:00, 23.62sample/s]
Epoch 10/10 (Eval): 100%|███████████████████████████████████████████████|
55/55 [00:00<00:00, 150.14sample/s]
```

```
Epoch 00010: reducing learning rate of group 0 to 1.0000e-06.

Epoch 10/10, Training Loss: 0.2245840939867776
mAP: 0.1761827571779816
Тренировка завершена!
(Eval): 100%|████████████████████████████████████████████████████████████|
55/55 [00:00<00:00, 131.72sample/s]
Test mAP: 0.1761827571779816


        ----------------------------------------------------------------------

        Выбранная модель: convnext_tiny
        Пользовательское название модели: convnext_tiny_Exp1
        Выбранный оптимизатор: SGD

        ----------------------------------------------------------------------


Epoch 1/10 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:24<00:00, 10.26sample/s]
Epoch 1/10 (Eval): 100%|████████████████████████████████████████████████|
55/55 [00:00<00:00, 70.25sample/s]
Epoch 1/10, Training Loss: 0.2420376272420981
mAP: 0.3592120204436654
Epoch 2/10 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:25<00:00, 10.07sample/s]
Epoch 2/10 (Eval): 100%|████████████████████████████████████████████████|
55/55 [00:00<00:00, 71.90sample/s]
Epoch 2/10, Training Loss: 0.20725657729781233
mAP: 0.3664644670376315
Epoch 3/10 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:25<00:00, 10.03sample/s]
Epoch 3/10 (Eval): 100%|████████████████████████████████████████████████|
55/55 [00:00<00:00, 71.12sample/s]
Epoch 3/10, Training Loss: 0.19287486592656933
mAP: 0.41173657685577525
Epoch 4/10 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.91sample/s]
Epoch 4/10 (Eval): 100%|████████████████████████████████████████████████|
55/55 [00:00<00:00, 68.67sample/s]
Epoch 00004: reducing learning rate of group 0 to 1.0000e-04.

Epoch 4/10, Training Loss: 0.1787556218187092
mAP: 0.42517025543804376
Epoch 5/10 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:27<00:00,  9.21sample/s]
Epoch 5/10 (Eval): 100%|████████████████████████████████████████████████|
55/55 [00:00<00:00, 58.57sample/s]
Epoch 5/10, Training Loss: 0.1659707118960796
mAP: 0.4420990862162339
```

```
Epoch 6/10 (Train): 100%|██████████| 256/256 [00:30<00:00,  8.51sample/s]
Epoch 6/10 (Eval): 100%|██████████| 55/55 [00:00<00:00, 64.92sample/s]
Epoch 6/10, Training Loss: 0.16874152292439248
mAP: 0.4434876584257468
Epoch 7/10 (Train): 100%|██████████| 256/256 [00:26<00:00,  9.63sample/s]
Epoch 7/10 (Eval): 100%|██████████| 55/55 [00:00<00:00, 59.92sample/s]
Epoch 00007: reducing learning rate of group 0 to 1.0000e-05.

Epoch 7/10, Training Loss: 0.16720569122117013
mAP: 0.442552414142927
Epoch 8/10 (Train): 100%|██████████| 256/256 [00:27<00:00,  9.45sample/s]
Epoch 8/10 (Eval): 100%|██████████| 55/55 [00:00<00:00, 71.91sample/s]
Epoch 8/10, Training Loss: 0.15834165006526746
mAP: 0.44384679916988856
Epoch 9/10 (Train): 100%|██████████| 256/256 [00:26<00:00,  9.84sample/s]
Epoch 9/10 (Eval): 100%|██████████| 55/55 [00:00<00:00, 67.21sample/s]
Epoch 9/10, Training Loss: 0.16524921772361267
mAP: 0.44571988223257625
Epoch 10/10 (Train): 100%|██████████| 256/256 [00:26<00:00,  9.54sample/s]
Epoch 10/10 (Eval): 100%|██████████| 55/55 [00:00<00:00, 58.18sample/s]
Epoch 00010: reducing learning rate of group 0 to 1.0000e-06.

Epoch 10/10, Training Loss: 0.16010307699616533
mAP: 0.4465276441102752
Тренировка завершена!
(Eval): 100%|██████████| 55/55 [00:00<00:00, 61.30sample/s]
Test mAP: 0.4465276441102752


        ----------------------------------------------------------------------

        Выбранная модель: densenet121
        Пользовательское название модели: densenet121_Exp1
        Выбранный оптимизатор: SGD

        ----------------------------------------------------------------------


Epoch 1/10 (Train): 100%|██████████| 256/256 [01:00<00:00,  4.26sample/s]
Epoch 1/10 (Eval): 100%|██████████| 55/55 [00:01<00:00, 33.53sample/s]
```

```
Epoch 1/10, Training Loss: 0.45255943820689026
mAP: 0.11211126853819527
Epoch 2/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.75sample/s]
Epoch 2/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 32.85sample/s]
Epoch 2/10, Training Loss: 0.30936313100392
mAP: 0.11345065054626155
Epoch 3/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:54<00:00,  4.70sample/s]
Epoch 3/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 34.33sample/s]
Epoch 3/10, Training Loss: 0.2654756532574538
mAP: 0.10876011500995887
Epoch 4/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.76sample/s]
Epoch 4/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 34.03sample/s]
Epoch 4/10, Training Loss: 0.2520573773654178
mAP: 0.11742499691248924
Epoch 5/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.78sample/s]
Epoch 5/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 34.59sample/s]
Epoch 5/10, Training Loss: 0.24917301448294893
mAP: 0.1248071840936929
Epoch 6/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.79sample/s]
Epoch 6/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 34.13sample/s]
Epoch 00006: reducing learning rate of group 0 to 1.0000e-04.

Epoch 6/10, Training Loss: 0.24496416526380926
mAP: 0.12579843926683185
Epoch 7/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.81sample/s]
Epoch 7/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 35.38sample/s]
Epoch 7/10, Training Loss: 0.24577839352423325
mAP: 0.12780814555908623
Epoch 8/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.78sample/s]
Epoch 8/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 34.43sample/s]
Epoch 8/10, Training Loss: 0.2460851634386927
mAP: 0.12438045531944177
Epoch 9/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:54<00:00,  4.67sample/s]
Epoch 9/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 33.02sample/s]
Epoch 00009: reducing learning rate of group 0 to 1.0000e-05.

Epoch 9/10, Training Loss: 0.24497468868503347
mAP: 0.1299637832408077
```

```
Epoch 10/10 (Train): 100%|████████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.74sample/s]
Epoch 10/10 (Eval): 100%|███████████████████████████████████████████████████████|
55/55 [00:01<00:00, 33.65sample/s]
Epoch 10/10, Training Loss: 0.2440913062891923
mAP: 0.12993182293676966
Тренировка завершена!
(Eval): 100%|███████████████████████████████████████████████████████████████████|
55/55 [00:01<00:00, 35.43sample/s]
Test mAP: 0.12993182293676966


        ----------------------------------------------------------------------

        Выбранная модель: densenet169
        Пользовательское название модели: densenet169_Exp1
        Выбранный оптимизатор: SGD

        ----------------------------------------------------------------------


Epoch 1/10 (Train): 100%|████████████████████████████████████████████████████| 2
56/256 [01:13<00:00,  3.48sample/s]
Epoch 1/10 (Eval): 100%|███████████████████████████████████████████████████████|
55/55 [00:02<00:00, 25.62sample/s]
Epoch 1/10, Training Loss: 0.3411007006889122
mAP: 0.10888243077644602
Epoch 2/10 (Train): 100%|████████████████████████████████████████████████████| 2
56/256 [01:13<00:00,  3.48sample/s]
Epoch 2/10 (Eval): 100%|███████████████████████████████████████████████████████|
55/55 [00:02<00:00, 25.06sample/s]
Epoch 2/10, Training Loss: 0.2689396054192912
mAP: 0.12255411683037556
Epoch 3/10 (Train): 100%|████████████████████████████████████████████████████| 2
56/256 [01:13<00:00,  3.50sample/s]
Epoch 3/10 (Eval): 100%|███████████████████████████████████████████████████████|
55/55 [00:02<00:00, 25.64sample/s]
Epoch 3/10, Training Loss: 0.25156891683582217
mAP: 0.12801575143837463
Epoch 4/10 (Train): 100%|████████████████████████████████████████████████████| 2
56/256 [01:12<00:00,  3.54sample/s]
Epoch 4/10 (Eval): 100%|███████████████████████████████████████████████████████|
55/55 [00:02<00:00, 25.89sample/s]
Epoch 00004: reducing learning rate of group 0 to 1.0000e-04.

Epoch 4/10, Training Loss: 0.24606041179504246
mAP: 0.13229240675238538
Epoch 5/10 (Train): 100%|████████████████████████████████████████████████████| 2
56/256 [01:13<00:00,  3.46sample/s]
Epoch 5/10 (Eval): 100%|███████████████████████████████████████████████████████|
55/55 [00:02<00:00, 25.17sample/s]
Epoch 5/10, Training Loss: 0.24398436798946932
mAP: 0.13520635834924832
```

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Epoch 6/10 (Train): 100%|██████████████████████████████████████████████████████| 2
56/256 [01:12<00:00,  3.51sample/s]
Epoch 6/10 (Eval): 100%|████████████████████████████████████████████████████████|
55/55 [00:02<00:00, 26.05sample/s]
Epoch 6/10, Training Loss: 0.24521717394236475
mAP: 0.13410158913243891
Epoch 7/10 (Train): 100%|██████████████████████████████████████████████████████| 2
56/256 [01:12<00:00,  3.54sample/s]
Epoch 7/10 (Eval): 100%|████████████████████████████████████████████████████████|
55/55 [00:02<00:00, 25.47sample/s]
Epoch 00007: reducing learning rate of group 0 to 1.0000e-05.

Epoch 7/10, Training Loss: 0.2469468319322914
mAP: 0.13567288851899695
Epoch 8/10 (Train): 100%|██████████████████████████████████████████████████████| 2
56/256 [01:12<00:00,  3.53sample/s]
Epoch 8/10 (Eval): 100%|████████████████████████████████████████████████████████|
55/55 [00:02<00:00, 25.28sample/s]
Epoch 8/10, Training Loss: 0.24437719851266593
mAP: 0.12862555131842446
Epoch 9/10 (Train): 100%|██████████████████████████████████████████████████████| 2
56/256 [01:12<00:00,  3.53sample/s]
Epoch 9/10 (Eval): 100%|████████████████████████████████████████████████████████|
55/55 [00:02<00:00, 26.52sample/s]
Epoch 9/10, Training Loss: 0.24364256649278104
mAP: 0.13300804481571513
Epoch 10/10 (Train): 100%|█████████████████████████████████████████████████████| 2
56/256 [01:11<00:00,  3.56sample/s]
Epoch 10/10 (Eval): 100%|███████████████████████████████████████████████████████|
55/55 [00:02<00:00, 26.04sample/s]
Epoch 00010: reducing learning rate of group 0 to 1.0000e-06.

Epoch 10/10, Training Loss: 0.24283002462470904
mAP: 0.1353440010786441
Тренировка завершена!
(Eval): 100%|███████████████████████████████████████████████████████████████████|
55/55 [00:02<00:00, 25.11sample/s]
Test mAP: 0.1353440010786441


        ----------------------------------------------------------------------

        Выбранная модель: densenet201
        Пользовательское название модели: densenet201_Exp1
        Выбранный оптимизатор: SGD

        ----------------------------------------------------------------------


Epoch 1/10 (Train): 100%|██████████████████████████████████████████████████████| 2
56/256 [01:26<00:00,  2.96sample/s]
Epoch 1/10 (Eval): 100%|████████████████████████████████████████████████████████|
55/55 [00:03<00:00, 17.99sample/s]
```

```
Epoch 1/10, Training Loss: 0.3160666330186359
mAP: 0.11660693069963368
Epoch 2/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [01:25<00:00,  3.01sample/s]
Epoch 2/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:02<00:00, 21.82sample/s]
Epoch 2/10, Training Loss: 0.27313150046393275
mAP: 0.11903601507097364
Epoch 3/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [01:25<00:00,  3.00sample/s]
Epoch 3/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:02<00:00, 22.74sample/s]
Epoch 3/10, Training Loss: 0.2555298166698776
mAP: 0.12120660896055012
Epoch 4/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [01:24<00:00,  3.03sample/s]
Epoch 4/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:02<00:00, 22.95sample/s]
Epoch 00004: reducing learning rate of group 0 to 1.0000e-04.

Epoch 4/10, Training Loss: 0.24595513253007084
mAP: 0.12333498081814272
Epoch 5/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [01:25<00:00,  2.99sample/s]
Epoch 5/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:02<00:00, 22.07sample/s]
Epoch 5/10, Training Loss: 0.24599000229500234
mAP: 0.12729650325152808
Epoch 6/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [01:25<00:00,  2.99sample/s]
Epoch 6/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:02<00:00, 22.66sample/s]
Epoch 6/10, Training Loss: 0.24656360049266368
mAP: 0.12487019021226968
Epoch 7/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [01:25<00:00,  2.99sample/s]
Epoch 7/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:02<00:00, 22.82sample/s]
Epoch 00007: reducing learning rate of group 0 to 1.0000e-05.

Epoch 7/10, Training Loss: 0.24745091702789068
mAP: 0.12789069393264518
Epoch 8/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [01:25<00:00,  3.00sample/s]
Epoch 8/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:02<00:00, 22.50sample/s]
Epoch 8/10, Training Loss: 0.24654953071149066
mAP: 0.13026280515054692
Epoch 9/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [01:25<00:00,  2.99sample/s]
Epoch 9/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:02<00:00, 22.79sample/s]
Epoch 9/10, Training Loss: 0.2446863838704303
mAP: 0.12747252676224138
```

```
Epoch 10/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [01:25<00:00,  2.98sample/s]
Epoch 10/10 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:02<00:00, 22.36sample/s]
Epoch 00010: reducing learning rate of group 0 to 1.0000e-06.

Epoch 10/10, Training Loss: 0.24762803077464923
mAP: 0.12931359151659116
Тренировка завершена!
(Eval): 100%|███████████████████████████████████████████████████████████████|
55/55 [00:02<00:00, 22.42sample/s]
Test mAP: 0.12931359151659116


        ----------------------------------------------------------------------

        Выбранная модель: efficientnet_b0
        Пользовательское название модели: efficientnet_b0_Exp1
        Выбранный оптимизатор: AdamW

        ----------------------------------------------------------------------


Epoch 1/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:40<00:00,  6.28sample/s]
Epoch 1/10 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:01<00:00, 46.07sample/s]
Epoch 1/10, Training Loss: 0.5827267359563848
mAP: 0.084086486303743
Epoch 2/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:40<00:00,  6.25sample/s]
Epoch 2/10 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 56.78sample/s]
Epoch 2/10, Training Loss: 0.33858114524628036
mAP: 0.08321431483916038
Epoch 3/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:39<00:00,  6.44sample/s]
Epoch 3/10 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 55.96sample/s]
Epoch 3/10, Training Loss: 0.3313619687614846
mAP: 0.09694534873915486
Epoch 4/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:40<00:00,  6.39sample/s]
Epoch 4/10 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 56.99sample/s]
Epoch 4/10, Training Loss: 0.36889249717933126
mAP: 0.07787748963843129
Epoch 5/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:40<00:00,  6.34sample/s]
Epoch 5/10 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 56.43sample/s]
Epoch 5/10, Training Loss: 0.42037881313081016
mAP: 0.07915576811556707
```

```
Epoch 6/10 (Train): 100%|██████████| 256/256 [00:40<00:00,  6.38sample/s]
Epoch 6/10 (Eval): 100%|██████████| 55/55 [00:00<00:00, 57.84sample/s]
Epoch 6/10, Training Loss: 0.4558310771633753
mAP: 0.08025297436554657
Epoch 7/10 (Train): 100%|██████████| 256/256 [00:40<00:00,  6.38sample/s]
Epoch 7/10 (Eval): 100%|██████████| 55/55 [00:00<00:00, 57.85sample/s]
Epoch 00007: reducing learning rate of group 0 to 1.0000e-04.

Epoch 7/10, Training Loss: 0.4735697199729718
mAP: 0.08086134164836473
Epoch 8/10 (Train): 100%|██████████| 256/256 [00:39<00:00,  6.43sample/s]
Epoch 8/10 (Eval): 100%|██████████| 55/55 [00:00<00:00, 57.97sample/s]
Epoch 8/10, Training Loss: 0.37603193145332625
mAP: 0.08433570744046022
Epoch 9/10 (Train): 100%|██████████| 256/256 [00:40<00:00,  6.38sample/s]
Epoch 9/10 (Eval): 100%|██████████| 55/55 [00:00<00:00, 57.22sample/s]
Epoch 9/10, Training Loss: 0.36564307752632885
mAP: 0.08771931951235838
Epoch 10/10 (Train): 100%|██████████| 256/256 [00:40<00:00,  6.40sample/s]
Epoch 10/10 (Eval): 100%|██████████| 55/55 [00:00<00:00, 56.34sample/s]
Epoch 00010: reducing learning rate of group 0 to 1.0000e-05.

Epoch 10/10, Training Loss: 0.34644968470092863
mAP: 0.09166672512688483
Тренировка завершена!
(Eval): 100%|██████████| 55/55 [00:01<00:00, 47.89sample/s]
Test mAP: 0.09166672512688483


        ----------------------------------------------------------------------

        Выбранная модель: efficientnet_b1
        Пользовательское название модели: efficientnet_b1_Exp1
        Выбранный оптимизатор: AdamW

        ----------------------------------------------------------------------


Epoch 1/10 (Train): 100%|██████████| 256/256 [00:53<00:00,  4.82sample/s]
Epoch 1/10 (Eval): 100%|██████████| 55/55 [00:01<00:00, 38.01sample/s]
```

```
Epoch 1/10, Training Loss: 0.42254169465013547
mAP: 0.08821747443343873
Epoch 2/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.83sample/s]
Epoch 2/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 43.99sample/s]
Epoch 2/10, Training Loss: 0.35572233450147905
mAP: 0.07076131857145755
Epoch 3/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.79sample/s]
Epoch 3/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 44.62sample/s]
Epoch 3/10, Training Loss: 0.37660503200413586
mAP: 0.0796629590045794
Epoch 4/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.80sample/s]
Epoch 4/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 44.77sample/s]
Epoch 4/10, Training Loss: 0.4137173338485809
mAP: 0.0789718454891387
Epoch 5/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.77sample/s]
Epoch 5/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 44.10sample/s]
Epoch 00005: reducing learning rate of group 0 to 1.0000e-04.

Epoch 5/10, Training Loss: 5.716999918244312
mAP: 0.07710725803359961
Epoch 6/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.78sample/s]
Epoch 6/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 42.36sample/s]
Epoch 6/10, Training Loss: 0.3725875879766818
mAP: 0.07341725404782788
Epoch 7/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.80sample/s]
Epoch 7/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 39.38sample/s]
Epoch 7/10, Training Loss: 0.3364571274651098
mAP: 0.07137744342695201
Epoch 8/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:54<00:00,  4.72sample/s]
Epoch 8/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 42.28sample/s]
Epoch 8/10, Training Loss: 0.34347550584061537
mAP: 0.07045111578865859
Epoch 9/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.77sample/s]
Epoch 9/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 42.51sample/s]
Epoch 9/10, Training Loss: 0.34887090415577404
mAP: 0.07271640120041005
```

```
Epoch 10/10 (Train): 100%|████████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.78sample/s]
Epoch 10/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 43.00sample/s]
Epoch 10/10, Training Loss: 0.3466335145931225
mAP: 0.0698564778480628
Тренировка завершена!
(Eval): 100%|████████████████████████████████████████████████████████████████|
55/55 [00:01<00:00, 42.78sample/s]
Test mAP: 0.0698564778480628
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        ----------------------------------------------------------------------

        Выбранная модель: efficientnet_b2
        Пользовательское название модели: efficientnet_b2_Exp1
        Выбранный оптимизатор: AdamW

        ----------------------------------------------------------------------
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```
Epoch 1/10 (Train): 100%|████████████████████████████████████████████████████| 2
56/256 [00:54<00:00,  4.71sample/s]
Epoch 1/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 35.51sample/s]
Epoch 1/10, Training Loss: 0.36381751848966815
mAP: 0.08931924872254692
Epoch 2/10 (Train): 100%|████████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.81sample/s]
Epoch 2/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 43.58sample/s]
Epoch 2/10, Training Loss: 0.4105479332406503
mAP: 0.07902215705570156
Epoch 3/10 (Train): 100%|████████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.74sample/s]
Epoch 3/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 42.83sample/s]
Epoch 3/10, Training Loss: 0.4151932521690469
mAP: 0.076681110408034
Epoch 4/10 (Train): 100%|████████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.78sample/s]
Epoch 4/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 44.09sample/s]
Epoch 4/10, Training Loss: 1.3013690435394665
mAP: 0.0791342887271379
Epoch 5/10 (Train): 100%|████████████████████████████████████████████████████| 2
56/256 [00:54<00:00,  4.73sample/s]
Epoch 5/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 44.92sample/s]
Epoch 5/10, Training Loss: 25.41440489494653
mAP: 0.078939076568779
```

```
Epoch 6/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.75sample/s]
Epoch 6/10 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:01<00:00, 41.79sample/s]
Epoch 6/10, Training Loss: 165.80045519357253
mAP: 0.07432254750982063
Epoch 7/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.75sample/s]
Epoch 7/10 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:01<00:00, 42.85sample/s]
Epoch 7/10, Training Loss: 0.3736138234767168
mAP: 0.07756069271837734
Epoch 8/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.75sample/s]
Epoch 8/10 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:01<00:00, 43.17sample/s]
Epoch 8/10, Training Loss: 0.3696438290644437
mAP: 0.07526036523450923
Epoch 9/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:54<00:00,  4.72sample/s]
Epoch 9/10 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:01<00:00, 41.39sample/s]
Epoch 00009: reducing learning rate of group 0 to 1.0000e-04.

Epoch 9/10, Training Loss: 0.36805771898070816
mAP: 0.07616013990385419
Epoch 10/10 (Train): 100%|█████████████████████████████████████████████████| 2
56/256 [00:53<00:00,  4.78sample/s]
Epoch 10/10 (Eval): 100%|██████████████████████████████████████████████████|
55/55 [00:01<00:00, 43.01sample/s]
Epoch 10/10, Training Loss: 0.341026983136544
mAP: 0.07311735273692678
Тренировка завершена!
(Eval): 100%|██████████████████████████████████████████████████████████████|
55/55 [00:01<00:00, 42.98sample/s]
Test mAP: 0.07311735273692678


        ----------------------------------------------------------------------

        Выбранная модель: efficientnet_b3
        Пользовательское название модели: efficientnet_b3_Exp1
        Выбранный оптимизатор: AdamW

        ----------------------------------------------------------------------


Epoch 1/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [01:00<00:00,  4.23sample/s]
Epoch 1/10 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:01<00:00, 29.95sample/s]
Epoch 1/10, Training Loss: 0.5361012887151446
mAP: 0.07337378947030539
```

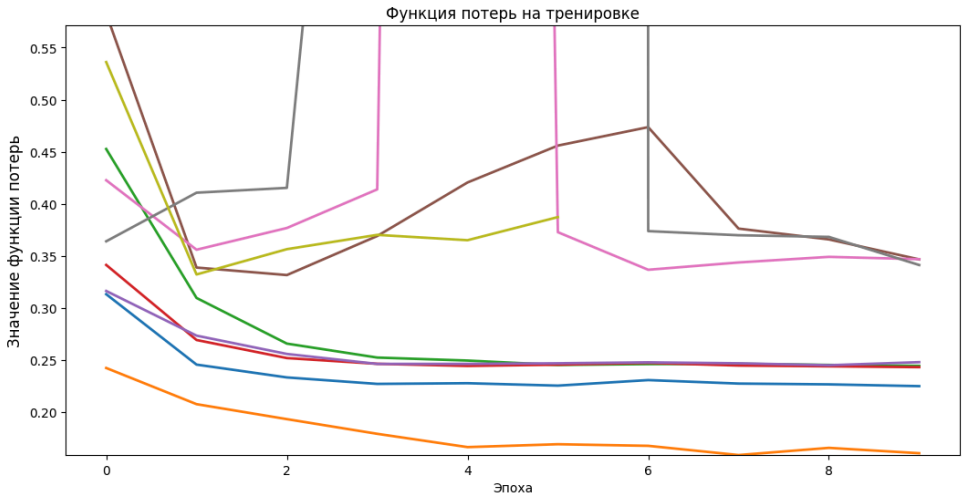
```
Epoch 2/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [01:00<00:00,  4.24sample/s]
Epoch 2/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 38.93sample/s]
Epoch 2/10, Training Loss: 0.3319463457810343
mAP: 0.0958390014549142
Epoch 3/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:59<00:00,  4.29sample/s]
Epoch 3/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 38.11sample/s]
Epoch 3/10, Training Loss: 0.3562496845552232
mAP: 0.07893954915517533
Epoch 4/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:59<00:00,  4.28sample/s]
Epoch 4/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 38.69sample/s]
Epoch 4/10, Training Loss: 0.3700025456473668
mAP: 0.07149263587821214
Epoch 5/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:59<00:00,  4.30sample/s]
Epoch 5/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 38.20sample/s]
Epoch 5/10, Training Loss: 0.36487104022398853
mAP: 0.07922940727186167
Epoch 6/10 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:59<00:00,  4.30sample/s]
Epoch 6/10 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:01<00:00, 39.77sample/s]
Epoch 6/10, Training Loss: 0.3870895737381943
mAP: 0.07315862751750947
Epoch 7/10 (Train):   9%|████▉                                              |
24/256 [00:05<00:54,  4.26sample/s]
```

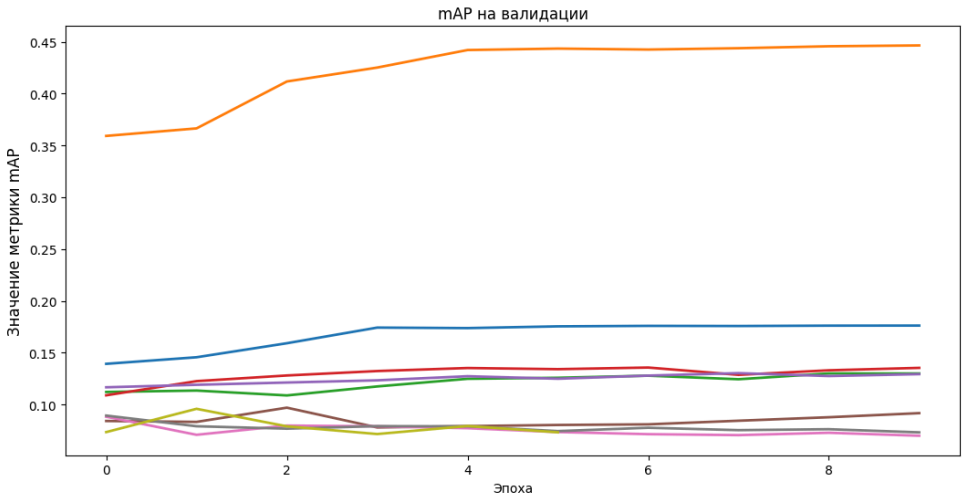
```
Traceback (most recent call last)
in <module>:1

❱ 1 graduate_pipeline.graduate()
  2

in graduate:48

  47                                  seed=seed)
❱ 48                  train.graduate()
  49              except Exception as ex:

in graduate:105

  104          # Обучаем
❱ 105          self.train_model()
  106          # Тестируем

in train_model:226

  225                  inputs1, inputs2, labels = inputs1.cuda(), inputs2.cu
❱ 226                  self.optimizer.zero_grad()
  227                  emb1, emb2 = self.model(inputs1, inputs2)

c:\users\nightmare\pycharmprojects\cvtools\venv\lib\site-packages\torch\opt

  248                                  if (not foreach or p.grad.is_sparse):
❱ 249                                      p.grad.zero_()
  250                                  else:
```

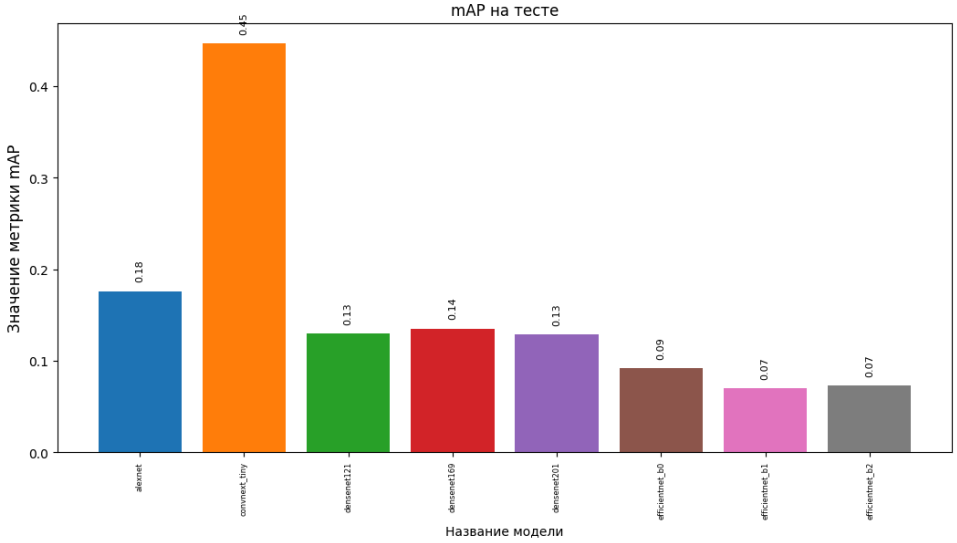
KeyboardInterrupt

In [312…

```
metrics_visualizer = MetricsVisualizer(prefix="Exp1",
                                       path_to_metrics_train="./metrics_train",
                                       path_to_metrics_test="./metrics_test",
                                       path_to_save_plots="./plot")
```







Обучение лучшего бекбона

In [319…

```
graduate_pipeline = validate_with_pydantic(EntryGraduateModel)(GraduateModelPipelin
    entry = {
        "prefix": "Exp2",
        "models": ["convnext_tiny"],
        "name_optimizers": ["SGD"],
```

```
        "size_img": (128, 128),
        "batch_size": 5,
        "num_epochs": 50,
        "emb_size": 512,
        "is_use_imagenet_weights": True,
        "use_device": "cuda",
        "start_learning_rate": 0.00055,
        "num_workers": 0,
        "pin_memory": True,
        "seed": 17
    }
)
```

In [320…

```
graduate_pipeline.graduate()
```

```
----------------------------------------------------------------------

Выбранная модель: convnext_tiny
Пользовательское название модели: convnext_tiny_Exp2
Выбранный оптимизатор: SGD

----------------------------------------------------------------------
```

```
Epoch 1/50 (Train): 100%|██████████| 256/256 [00:25<00:00, 10.08sample/s]
Epoch 1/50 (Eval): 100%|██████████| 55/55 [00:00<00:00, 73.39sample/s]
Epoch 1/50, Training Loss: 0.2420376272420981
mAP: 0.3592120204436654
Epoch 2/50 (Train): 100%|██████████| 256/256 [00:25<00:00,  9.85sample/s]
Epoch 2/50 (Eval): 100%|██████████| 55/55 [00:00<00:00, 64.26sample/s]
Epoch 2/50, Training Loss: 0.20725657729781233
mAP: 0.3664644670376315
Epoch 3/50 (Train): 100%|██████████| 256/256 [00:26<00:00,  9.77sample/s]
Epoch 3/50 (Eval): 100%|██████████| 55/55 [00:00<00:00, 64.56sample/s]
Epoch 3/50, Training Loss: 0.19287486592656933
mAP: 0.41173657685577525
Epoch 4/50 (Train): 100%|██████████| 256/256 [00:25<00:00, 10.00sample/s]
Epoch 4/50 (Eval): 100%|██████████| 55/55 [00:00<00:00, 73.11sample/s]
Epoch 00004: reducing learning rate of group 0 to 1.0000e-04.

Epoch 4/50, Training Loss: 0.1787556218187092
mAP: 0.42517025543804376
Epoch 5/50 (Train): 100%|██████████| 256/256 [00:25<00:00,  9.91sample/s]
Epoch 5/50 (Eval): 100%|██████████| 55/55 [00:00<00:00, 71.00sample/s]
```

```
Epoch 5/50, Training Loss: 0.1659707118960796
mAP: 0.4420990862162339
Epoch 6/50 (Train): 100%|██████████| 256/256 [00:25<00:00,  9.97sample/s]
Epoch 6/50 (Eval): 100%|██████████| 55/55 [00:00<00:00, 67.61sample/s]
Epoch 6/50, Training Loss: 0.16874152292439248
mAP: 0.4434876584257468
Epoch 7/50 (Train): 100%|██████████| 256/256 [00:25<00:00,  9.97sample/s]
Epoch 7/50 (Eval): 100%|██████████| 55/55 [00:00<00:00, 68.47sample/s]
Epoch 00007: reducing learning rate of group 0 to 1.0000e-05.

Epoch 7/50, Training Loss: 0.16720569122117013
mAP: 0.442552414142927
Epoch 8/50 (Train): 100%|██████████| 256/256 [00:26<00:00,  9.83sample/s]
Epoch 8/50 (Eval): 100%|██████████| 55/55 [00:00<00:00, 65.48sample/s]
Epoch 8/50, Training Loss: 0.15834165006526746
mAP: 0.44384679916988856
Epoch 9/50 (Train): 100%|██████████| 256/256 [00:26<00:00,  9.77sample/s]
Epoch 9/50 (Eval): 100%|██████████| 55/55 [00:00<00:00, 68.48sample/s]
Epoch 9/50, Training Loss: 0.16524921772361267
mAP: 0.44571988223257625
Epoch 10/50 (Train): 100%|██████████| 256/256 [00:25<00:00,  9.94sample/s]
Epoch 10/50 (Eval): 100%|██████████| 55/55 [00:00<00:00, 73.97sample/s]
Epoch 00010: reducing learning rate of group 0 to 1.0000e-06.

Epoch 10/50, Training Loss: 0.16010307699616533
mAP: 0.4465276441102752
Epoch 11/50 (Train): 100%|██████████| 256/256 [00:25<00:00, 10.04sample/s]
Epoch 11/50 (Eval): 100%|██████████| 55/55 [00:00<00:00, 66.75sample/s]
Epoch 11/50, Training Loss: 0.16429043143580202
mAP: 0.4465770137187088
Epoch 12/50 (Train): 100%|██████████| 256/256 [00:25<00:00,  9.87sample/s]
Epoch 12/50 (Eval): 100%|██████████| 55/55 [00:00<00:00, 66.05sample/s]
Epoch 12/50, Training Loss: 0.1575220394297503
mAP: 0.4466737770922175
Epoch 13/50 (Train): 100%|██████████| 256/256 [00:26<00:00,  9.81sample/s]
Epoch 13/50 (Eval): 100%|██████████| 55/55 [00:00<00:00, 68.55sample/s]
```

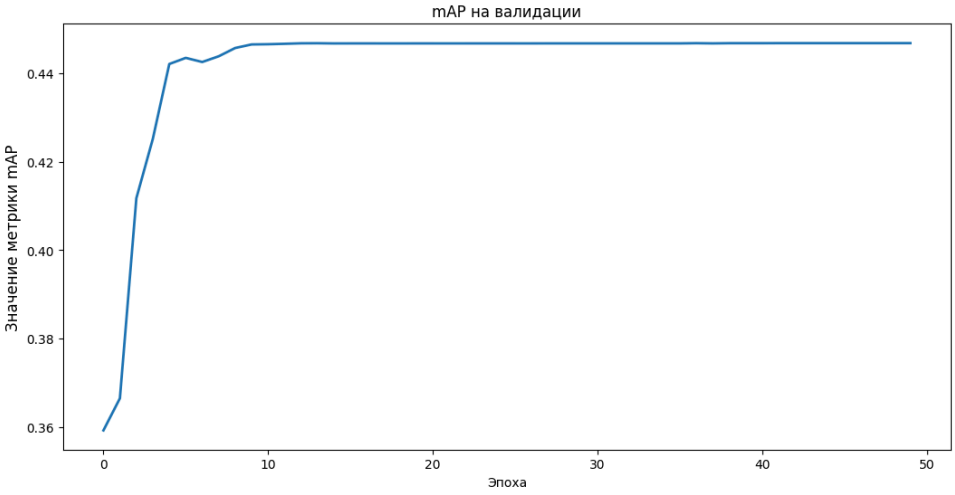
```
Epoch 00013: reducing learning rate of group 0 to 1.0000e-07.

Epoch 13/50, Training Loss: 0.1670254795026267
mAP: 0.4467828245874687
Epoch 14/50 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.97sample/s]
Epoch 14/50 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:00<00:00, 69.19sample/s]
Epoch 14/50, Training Loss: 0.16266824321064632
mAP: 0.44679765020278916
Epoch 15/50 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.93sample/s]
Epoch 15/50 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:00<00:00, 73.65sample/s]
Epoch 15/50, Training Loss: 0.16392514409380965
mAP: 0.44675489057826245
Epoch 16/50 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.88sample/s]
Epoch 16/50 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:00<00:00, 69.62sample/s]
Epoch 00016: reducing learning rate of group 0 to 1.0000e-08.

Epoch 16/50, Training Loss: 0.1636621972429566
mAP: 0.4467612678414148
Epoch 17/50 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.95sample/s]
Epoch 17/50 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:00<00:00, 74.79sample/s]
Epoch 17/50, Training Loss: 0.16207172245776746
mAP: 0.4467624892602149
Epoch 18/50 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:26<00:00,  9.83sample/s]
Epoch 18/50 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:00<00:00, 68.76sample/s]
Epoch 18/50, Training Loss: 0.16266629767778795
mAP: 0.4467599092501 33
Epoch 19/50 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.86sample/s]
Epoch 19/50 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:00<00:00, 70.62sample/s]
Epoch 19/50, Training Loss: 0.16035535365517717
mAP: 0.44675849309653204
Epoch 20/50 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.85sample/s]
Epoch 20/50 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:00<00:00, 69.35sample/s]
Epoch 20/50, Training Loss: 0.16160017550282646
mAP: 0.4467616735217331
Epoch 21/50 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.90sample/s]
Epoch 21/50 (Eval): 100%|████████████████████████████████████████████████████|
55/55 [00:00<00:00, 67.95sample/s]
Epoch 21/50, Training Loss: 0.16087209740362596
mAP: 0.4467609222472446
```

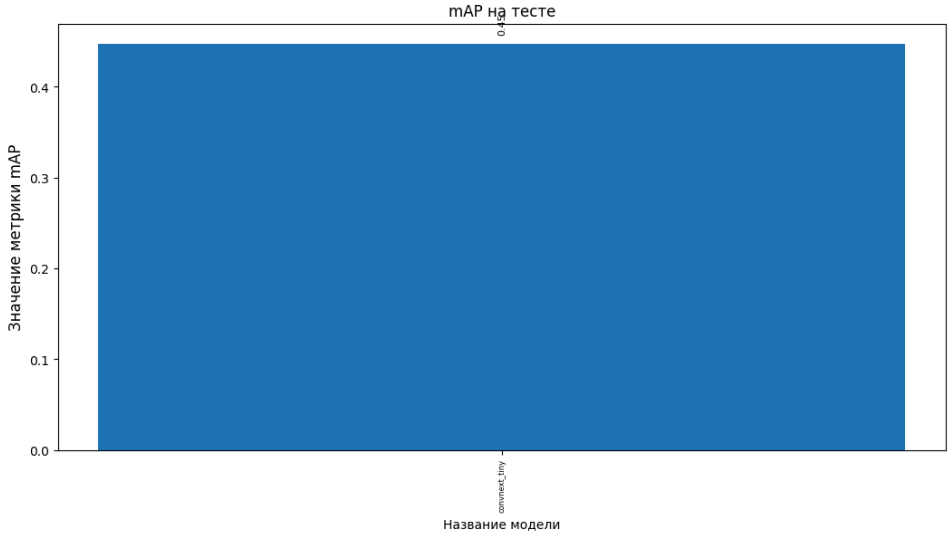
```
Epoch 22/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.86sample/s]
Epoch 22/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 70.14sample/s]
Epoch 22/50, Training Loss: 0.1649194652782171
mAP: 0.4467610429495194
Epoch 23/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:26<00:00,  9.74sample/s]
Epoch 23/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 72.25sample/s]
Epoch 23/50, Training Loss: 0.1668139589310158
mAP: 0.4467610429495194
Epoch 24/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.90sample/s]
Epoch 24/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 68.76sample/s]
Epoch 24/50, Training Loss: 0.16489204834215343
mAP: 0.4467617247384278
Epoch 25/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:26<00:00,  9.66sample/s]
Epoch 25/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 65.15sample/s]
Epoch 25/50, Training Loss: 0.1604703206612612
mAP: 0.4467617247384278
Epoch 26/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:26<00:00,  9.67sample/s]
Epoch 26/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 68.09sample/s]
Epoch 26/50, Training Loss: 0.15791616769274697
mAP: 0.44676137813514005
Epoch 27/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:26<00:00,  9.72sample/s]
Epoch 27/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 67.59sample/s]
Epoch 27/50, Training Loss: 0.15650415511663596
mAP: 0.44676137813514005
Epoch 28/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:26<00:00,  9.83sample/s]
Epoch 28/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 72.11sample/s]
Epoch 28/50, Training Loss: 0.1582480807992397
mAP: 0.4467663092256857
Epoch 29/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:26<00:00,  9.78sample/s]
Epoch 29/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 68.26sample/s]
Epoch 29/50, Training Loss: 0.15746389178093523
mAP: 0.44676842213981693
Epoch 30/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:26<00:00,  9.81sample/s]
Epoch 30/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 73.15sample/s]
```

```
Epoch 30/50, Training Loss: 0.1608539401786402
mAP: 0.4467687282099101
Epoch 31/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.90sample/s]
Epoch 31/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 69.82sample/s]
Epoch 31/50, Training Loss: 0.16239644105371553
mAP: 0.4467665957352403
Epoch 32/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.89sample/s]
Epoch 32/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 68.34sample/s]
Epoch 32/50, Training Loss: 0.16485398406803142
mAP: 0.4467663361219166
Epoch 33/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.92sample/s]
Epoch 33/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 69.18sample/s]
Epoch 33/50, Training Loss: 0.16768508835957618
mAP: 0.44676581315312736
Epoch 34/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:26<00:00,  9.72sample/s]
Epoch 34/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 64.57sample/s]
Epoch 34/50, Training Loss: 0.16004691197304055
mAP: 0.44676581315312736
Epoch 35/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:26<00:00,  9.78sample/s]
Epoch 35/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 65.83sample/s]
Epoch 35/50, Training Loss: 0.16067874059081078
mAP: 0.4467659653328855
Epoch 36/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:26<00:00,  9.81sample/s]
Epoch 36/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 69.32sample/s]
Epoch 36/50, Training Loss: 0.1678225316281896
mAP: 0.4467653591162741
Epoch 37/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:26<00:00,  9.76sample/s]
Epoch 37/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 69.60sample/s]
Epoch 37/50, Training Loss: 0.16308432357618585
mAP: 0.44680738281033383
Epoch 38/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.86sample/s]
Epoch 38/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 67.51sample/s]
Epoch 38/50, Training Loss: 0.16306274928501807
mAP: 0.4467655632177731
```

```
Epoch 39/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:26<00:00,  9.84sample/s]
Epoch 39/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 74.26sample/s]
Epoch 39/50, Training Loss: 0.16177241510013118
mAP: 0.4468063101330091
Epoch 40/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.96sample/s]
Epoch 40/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 72.39sample/s]
Epoch 40/50, Training Loss: 0.16165091173024848
mAP: 0.446806812125921
Epoch 41/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:25<00:00, 10.01sample/s]
Epoch 41/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 66.93sample/s]
Epoch 41/50, Training Loss: 0.1629571228986606
mAP: 0.44680679660648837
Epoch 42/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.94sample/s]
Epoch 42/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 70.96sample/s]
Epoch 42/50, Training Loss: 0.1587576158635784
mAP: 0.446817009024789
Epoch 43/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:26<00:00,  9.82sample/s]
Epoch 43/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 70.60sample/s]
Epoch 43/50, Training Loss: 0.1553029241913464
mAP: 0.44681729484428556
Epoch 44/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:26<00:00,  9.79sample/s]
Epoch 44/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 69.49sample/s]
Epoch 44/50, Training Loss: 0.162326600162487
mAP: 0.44681685363202805
Epoch 45/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.86sample/s]
Epoch 45/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 68.32sample/s]
Epoch 45/50, Training Loss: 0.16226276454108302
mAP: 0.4468172053619942
Epoch 46/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.99sample/s]
Epoch 46/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 68.35sample/s]
Epoch 46/50, Training Loss: 0.16452833160292357
mAP: 0.44681760141579024
Epoch 47/50 (Train): 100%|████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.98sample/s]
Epoch 47/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 69.86sample/s]
```

```
Epoch 47/50, Training Loss: 0.16955844548647292
mAP: 0.4468172404013038
Epoch 48/50 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.92sample/s]
Epoch 48/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 67.92sample/s]
Epoch 48/50, Training Loss: 0.17294169546221383
mAP: 0.4468179816252127
Epoch 49/50 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.93sample/s]
Epoch 49/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 71.34sample/s]
Epoch 49/50, Training Loss: 0.16394865793699864
mAP: 0.44682093003238094
Epoch 50/50 (Train): 100%|██████████████████████████████████████████████████| 2
56/256 [00:25<00:00,  9.98sample/s]
Epoch 50/50 (Eval): 100%|███████████████████████████████████████████████████|
55/55 [00:00<00:00, 70.77sample/s]
Epoch 50/50, Training Loss: 0.16437969049729872
mAP: 0.44682116477425116
Тренировка завершена!
(Eval): 100%|███████████████████████████████████████████████████████████████|
55/55 [00:00<00:00, 72.32sample/s]
Test mAP: 0.44682116477425116
```





convnext_tiny

Тестирование модели

Инференс моделью

Класс InferenceModel

In [370…

```python
class InferenceModel:
    def __init__(self,
                 model_name: str,
                 emb_size: int,
                 path_to_weights: str,
                 image_paths: list[str],
                 prefix: str = "",
                 use_device: str = None,
                 size_img: tuple = (128, 128),
                 labels: bool = False):
        self.name_model = model_name
        self.emb_size = emb_size
        self.path_to_weights = path_to_weights
        self.image_paths = image_paths
        self.size_img = size_img
        self.use_device = use_device
        self.device = None
        self.prefix = prefix
        self.labels = labels

        # Перемещение модели на GPU, если CUDA доступен
        if self.use_device == None:
            self.device = torch.device("cuda" if torch.cuda.is_available() else "cp
        elif self.use_device == "cuda":
            self.device = torch.device("cuda")
        elif self.use_device == "cpu":
            self.device = torch.device("cpu")
        else:
            raise Exception("use_device must be cuda or cpu or None")

    def get_model(self):
        # Инициализация сиамской сети
        self.model = SiameseNetwork(self.name_model, self.emb_size, False).to(self.

    def load_model(self):
        if self.prefix != "":
            path = os.path.join(self.path_to_weights, f"{self.name_model}_{self.pre
        else:
            path = os.path.join(self.path_to_weights, f"{self.name_model}.pt")
        try:
            if self.name_model in models.__dict__:
                if os.path.exists(path):
                    state_dict = torch.load(path, map_location=self.device)['model_
                    self.model.load_state_dict(state_dict)
                    log.info(f"Модель {self.name_model} успешно загружена с весами
                else:
                    log.warning(f"Модель {self.name_model} не найдена.")
        except Exception as e:
            log.exception(f"Ошибка при загрузке модели: {e}")

    def get_transform(self):
        return transforms.Compose([
                ResizeWithPadding(self.size_img[0]),
                transforms.ToTensor(),
                transforms.Normalize(
```

```
                    mean=[0.485, 0.456, 0.406],  # Средние значения для ImageNet
                    std=[0.229, 0.224, 0.225]    # Стандартные отклонения для Image
                )
            ])

    def predict(self):
        self.get_model()
        self.load_model()
        self.model.eval()
        all_embeds = []
        all_labels = []

        with torch.no_grad():
            for image_path in tqdm(self.image_paths, desc="Extract Feature", unit="
                try:
                    # Load image
                    image = Image.open(image_path).convert('RGB')
                    image = self.get_transform()(image).unsqueeze(0).to(self.device

                    # Get label
                    if self.labels:
                        label = int(os.path.basename(image_path).split(".png")[0].s
                    else:
                        label = None

                    all_labels.append(label)

                    # Inference
                    outputs = self.model.forward_once(image)
                    all_embeds += [out.cpu().numpy() for out in outputs]

                except Exception as e:
                    print(f"Ошибка при обработке изображения {image_path}: {e}")
                    all_embeds.append(None)

        if self.labels:
            mAP = calculate_map(all_embeds, all_labels)  # Вычисление mAP
            print(f"Inference mAP: {mAP}")

        return all_embeds, all_labels
```

InferenceModelPipeline

In [371…

```
@dataclass
class InferenceModelPipeline:
    entry: EntryInferenceModel

    def __post_init__(self):
        pass

    def inference(self):

        prefix = self.entry.Prefix
        name_model = self.entry.NameModel
        path_to_weights = self.entry.PathWeights
```

```
        image_path_list = self.entry.ImageList
        emb_size = self.entry.EmbSize
        use_device = self.entry.UseDevice
        size_img = self.entry.SizeImg
        labels = self.entry.Labels

        try:
            inference_model = InferenceModel(
                        prefix=prefix,
                        model_name=name_model,
                        path_to_weights=path_to_weights,
                        image_paths=image_path_list,
                        emb_size=emb_size,
                        use_device=use_device,
                        size_img=size_img,
                        labels=labels)
            all_embeds, all_labels = inference_model.predict()
            return all_embeds, all_labels
        except Exception as ex:
            log.exception("InferenceModel\n", exc_info=ex)
```

Инференс

In [372…

```
path_to_images = "./images"

# Получаем ембеддинги
inference_pipeline = validate_with_pydantic(EntryInferenceModel)(InferenceModelPipe
    entry = {
        "prefix": "Exp2",
        "name_model": "convnext_tiny",
        "image_path_list": [os.path.join(path_to_images, path) for path in os.listd
        "size_img": (128, 128),
        "emb_size": 512,
        "use_device": "cuda",
        "path_to_weights": "./weights",
        "labels": True
    }
)
```

In [373…

```
all_embeds, all_labels = inference_pipeline.inference()
```

```
12:54:27-416027 INFO     Модель convnext_tiny успешно загружена с весами из .

Extract Feature: 100%|██████████████████████████████████████████████████████████|
272/272 [00:03<00:00, 88.97image/s]
Inference mAP: 0.5605753437519935
```

Визуализация ембеддингов через TSNE

Класс Emb2Tsne

In [375…

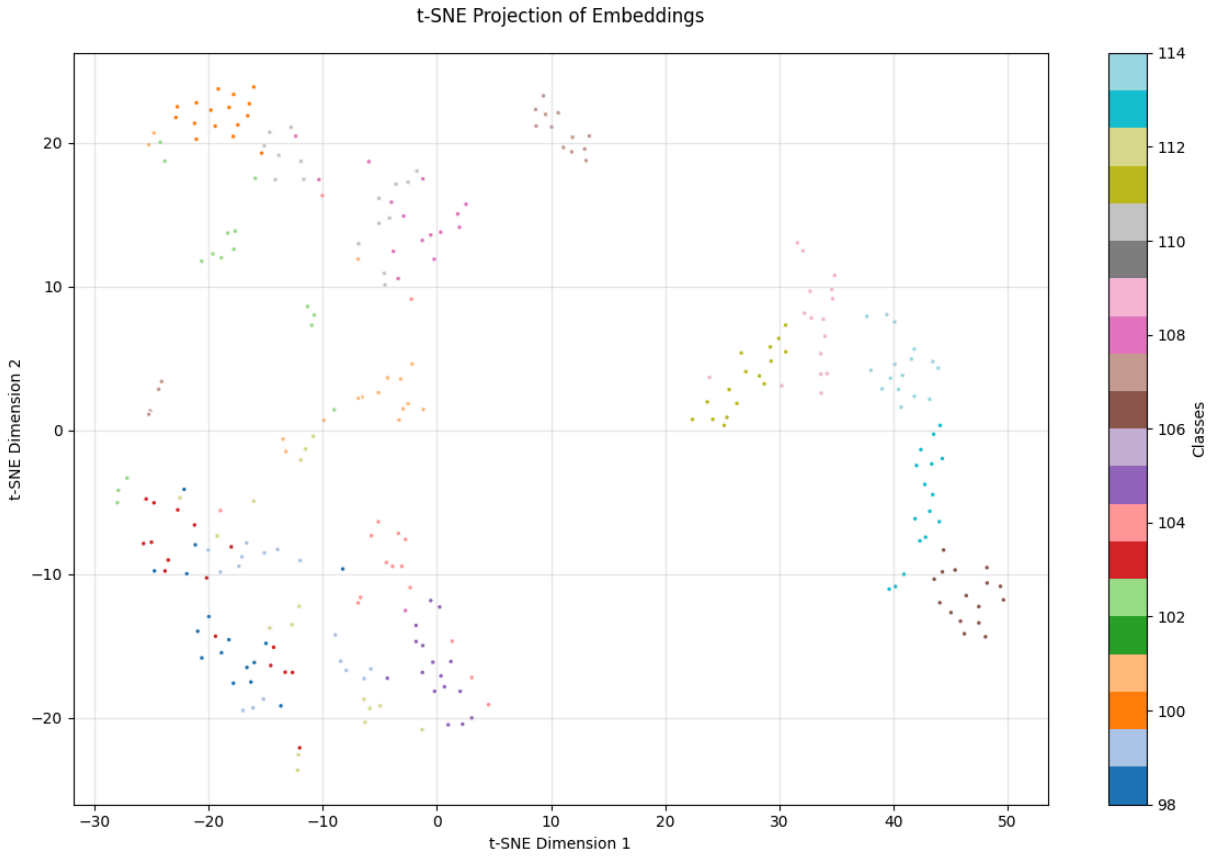
```python
class Emb2Tsne:
    def __init__(
        self,
        embeddings: list[np.ndarray],
        labels: List[int],
        n_components: int = 2,
        perplexity: float = 30.0,
        random_state: int = 42,
        **tsne_kwargs
    ):
        """
        Инициализация визуализатора t-SNE

        :param embeddings: Массив эмбеддингов (N_samples, 512)
        :param labels: Список меток классов для каждого эмбеддинга
        :param n_components: Размерность выходного пространства
        :param perplexity: Параметр перплексии для t-SNE
        :param random_state: Seed для воспроизводимости
        :param tsne_kwargs: Дополнительные параметры для TSNE
        """
        self.embeddings = np.stack(embeddings)
        self.labels = np.array(labels)
        self.n_components = n_components
        self.tsne = TSNE(
            n_components=n_components,
            perplexity=perplexity,
            random_state=random_state,
            **tsne_kwargs
        )
        self.tsne_results: Optional[np.ndarray] = None

    def fit_transform(self) -> np.ndarray:
        """Применяет t-SNE к эмбеддингам и возвращает 2D проекцию"""
        self.tsne_results = self.tsne.fit_transform(self.embeddings)
        return self.tsne_results

    def plot(
        self,
        figsize: tuple = (12, 8),
        title: str = "t-SNE Visualization",
        alpha: float = 0.7,
        marker_size: float = 8.0
    ) -> None:
        """Визуализирует 2D проекцию с цветовой маркировкой классов"""
        if self.tsne_results is None:
            raise ValueError("Сначала выполните fit_transform()")

        if len(np.unique(self.labels)) > 10:
            cmap = plt.cm.get_cmap('tab20')
        else:
            cmap = plt.cm.get_cmap('tab10')

        plt.figure(figsize=figsize)
        scatter = plt.scatter(
            self.tsne_results[:, 0],
```

```
            self.tsne_results[:, 1],
            c=self.labels,
            cmap=cmap,
            alpha=alpha,
            s=marker_size,
            edgecolors='w'
        )

        plt.title(title, pad=20)
        plt.xlabel("t-SNE Dimension 1")
        plt.ylabel("t-SNE Dimension 2")
        plt.colorbar(scatter, label="Classes")
        plt.grid(alpha=0.3)
        plt.tight_layout()
        plt.show()
```

Визуализируем

In [378…

```
# Создание и использование визуализатора
visualizer = Emb2Tsne(
    embeddings=all_embeds,
    labels=all_labels,
    perplexity=15,
    random_state=42
)

# Получение 2D проекции
projection = visualizer.fit_transform()

# Визуализация
visualizer.plot(
    title="t-SNE Projection of Embeddings",
    alpha=1.0,
    marker_size=12.0
)
```



Реализация нейросетевого трекинга

Класс Tracker

In [390…

```python
TrackElement = collections.namedtuple(
    typename='TrackElement',
    field_names=['t', 'track_id', 'mask', 'mean', 'embed'],
)

def center_mass(masks: np.ndarray) -> np.ndarray:
    N = masks.shape[0]
    center_mass = np.empty((N, 2), dtype=np.float32)
    for i in range(N):
        xs, ys = masks[i].nonzero()
        center_mass[i, 0] = xs.mean()
        center_mass[i, 1] = ys.mean()
    return center_mass

class Tracker:
    def __init__(
        self,
        alive_threshold: int,
        association_threshold: float = 0.1,
        means_threshold: float = 150,
        min_value: float = -1e9,
        euclidean_scale: float = 1.344,
```

```
        euclidean_offset: float = 9.447,
        iou_scale: float = 0.402,
    ) -> None:
        self.alive_threshold = alive_threshold
        self.means_threshold = means_threshold
        self.association_threshold = association_threshold
        self.min_value = min_value
        self.active_tracks = []
        self.next_inst_id = None
        self.euclidean_scale = euclidean_scale
        self.euclidean_offset = euclidean_offset
        self.iou_scale = iou_scale

    def update_active_track(self, frame_count: int) -> None:
        self.active_tracks = [
            track for track in self.active_tracks
            if track.t >= frame_count - self.alive_threshold
        ]

    def tracking(
        self,
        frame_count: int,
        masks: np.ndarray,
        embeds: np.ndarray,
    ) -> list:
        if self.next_inst_id is None:
            self.next_inst_id = 1
        else:
            self.update_active_track(frame_count)

        n, k = len(masks), len(self.active_tracks)
        result = np.zeros(n, dtype=np.int32)

        if n < 1:
            return result

        means = center_mass(masks)
        masks = [maskUtils.encode(np.asfortranarray(mask)) for mask in masks.astype

        if k == 0:
            for i in range(n):
                self.active_tracks.append(
                    TrackElement(
                        t=frame_count,
                        mask=masks[i],
                        track_id=self.next_inst_id,
                        mean=means[i],
                        embed=embeds[i],
                    )
                )
                result[i] = self.next_inst_id
                self.next_inst_id += 1
            return result

        detections_unassigned = np.ones(n, dtype=bool)
```

```
        # Получаем эмбеддинги из памяти
        last_reids = np.array([track.embed for track in self.active_tracks])

        # Вычисляем матрицу схожести для эмбеддингов
        distance_matrix = cdist(embeds, last_reids, metric='euclidean')
        embed_sim = self.euclidean_scale * (self.euclidean_offset - distance_matrix

        # Вычисляем матрицу IoU
        memory_masks = [track.mask for track in self.active_tracks]
        iou_matrix = maskUtils.iou(masks, memory_masks, [False]*len(memory_masks))
        asso_sim = embed_sim + self.iou_scale * iou_matrix

        # Применяем пороги и решаем задачу назначения
        asso_sim[asso_sim <= self.association_threshold] = self.min_value
        rows, cols = linear_sum_assignment(asso_sim, maximize=True)

        for row, column in zip(rows, cols):
            if np.linalg.norm(self.active_tracks[column].mean - means[row]) < self.
                self.active_tracks[column] = TrackElement(
                    t=frame_count,
                    mask=masks[row],
                    track_id=self.active_tracks[column].track_id,
                    mean=means[row],
                    embed=embeds[row],
                )
                result[row] = self.active_tracks[column].track_id
            else:
                self.active_tracks.append(TrackElement(
                    t=frame_count,
                    mask=masks[row],
                    track_id=self.next_inst_id,
                    mean=means[row],
                    embed=embeds[row],
                ))
                result[row] = self.next_inst_id
                self.next_inst_id += 1
            detections_unassigned[row] = False

        for i in detections_unassigned.nonzero()[0]:
            self.active_tracks.append(TrackElement(
                t=frame_count,
                mask=masks[i],
                track_id=self.next_inst_id,
                mean=means[i],
                embed=embeds[i],
            ))
            result[i] = self.next_inst_id
            self.next_inst_id += 1

        return result
```

Функция для получения эмбеддингов по известным детекциям

In [391…

```python
def get_embeddings(
    img: np.ndarray,
    preds_bboxes: np.ndarray,
    emb_size: int,
    size_img: Tuple[int, int],
    prefix: str,
    name_model: str,
    path_to_weights: str = "./weights"
) -> np.ndarray:
    '''
    Функция для получения эмбеддингов изображения по bounding boxes.

    Входные параметры:
    img - np.ndarray: Входное изображение (H, W, C).
    preds_bboxes - np.ndarray: Предсказанные bounding boxes (N, 4) в формате [x1, y
    emb_size - int: Размер эмбеддингов модели.
    size_img - Tuple[int, int]: Размер изображений, на котором обучалась модель (H,
    prefix - str: Префикс модели.
    name_model - str: Название модели.
    path_to_weights - str: Путь к весам модели.

    Выходные параметры:
    np.ndarray: Эмбеддинги для каждого bounding box (N, emb_size).
    '''
    # Создаем временную директорию для сохранения обрезанных изображений
    with tempfile.TemporaryDirectory() as temp_dir:
        image_paths = []

        # Обрезаем изображение по каждому bounding box и сохраняем во временную дир
        for i, bbox in enumerate(preds_bboxes):
            x1, y1, x2, y2 = map(int, bbox)
            cropped_img = img[y1:y2, x1:x2]  # Обрезаем изображение
            cropped_img = Image.fromarray(cropped_img)  # Конвертируем в PIL Image
            cropped_img = cropped_img.resize(size_img)  # Ресайзим до нужного разме

            # Сохраняем обрезанное изображение во временную директорию
            temp_image_path = os.path.join(temp_dir, f"cropped_{i}.jpg")
            cropped_img.save(temp_image_path)
            image_paths.append(temp_image_path)

        # Инициализируем модель для получения эмбеддингов
        inference_model = InferenceModel(
            prefix=prefix,
            model_name=name_model,
            path_to_weights=path_to_weights,
            image_paths=image_paths,
            emb_size=emb_size,
            use_device="cuda",
            size_img=size_img,
            labels=False
        )

        # Получаем эмбеддинги
        all_embeds, all_labels = inference_model.predict()
```

```
    # Возвращаем эмбеддинги в виде numpy массива
    return np.stack(all_embeds, axis=0)
```

Скрипты для визуализации предсказаний

In [392…

```
# код с семинара
class YoloDetector:
    def __init__(self, conf_threshold: float = 0.3) -> None:
        self.model = YOLO("yolov8s-seg.pt")
        self.conf_threshold = conf_threshold

    def __call__(self, img_path: np.ndarray) -> np.ndarray:
        preds = self.model.predict(source=img_path, conf=self.conf_threshold, show=
        preds_masks = preds[0].masks.data.to(torch.bool).cpu().numpy()
        preds_bboxes = preds[0].boxes.xyxy.cpu().numpy()
        return preds_masks, preds_bboxes

def tracking_visualization(
    img: np.ndarray,
    preds: np.ndarray,
) -> np.ndarray:
    '''
    Входные данные:
    img: np.ndarray - изображение для отрисовки
    preds: np.ndarray - маски сегментации
    '''
    for i in range(len(preds)):
        # отрисуем ограничивающий прямоугольник
        cv2.rectangle(
            img,
            (int(preds[i][0]), int(preds[i][1])),
            (int(preds[i][2]), int(preds[i][3])),
            color=(0, 255, 0),
            thickness=1,
        )
        # отрисуем track_id
        cv2.putText(
            img,
            str(int(preds[i][4])),
            (int(preds[i][0] + 10), int(preds[i][1] + 10)),
            cv2.FONT_HERSHEY_SIMPLEX,
            fontScale=1,
            color=(0, 255, 0),
            thickness=1,
        )

    return img
```

Оцениваем работу трекера

Скачиваем видео

In [384…

```
# скачаем тестовое видео
!wget https://motchallenge.net/sequenceVideos/MOT20-02-raw.mp4 -O mot20.mp4
```

```
--2025-02-25 13:17:11--  https://motchallenge.net/sequenceVideos/MOT20-02-raw.mp4
Resolving motchallenge.net (motchallenge.net)... 131.159.19.34, 2a09:80c0:18::1034
Connecting to motchallenge.net (motchallenge.net)|131.159.19.34|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 76066403 (73M) [video/mp4]
Saving to: 'mot20.mp4'

     0K .......... .......... .......... .......... ..........  0%  453K 2m44s
    50K .......... .......... .......... .......... ..........  0%  960K 2m1s
   100K .......... .......... .......... .......... ..........  0% 33,6M 81s
   150K .......... .......... .......... .......... ..........  0% 20,5M 62s
   200K .......... .......... .......... .......... ..........  0% 1023K 64s
   250K .......... .......... .......... .......... ..........  0%  926K 66s
   300K .......... .......... .......... .......... ..........  0% 91,5M 57s
   350K .......... .......... .......... .......... ..........  0%  678M 50s
   400K .......... .......... .......... .......... ..........  0%  700M 44s
   450K .......... .......... .......... .......... ..........  0%  937K 48s
   500K .......... .......... .......... .......... ..........  0% 19,9M 44s
   550K .......... .......... .......... .......... ..........  0% 21,6M 40s
   600K .......... .......... .......... .......... ..........  0%  739M 37s
   650K .......... .......... .......... .......... ..........  0% 1,01M 40s
   700K .......... .......... .......... .......... ..........  1%  929K 42s
   750K .......... .......... .......... .......... ..........  1%  923K 45s
   800K .......... .......... .......... .......... ..........  1%  227M 42s
   850K .......... .......... .......... .......... ..........  1%  886M 40s
   900K .......... .......... .......... .......... ..........  1% 9,04M 38s
   950K .......... .......... .......... .......... ..........  1% 1,03M 39s
  1000K .......... .......... .......... .......... ..........  1% 4,80M 38s
  1050K .......... .......... .......... .......... ..........  1% 6,18M 37s
  1100K .......... .......... .......... .......... ..........  1% 1,35M 38s
  1150K .......... .......... .......... .......... ..........  1% 5,49M 37s
  1200K .......... .......... .......... .......... ..........  1% 3,95M 36s
  1250K .......... .......... .......... .......... ..........  1% 1,38M 36s
  1300K .......... .......... .......... .......... ..........  1% 9,39M 35s
  1350K .......... .......... .......... .......... ..........  1% 3,95M 35s
  1400K .......... .......... .......... .......... ..........  1% 1,47M 35s
  1450K .......... .......... .......... .......... ..........  2% 6,61M 34s
  1500K .......... .......... .......... .......... ..........  2% 3,97M 34s
  1550K .......... .......... .......... .......... ..........  2% 1,47M 34s
  1600K .......... .......... .......... .......... ..........  2% 4,34M 34s
  1650K .......... .......... .......... .......... ..........  2% 5,38M 33s
  1700K .......... .......... .......... .......... ..........  2% 1,38M 33s
  1750K .......... .......... .......... .......... ..........  2% 10,6M 33s
  1800K .......... .......... .......... .......... ..........  2% 3,91M 32s
  1850K .......... .......... .......... .......... ..........  2% 1,48M 33s
  1900K .......... .......... .......... .......... ..........  2% 6,82M 32s
  1950K .......... .......... .......... .......... ..........  2% 3,84M 32s
  2000K .......... .......... .......... .......... ..........  2% 1,36M 32s
  2050K .......... .......... .......... .......... ..........  2% 9,51M 32s
  2100K .......... .......... .......... .......... ..........  2% 4,08M 31s
  2150K .......... .......... .......... .......... ..........  2% 1,50M 32s
  2200K .......... .......... .......... .......... ..........  3% 6,14M 31s
  2250K .......... .......... .......... .......... ..........  3% 5,32M 31s
  2300K .......... .......... .......... .......... ..........  3% 1,45M 31s
  2350K .......... .......... .......... .......... ..........  3% 5,71M 31s
  2400K .......... .......... .......... .......... ..........  3% 3,52M 30s
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2450K .......... .......... .......... .......... ..........  3% 1,58M 31s
2500K .......... .......... .......... .......... ..........  3% 4,44M 30s
2550K .......... .......... .......... .......... ..........  3% 8,68M 30s
2600K .......... .......... .......... .......... ..........  3% 1,33M 30s
2650K .......... .......... .......... .......... ..........  3% 6,97M 30s
2700K .......... .......... .......... .......... ..........  3% 5,30M 30s
2750K .......... .......... .......... .......... ..........  3% 5,79M 29s
2800K .......... .......... .......... .......... ..........  3% 1,42M 30s
2850K .......... .......... .......... .......... ..........  3% 4,90M 29s
2900K .......... .......... .......... .......... ..........  3% 5,79M 29s
2950K .......... .......... .......... .......... ..........  4% 1,41M 29s
3000K .......... .......... .......... .......... ..........  4% 7,69M 29s
3050K .......... .......... .......... .......... ..........  4% 5,36M 29s
3100K .......... .......... .......... .......... ..........  4% 1,46M 29s
3150K .......... .......... .......... .......... ..........  4% 5,62M 29s
3200K .......... .......... .......... .......... ..........  4% 4,20M 28s
3250K .......... .......... .......... .......... ..........  4% 1,48M 29s
3300K .......... .......... .......... .......... ..........  4% 6,01M 28s
3350K .......... .......... .......... .......... ..........  4% 6,45M 28s
3400K .......... .......... .......... .......... ..........  4% 1,33M 28s
3450K .......... .......... .......... .......... ..........  4% 10,5M 28s
3500K .......... .......... .......... .......... ..........  4% 5,46M 28s
3550K .......... .......... .......... .......... ..........  4% 5,87M 28s
3600K .......... .......... .......... .......... ..........  4% 1,37M 28s
3650K .......... .......... .......... .......... ..........  4% 5,43M 28s
3700K .......... .......... .......... .......... ..........  5% 5,74M 28s
3750K .......... .......... .......... .......... ..........  5% 1,37M 28s
3800K .......... .......... .......... .......... ..........  5% 6,58M 28s
3850K .......... .......... .......... .......... ..........  5% 2,90M 27s
3900K .......... .......... .......... .......... ..........  5% 1,87M 28s
3950K .......... .......... .......... .......... ..........  5% 5,86M 27s
4000K .......... .......... .......... .......... ..........  5% 3,35M 27s
4050K .......... .......... .......... .......... ..........  5% 1,63M 27s
4100K .......... .......... .......... .......... ..........  5% 15,0M 27s
4150K .......... .......... .......... .......... ..........  5% 3,04M 27s
4200K .......... .......... .......... .......... ..........  5% 4,24M 27s
4250K .......... .......... .......... .......... ..........  5% 1,97M 27s
4300K .......... .......... .......... .......... ..........  5% 5,27M 27s
4350K .......... .......... .......... .......... ..........  5% 3,29M 27s
4400K .......... .......... .......... .......... ..........  5% 1,70M 27s
4450K .......... .......... .......... .......... ..........  6% 8,09M 27s
4500K .......... .......... .......... .......... ..........  6% 2,69M 27s
4550K .......... .......... .......... .......... ..........  6% 1,85M 27s
4600K .......... .......... .......... .......... ..........  6% 5,34M 27s
4650K .......... .......... .......... .......... ..........  6% 4,85M 26s
4700K .......... .......... .......... .......... ..........  6% 3,07M 26s
4750K .......... .......... .......... .......... ..........  6% 2,10M 26s
4800K .......... .......... .......... .......... ..........  6% 4,33M 26s
4850K .......... .......... .......... .......... ..........  6% 3,41M 26s
4900K .......... .......... .......... .......... ..........  6% 1,80M 26s
4950K .......... .......... .......... .......... ..........  6% 9,09M 26s
5000K .......... .......... .......... .......... ..........  6% 2,91M 26s
5050K .......... .......... .......... .......... ..........  6% 2,03M 26s
5100K .......... .......... .......... .......... ..........  6% 7,07M 26s
5150K .......... .......... .......... .......... ..........  7% 4,36M 26s
5200K .......... .......... .......... .......... ..........  7% 2,72M 26s
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5250K .......... .......... .......... .......... ..........  7% 2,00M 26s
5300K .......... .......... .......... .......... ..........  7% 5,76M 26s
5350K .......... .......... .......... .......... ..........  7% 2,96M 26s
5400K .......... .......... .......... .......... ..........  7% 1,78M 26s
5450K .......... .......... .......... .......... ..........  7% 13,8M 25s
5500K .......... .......... .......... .......... ..........  7% 2,80M 25s
5550K .......... .......... .......... .......... ..........  7% 6,87M 25s
5600K .......... .......... .......... .......... ..........  7% 1,72M 25s
5650K .......... .......... .......... .......... ..........  7% 4,21M 25s
5700K .......... .......... .......... .......... ..........  7% 3,84M 25s
5750K .......... .......... .......... .......... ..........  7% 1,92M 25s
5800K .......... .......... .......... .......... ..........  7% 12,9M 25s
5850K .......... .......... .......... .......... ..........  7% 2,94M 25s
5900K .......... .......... .......... .......... ..........  8% 1,61M 25s
5950K .......... .......... .......... .......... ..........  8% 8,76M 25s
6000K .......... .......... .......... .......... ..........  8% 2,84M 25s
6050K .......... .......... .......... .......... ..........  8% 5,85M 25s
6100K .......... .......... .......... .......... ..........  8% 2,02M 25s
6150K .......... .......... .......... .......... ..........  8% 6,11M 25s
6200K .......... .......... .......... .......... ..........  8% 2,84M 25s
6250K .......... .......... .......... .......... ..........  8% 1,78M 25s
6300K .......... .......... .......... .......... ..........  8% 8,13M 25s
6350K .......... .......... .......... .......... ..........  8% 5,08M 25s
6400K .......... .......... .......... .......... ..........  8% 1,39M 25s
6450K .......... .......... .......... .......... ..........  8% 18,7M 25s
6500K .......... .......... .......... .......... ..........  8% 4,63M 24s
6550K .......... .......... .......... .......... ..........  8% 2,71M 24s
6600K .......... .......... .......... .......... ..........  8% 2,04M 24s
6650K .......... .......... .......... .......... ..........  9% 7,88M 24s
6700K .......... .......... .......... .......... ..........  9% 2,82M 24s
6750K .......... .......... .......... .......... ..........  9% 1,86M 24s
6800K .......... .......... .......... .......... ..........  9% 5,33M 24s
6850K .......... .......... .......... .......... ..........  9% 5,18M 24s
6900K .......... .......... .......... .......... ..........  9% 3,71M 24s
6950K .......... .......... .......... .......... ..........  9% 1,95M 24s
7000K .......... .......... .......... .......... ..........  9% 8,30M 24s
7050K .......... .......... .......... .......... ..........  9% 2,80M 24s
7100K .......... .......... .......... .......... ..........  9% 1,86M 24s
7150K .......... .......... .......... .......... ..........  9% 5,16M 24s
7200K .......... .......... .......... .......... ..........  9% 2,93M 24s
7250K .......... .......... .......... .......... ..........  9% 1,82M 24s
7300K .......... .......... .......... .......... ..........  9% 15,2M 24s
7350K .......... .......... .......... .......... ..........  9% 7,21M 24s
7400K .......... .......... .......... .......... .......... 10% 2,88M 24s
7450K .......... .......... .......... .......... .......... 10% 1,71M 24s
7500K .......... .......... .......... .......... .......... 10% 7,07M 24s
7550K .......... .......... .......... .......... .......... 10% 5,76M 24s
7600K .......... .......... .......... .......... .......... 10% 1,39M 24s
7650K .......... .......... .......... .......... .......... 10% 13,7M 24s
7700K .......... .......... .......... .......... .......... 10% 5,72M 23s
7750K .......... .......... .......... .......... .......... 10% 3,26M 23s
7800K .......... .......... .......... .......... .......... 10% 1,73M 23s
7850K .......... .......... .......... .......... .......... 10% 7,15M 23s
7900K .......... .......... .......... .......... .......... 10% 2,71M 23s
7950K .......... .......... .......... .......... .......... 10% 7,19M 23s
8000K .......... .......... .......... .......... .......... 10% 2,22M 23s
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 8050K ..........  .......... .......... .......... .......... 10% 6,01M 23s
 8100K ..........  .......... .......... .......... .......... 10% 2,50M 23s
 8150K ..........  .......... .......... .......... .......... 11% 2,02M 23s
 8200K ..........  .......... .......... .......... .......... 11% 6,72M 23s
 8250K ..........  .......... .......... .......... .......... 11% 2,66M 23s
 8300K ..........  .......... .......... .......... .......... 11%  638K 24s
 8350K ..........  .......... .......... .......... .......... 11%  199M 23s
 8400K ..........  .......... .......... .......... .......... 11% 5,30M 23s
 8450K ..........  .......... .......... .......... .......... 11%  538M 23s
 8500K ..........  .......... .......... .......... .......... 11%  757M 23s
 8550K ..........  .......... .......... .......... .......... 11% 2,65M 23s
 8600K ..........  .......... .......... .......... .......... 11% 1,79M 23s
 8650K ..........  .......... .......... .......... .......... 11% 6,07M 23s
 8700K ..........  .......... .......... .......... .......... 11% 1,15M 23s
 8750K ..........  .......... .......... .......... .......... 11% 5,15M 23s
 8800K ..........  .......... .......... .......... .......... 11% 2,38M 23s
 8850K ..........  .......... .......... .......... .......... 11% 1,94M 23s
 8900K ..........  .......... .......... .......... .......... 12% 3,20M 23s
 8950K ..........  .......... .......... .......... .......... 12% 1,37M 23s
 9000K ..........  .......... .......... .......... .......... 12% 5,12M 23s
 9050K ..........  .......... .......... .......... .......... 12% 2,70M 23s
 9100K ..........  .......... .......... .......... .......... 12% 1,94M 23s
 9150K ..........  .......... .......... .......... .......... 12% 4,65M 23s
 9200K ..........  .......... .......... .......... .......... 12% 2,63M 23s
 9250K ..........  .......... .......... .......... .......... 12% 1,36M 23s
 9300K ..........  .......... .......... .......... .......... 12% 9,86M 23s
 9350K ..........  .......... .......... .......... .......... 12% 1,37M 23s
 9400K ..........  .......... .......... .......... .......... 12% 3,98M 23s
 9450K ..........  .......... .......... .......... .......... 12% 2,95M 23s
 9500K ..........  .......... .......... .......... .......... 12% 1,99M 23s
 9550K ..........  .......... .......... .......... .......... 12% 3,87M 23s
 9600K ..........  .......... .......... .......... .......... 12% 2,72M 23s
 9650K ..........  .......... .......... .......... .......... 13% 1,68M 23s
 9700K ..........  .......... .......... .......... .......... 13% 4,73M 23s
 9750K ..........  .......... .......... .......... .......... 13% 1,34M 23s
 9800K ..........  .......... .......... .......... .......... 13% 4,03M 23s
 9850K ..........  .......... .......... .......... .......... 13% 10,2M 23s
 9900K ..........  .......... .......... .......... .......... 13% 1,35M 23s
 9950K ..........  .......... .......... .......... .......... 13% 4,10M 23s
10000K ..........  .......... .......... .......... .......... 13% 2,49M 23s
10050K ..........  .......... .......... .......... .......... 13% 2,23M 23s
10100K ..........  .......... .......... .......... .......... 13% 4,26M 23s
10150K ..........  .......... .......... .......... .......... 13% 2,48M 23s
10200K ..........  .......... .......... .......... .......... 13% 2,22M 23s
10250K ..........  .......... .......... .......... .......... 13% 3,06M 23s
10300K ..........  .......... .......... .......... .......... 13% 3,25M 23s
10350K ..........  .......... .......... .......... .......... 14% 2,26M 23s
10400K ..........  .......... .......... .......... .......... 14% 2,77M 23s
10450K ..........  .......... .......... .......... .......... 14% 3,40M 23s
10500K ..........  .......... .......... .......... .......... 14% 1,98M 23s
10550K ..........  .......... .......... .......... .......... 14% 3,55M 23s
10600K ..........  .......... .......... .......... .......... 14% 3,33M 23s
10650K ..........  .......... .......... .......... .......... 14% 2,29M 23s
10700K ..........  .......... .......... .......... .......... 14% 2,88M 23s
10750K ..........  .......... .......... .......... .......... 14% 3,35M 23s
10800K ..........  .......... .......... .......... .......... 14% 1,89M 23s
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10850K .......... .......... .......... .......... .......... 14% 4,12M 23s
10900K .......... .......... .......... .......... .......... 14% 3,16M 23s
10950K .......... .......... .......... .......... .......... 14%  673K 23s
11000K .......... .......... .......... .......... .......... 14%  700M 23s
11050K .......... .......... .......... .......... .......... 14%  547M 23s
11100K .......... .......... .......... .......... .......... 15% 18,0M 23s
11150K .......... .......... .......... .......... .......... 15% 1,72M 23s
11200K .......... .......... .......... .......... .......... 15% 1,49M 23s
11250K .......... .......... .......... .......... .......... 15% 2,39M 23s
11300K .......... .......... .......... .......... .......... 15% 2,00M 23s
11350K .......... .......... .......... .......... .......... 15% 1,69M 23s
11400K .......... .......... .......... .......... .......... 15% 2,26M 23s
11450K .......... .......... .......... .......... .......... 15% 1,61M 23s
11500K .......... .......... .......... .......... .......... 15% 2,17M 23s
11550K .......... .......... .......... .......... .......... 15% 3,17M 23s
11600K .......... .......... .......... .......... .......... 15% 1,30M 23s
11650K .......... .......... .......... .......... .......... 15% 3,20M 23s
11700K .......... .......... .......... .......... .......... 15% 2,73M 23s
11750K .......... .......... .......... .......... .......... 15% 1,58M 23s
11800K .......... .......... .......... .......... .......... 15% 2,26M 23s
11850K .......... .......... .......... .......... .......... 16% 2,14M 23s
11900K .......... .......... .......... .......... .......... 16% 1,64M 23s
11950K .......... .......... .......... .......... .......... 16%  673K 23s
12000K .......... .......... .......... .......... .......... 16%  397M 23s
12050K .......... .......... .......... .......... .......... 16%  770M 23s
12100K .......... .......... .......... .......... .......... 16%  945K 23s
12150K .......... .......... .......... .......... .......... 16% 4,80M 23s
12200K .......... .......... .......... .......... .......... 16% 1,14M 23s
12250K .......... .......... .......... .......... .......... 16% 1,56M 23s
12300K .......... .......... .......... .......... .......... 16% 2,20M 23s
12350K .......... .......... .......... .......... .......... 16%  954K 23s
12400K .......... .......... .......... .......... .......... 16% 4,98M 23s
12450K .......... .......... .......... .......... .......... 16% 1,12M 23s
12500K .......... .......... .......... .......... .......... 16% 1,60M 23s
12550K .......... .......... .......... .......... .......... 16% 2,17M 23s
12600K .......... .......... .......... .......... .......... 17%  952K 24s
12650K .......... .......... .......... .......... .......... 17% 5,07M 24s
12700K .......... .......... .......... .......... .......... 17% 1,12M 24s
12750K .......... .......... .......... .......... .......... 17% 5,31M 24s
12800K .......... .......... .......... .......... .......... 17% 1,10M 24s
12850K .......... .......... .......... .......... .......... 17%  816K 24s
12900K .......... .......... .......... .......... .......... 17%  284M 24s
12950K .......... .......... .......... .......... .......... 17%  949K 24s
13000K .......... .......... .......... .......... .......... 17% 1,10M 24s
13050K .......... .......... .......... .......... .......... 17%  935K 24s
13100K .......... .......... .......... .......... .......... 17% 1,60M 24s
13150K .......... .......... .......... .......... .......... 17% 1,56M 24s
13200K .......... .......... .......... .......... .......... 17%  934K 24s
13250K .......... .......... .......... .......... .......... 17% 1,09M 24s
13300K .......... .......... .......... .......... .......... 17%  941K 25s
13350K .......... .......... .......... .......... .......... 18% 4,44M 24s
13400K .......... .......... .......... .......... .......... 18%  982K 25s
13450K .......... .......... .......... .......... .......... 18% 1,10M 25s
13500K .......... .......... .......... .......... .......... 18% 1,58M 25s
13550K .......... .......... .......... .......... .......... 18% 1,54M 25s
13600K .......... .......... .......... .......... .......... 18%  937K 25s
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13650K .......... .......... .......... .......... .......... 18% 1,10M 25s
13700K .......... .......... .......... .......... .......... 18%  940K 25s
13750K .......... .......... .......... .......... .......... 18% 5,33M 25s
13800K .......... .......... .......... .......... .......... 18%  944K 25s
13850K .......... .......... .......... .......... .......... 18% 1,10M 25s
13900K .......... .......... .......... .......... .......... 18% 1,49M 25s
13950K .......... .......... .......... .......... .......... 18% 1,68M 25s
14000K .......... .......... .......... .......... .......... 18%  873K 25s
14050K .......... .......... .......... .......... .......... 18% 1,23M 25s
14100K .......... .......... .......... .......... .......... 19% 3,52M 25s
14150K .......... .......... .......... .......... .......... 19%  941K 25s
14200K .......... .......... .......... .......... .......... 19% 1,25M 26s
14250K .......... .......... .......... .......... .......... 19% 3,49M 25s
14300K .......... .......... .......... .......... .......... 19% 1,24M 26s
14350K .......... .......... .......... .......... .......... 19% 1,46M 26s
14400K .......... .......... .......... .......... .......... 19%  964K 26s
14450K .......... .......... .......... .......... .......... 19% 1,42M 26s
14500K .......... .......... .......... .......... .......... 19% 1,24M 26s
14550K .......... .......... .......... .......... .......... 19% 3,39M 26s
14600K .......... .......... .......... .......... .......... 19% 1,26M 26s
14650K .......... .......... .......... .......... .......... 19% 1,52M 26s
14700K .......... .......... .......... .......... .......... 19% 1,40M 26s
14750K .......... .......... .......... .......... .......... 19% 1,26M 26s
14800K .......... .......... .......... .......... .......... 19% 1,52M 26s
14850K .......... .......... .......... .......... .......... 20% 1,34M 26s
14900K .......... .......... .......... .......... .......... 20% 1,31M 26s
14950K .......... .......... .......... .......... .......... 20% 3,00M 26s
15000K .......... .......... .......... .......... .......... 20%  991K 26s
15050K .......... .......... .......... .......... .......... 20% 2,66M 26s
15100K .......... .......... .......... .......... .......... 20% 1,33M 26s
15150K .......... .......... .......... .......... .......... 20% 1,31M 26s
15200K .......... .......... .......... .......... .......... 20% 1,46M 26s
15250K .......... .......... .......... .......... .......... 20% 1,40M 26s
15300K .......... .......... .......... .......... .......... 20% 2,32M 26s
15350K .......... .......... .......... .......... .......... 20% 1,53M 26s
15400K .......... .......... .......... .......... .......... 20% 1,29M 26s
15450K .......... .......... .......... .......... .......... 20% 2,61M 26s
15500K .......... .......... .......... .......... .......... 20% 1,42M 26s
15550K .......... .......... .......... .......... .......... 21% 1,48M 26s
15600K .......... .......... .......... .......... .......... 21% 1,29M 26s
15650K .......... .......... .......... .......... .......... 21% 1,41M 26s
15700K .......... .......... .......... .......... .......... 21% 2,32M 26s
15750K .......... .......... .......... .......... .......... 21% 1,51M 26s
15800K .......... .......... .......... .......... .......... 21% 2,37M 26s
15850K .......... .......... .......... .......... .......... 21% 1,50M 26s
15900K .......... .......... .......... .......... .......... 21% 1,50M 26s
15950K .......... .......... .......... .......... .......... 21% 1,47M 26s
16000K .......... .......... .......... .......... .......... 21% 1,22M 26s
16050K .......... .......... .......... .......... .......... 21% 2,84M 26s
16100K .......... .......... .......... .......... .......... 21% 1,36M 26s
16150K .......... .......... .......... .......... .......... 21% 2,46M 26s
16200K .......... .......... .......... .......... .......... 21% 1,47M 26s
16250K .......... .......... .......... .......... .......... 21% 2,51M 26s
16300K .......... .......... .......... .......... .......... 22% 1,45M 26s
16350K .......... .......... .......... .......... .......... 22% 2,54M 26s
16400K .......... .......... .......... .......... .......... 22% 1,07M 26s
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16450K .......... .......... .......... .......... .......... 22% 2,41M 26s
16500K .......... .......... .......... .......... .......... 22% 1,28M 26s
16550K .......... .......... .......... .......... .......... 22% 3,31M 26s
16600K .......... .......... .......... .......... .......... 22% 1,29M 26s
16650K .......... .......... .......... .......... .......... 22% 3,27M 26s
16700K .......... .......... .......... .......... .......... 22% 1,28M 26s
16750K .......... .......... .......... .......... .......... 22% 3,28M 26s
16800K .......... .......... .......... .......... .......... 22% 1,27M 26s
16850K .......... .......... .......... .......... .......... 22% 1,36M 26s
16900K .......... .......... .......... .......... .......... 22% 2,87M 26s
16950K .......... .......... .......... .......... .......... 22% 1,36M 26s
17000K .......... .......... .......... .......... .......... 22% 2,86M 26s
17050K .......... .......... .......... .......... .......... 23% 1,36M 26s
17100K .......... .......... .......... .......... .......... 23% 2,86M 26s
17150K .......... .......... .......... .......... .......... 23% 1,35M 26s
17200K .......... .......... .......... .......... .......... 23% 2,84M 26s
17250K .......... .......... .......... .......... .......... 23% 1,36M 26s
17300K .......... .......... .......... .......... .......... 23% 2,77M 26s
17350K .......... .......... .......... .......... .......... 23% 1,38M 26s
17400K .......... .......... .......... .......... .......... 23% 2,18M 26s
17450K .......... .......... .......... .......... .......... 23% 1,60M 26s
17500K .......... .......... .......... .......... .......... 23% 2,52M 26s
17550K .......... .......... .......... .......... .......... 23% 3,48M 26s
17600K .......... .......... .......... .......... .......... 23% 1,17M 26s
17650K .......... .......... .......... .......... .......... 23% 1,55M 26s
17700K .......... .......... .......... .......... .......... 23% 2,26M 26s
17750K .......... .......... .......... .......... .......... 23% 1,52M 26s
17800K .......... .......... .......... .......... .......... 24% 2,66M 26s
17850K .......... .......... .......... .......... .......... 24% 3,93M 26s
17900K .......... .......... .......... .......... .......... 24% 1,50M 26s
17950K .......... .......... .......... .......... .......... 24% 2,39M 26s
18000K .......... .......... .......... .......... .......... 24% 1,19M 26s
18050K .......... .......... .......... .......... .......... 24% 3,92M 26s
18100K .......... .......... .......... .......... .......... 24% 1,50M 26s
18150K .......... .......... .......... .......... .......... 24% 2,38M 26s
18200K .......... .......... .......... .......... .......... 24% 1,51M 26s
18250K .......... .......... .......... .......... .......... 24% 4,18M 26s
18300K .......... .......... .......... .......... .......... 24% 1,58M 26s
18350K .......... .......... .......... .......... .......... 24% 2,24M 26s
18400K .......... .......... .......... .......... .......... 24% 1,57M 26s
18450K .......... .......... .......... .......... .......... 24% 2,21M 26s
18500K .......... .......... .......... .......... .......... 24% 1,59M 26s
18550K .......... .......... .......... .......... .......... 25% 2,22M 26s
18600K .......... .......... .......... .......... .......... 25% 4,58M 26s
18650K .......... .......... .......... .......... .......... 25% 1,61M 26s
18700K .......... .......... .......... .......... .......... 25% 2,12M 26s
18750K .......... .......... .......... .......... .......... 25% 1,63M 26s
18800K .......... .......... .......... .......... .......... 25% 2,21M 26s
18850K .......... .......... .......... .......... .......... 25% 1,58M 26s
18900K .......... .......... .......... .......... .......... 25% 3,93M 26s
18950K .......... .......... .......... .......... .......... 25% 1,56M 26s
19000K .......... .......... .......... .......... .......... 25% 2,24M 25s
19050K .......... .......... .......... .......... .......... 25% 4,06M 25s
19100K .......... .......... .......... .......... .......... 25% 1,70M 25s
19150K .......... .......... .......... .......... .......... 25% 2,05M 25s
19200K .......... .......... .......... .......... .......... 25% 1,67M 25s
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19250K .......... .......... .......... .......... .......... 25% 2,05M 25s
19300K .......... .......... .......... .......... .......... 26% 2,35M 25s
19350K .......... .......... .......... .......... .......... 26% 2,09M 25s
19400K .......... .......... .......... .......... .......... 26% 1,66M 25s
19450K .......... .......... .......... .......... .......... 26% 2,08M 25s
19500K .......... .......... .......... .......... .......... 26% 5,52M 25s
19550K .......... .......... .......... .......... .......... 26% 1,63M 25s
19600K .......... .......... .......... .......... .......... 26% 2,14M 25s
19650K .......... .......... .......... .......... .......... 26% 1,63M 25s
19700K .......... .......... .......... .......... .......... 26% 3,02M 25s
19750K .......... .......... .......... .......... .......... 26% 1,77M 25s
19800K .......... .......... .......... .......... .......... 26% 2,05M 25s
19850K .......... .......... .......... .......... .......... 26% 5,38M 25s
19900K .......... .......... .......... .......... .......... 26% 1,65M 25s
19950K .......... .......... .......... .......... .......... 26% 2,12M 25s
20000K .......... .......... .......... .......... .......... 26% 1,63M 25s
20050K .......... .......... .......... .......... .......... 27% 3,46M 25s
20100K .......... .......... .......... .......... .......... 27% 1,63M 25s
20150K .......... .......... .......... .......... .......... 27% 2,57M 25s
20200K .......... .......... .......... .......... .......... 27% 3,07M 25s
20250K .......... .......... .......... .......... .......... 27% 1,75M 25s
20300K .......... .......... .......... .......... .......... 27% 3,53M 25s
20350K .......... .......... .......... .......... .......... 27% 1,65M 25s
20400K .......... .......... .......... .......... .......... 27% 2,09M 25s
20450K .......... .......... .......... .......... .......... 27% 1,64M 25s
20500K .......... .......... .......... .......... .......... 27% 5,32M 25s
20550K .......... .......... .......... .......... .......... 27% 2,17M 25s
20600K .......... .......... .......... .......... .......... 27% 2,30M 25s
20650K .......... .......... .......... .......... .......... 27% 2,11M 25s
20700K .......... .......... .......... .......... .......... 27% 4,33M 25s
20750K .......... .......... .......... .......... .......... 28% 1,77M 25s
20800K .......... .......... .......... .......... .......... 28% 2,14M 25s
20850K .......... .......... .......... .......... .......... 28% 1,62M 25s
20900K .......... .......... .......... .......... .......... 28% 3,43M 25s
20950K .......... .......... .......... .......... .......... 28% 1,64M 25s
21000K .......... .......... .......... .......... .......... 28% 5,40M 24s
21050K .......... .......... .......... .......... .......... 28% 2,17M 24s
21100K .......... .......... .......... .......... .......... 28% 2,21M 24s
21150K .......... .......... .......... .......... .......... 28% 2,10M 24s
21200K .......... .......... .......... .......... .......... 28% 1,64M 24s
21250K .......... .......... .......... .......... .......... 28% 5,37M 24s
21300K .......... .......... .......... .......... .......... 28% 2,23M 24s
21350K .......... .......... .......... .......... .......... 28% 2,12M 24s
21400K .......... .......... .......... .......... .......... 28% 2,13M 24s
21450K .......... .......... .......... .......... .......... 28% 4,13M 24s
21500K .......... .......... .......... .......... .......... 29% 1,83M 24s
21550K .......... .......... .......... .......... .......... 29% 3,31M 24s
21600K .......... .......... .......... .......... .......... 29% 1,60M 24s
21650K .......... .......... .......... .......... .......... 29% 2,13M 24s
21700K .......... .......... .......... .......... .......... 29% 4,24M 24s
21750K .......... .......... .......... .......... .......... 29% 1,80M 24s
21800K .......... .......... .......... .......... .......... 29% 3,36M 24s
21850K .......... .......... .......... .......... .......... 29% 1,62M 24s
21900K .......... .......... .......... .......... .......... 29% 4,04M 24s
21950K .......... .......... .......... .......... .......... 29% 2,71M 24s
22000K .......... .......... .......... .......... .......... 29% 1,57M 24s
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22050K .......... .......... .......... .......... .......... 29% 3,39M 24s
22100K .......... .......... .......... .......... .......... 29% 4,46M 24s
22150K .......... .......... .......... .......... .......... 29% 1,56M 24s
22200K .......... .......... .......... .......... .......... 29% 2,71M 24s
22250K .......... .......... .......... .......... .......... 30% 2,12M 24s
22300K .......... .......... .......... .......... .......... 30% 2,79M 24s
22350K .......... .......... .......... .......... .......... 30% 3,93M 24s
22400K .......... .......... .......... .......... .......... 30% 1,54M 24s
22450K .......... .......... .......... .......... .......... 30% 3,32M 24s
22500K .......... .......... .......... .......... .......... 30% 1,64M 24s
22550K .......... .......... .......... .......... .......... 30% 3,75M 24s
22600K .......... .......... .......... .......... .......... 30% 4,45M 24s
22650K .......... .......... .......... .......... .......... 30% 1,66M 24s
22700K .......... .......... .......... .......... .......... 30% 3,85M 24s
22750K .......... .......... .......... .......... .......... 30% 2,90M 23s
22800K .......... .......... .......... .......... .......... 30% 1,53M 23s
22850K .......... .......... .......... .......... .......... 30% 3,35M 23s
22900K .......... .......... .......... .......... .......... 30% 4,86M 23s
22950K .......... .......... .......... .......... .......... 30% 1,52M 23s
23000K .......... .......... .......... .......... .......... 31% 4,46M 23s
23050K .......... .......... .......... .......... .......... 31% 1,62M 23s
23100K .......... .......... .......... .......... .......... 31% 4,19M 23s
23150K .......... .......... .......... .......... .......... 31% 4,22M 23s
23200K .......... .......... .......... .......... .......... 31% 1,61M 23s
23250K .......... .......... .......... .......... .......... 31% 2,23M 23s
23300K .......... .......... .......... .......... .......... 31% 7,51M 23s
23350K .......... .......... .......... .......... .......... 31% 1,99M 23s
23400K .......... .......... .......... .......... .......... 31% 2,25M 23s
23450K .......... .......... .......... .......... .......... 31% 7,00M 23s
23500K .......... .......... .......... .......... .......... 31% 1,56M 23s
23550K .......... .......... .......... .......... .......... 31% 3,36M 23s
23600K .......... .......... .......... .......... .......... 31% 1,59M 23s
23650K .......... .......... .......... .......... .......... 31% 4,31M 23s
23700K .......... .......... .......... .......... .......... 31% 2,82M 23s
23750K .......... .......... .......... .......... .......... 32% 2,00M 23s
23800K .......... .......... .......... .......... .......... 32% 4,39M 23s
23850K .......... .......... .......... .......... .......... 32% 2,79M 23s
23900K .......... .......... .......... .......... .......... 32% 1,99M 23s
23950K .......... .......... .......... .......... .......... 32% 4,26M 23s
24000K .......... .......... .......... .......... .......... 32% 2,79M 23s
24050K .......... .......... .......... .......... .......... 32% 1,55M 23s
24100K .......... .......... .......... .......... .......... 32% 3,53M 23s
24150K .......... .......... .......... .......... .......... 32% 6,25M 23s
24200K .......... .......... .......... .......... .......... 32% 1,56M 23s
24250K .......... .......... .......... .......... .......... 32% 3,58M 23s
24300K .......... .......... .......... .......... .......... 32% 5,02M 23s
24350K .......... .......... .......... .......... .......... 32% 1,65M 22s
24400K .......... .......... .......... .......... .......... 32% 3,34M 22s
24450K .......... .......... .......... .......... .......... 32% 1,57M 22s
24500K .......... .......... .......... .......... .......... 33% 4,14M 22s
24550K .......... .......... .......... .......... .......... 33% 5,13M 22s
24600K .......... .......... .......... .......... .......... 33% 1,55M 22s
24650K .......... .......... .......... .......... .......... 33% 4,12M 22s
24700K .......... .......... .......... .......... .......... 33% 5,11M 22s
24750K .......... .......... .......... .......... .......... 33% 1,54M 22s
24800K .......... .......... .......... .......... .......... 33% 3,09M 22s
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24850K .......... .......... .......... .......... .......... 33% 3,70M 22s
24900K .......... .......... .......... .......... .......... 33% 2,01M 22s
24950K .......... .......... .......... .......... .......... 33% 4,43M 22s
25000K .......... .......... .......... .......... .......... 33% 2,73M 22s
25050K .......... .......... .......... .......... .......... 33%  598K 22s
25100K .......... .......... .......... .......... .......... 33% 3,36M 22s
25150K .......... .......... .......... .......... .......... 33%  531M 22s
25200K .......... .......... .......... .......... .......... 33% 4,51M 22s
25250K .......... .......... .......... .......... .......... 34% 1,73M 22s
25300K .......... .......... .......... .......... .......... 34% 2,15M 22s
25350K .......... .......... .......... .......... .......... 34% 1,63M 22s
25400K .......... .......... .......... .......... .......... 34% 2,26M 22s
25450K .......... .......... .......... .......... .......... 34% 1,56M 22s
25500K .......... .......... .......... .......... .......... 34% 5,24M 22s
25550K .......... .......... .......... .......... .......... 34% 1,12M 22s
25600K .......... .......... .......... .......... .......... 34% 3,39M 22s
25650K .......... .......... .......... .......... .......... 34% 1,27M 22s
25700K .......... .......... .......... .......... .......... 34% 5,42M 22s
25750K .......... .......... .......... .......... .......... 34% 1,32M 22s
25800K .......... .......... .......... .......... .......... 34% 6,41M 22s
25850K .......... .......... .......... .......... .......... 34% 2,07M 22s
25900K .......... .......... .......... .......... .......... 34% 1,66M 22s
25950K .......... .......... .......... .......... .......... 35% 2,34M 22s
26000K .......... .......... .......... .......... .......... 35% 1,55M 22s
26050K .......... .......... .......... .......... .......... 35% 3,51M 22s
26100K .......... .......... .......... .......... .......... 35% 1,24M 22s
26150K .......... .......... .......... .......... .......... 35% 5,80M 22s
26200K .......... .......... .......... .......... .......... 35% 1,29M 22s
26250K .......... .......... .......... .......... .......... 35% 6,69M 22s
26300K .......... .......... .......... .......... .......... 35% 2,34M 22s
26350K .......... .......... .......... .......... .......... 35% 1,55M 22s
26400K .......... .......... .......... .......... .......... 35% 2,30M 22s
26450K .......... .......... .......... .......... .......... 35% 1,54M 22s
26500K .......... .......... .......... .......... .......... 35% 19,0M 22s
26550K .......... .......... .......... .......... .......... 35% 1,12M 22s
26600K .......... .......... .......... .......... .......... 35% 6,71M 21s
26650K .......... .......... .......... .......... .......... 35% 2,36M 21s
26700K .......... .......... .......... .......... .......... 36% 1,54M 21s
26750K .......... .......... .......... .......... .......... 36% 17,7M 21s
26800K .......... .......... .......... .......... .......... 36% 1,11M 21s
26850K .......... .......... .......... .......... .......... 36% 7,37M 21s
26900K .......... .......... .......... .......... .......... 36% 2,19M 21s
26950K .......... .......... .......... .......... .......... 36% 1,62M 21s
27000K .......... .......... .......... .......... .......... 36% 5,72M 21s
27050K .......... .......... .......... .......... .......... 36% 1,27M 21s
27100K .......... .......... .......... .......... .......... 36% 7,55M 21s
27150K .......... .......... .......... .......... .......... 36% 2,32M 21s
27200K .......... .......... .......... .......... .......... 36% 1,53M 21s
27250K .......... .......... .......... .......... .......... 36% 6,09M 21s
27300K .......... .......... .......... .......... .......... 36% 1,26M 21s
27350K .......... .......... .......... .......... .......... 36% 7,39M 21s
27400K .......... .......... .......... .......... .......... 36% 2,31M 21s
27450K .......... .......... .......... .......... .......... 37% 1,87M 21s
27500K .......... .......... .......... .......... .......... 37% 7,12M 21s
27550K .......... .......... .......... .......... .......... 37% 1,07M 21s
27600K .......... .......... .......... .......... .......... 37% 7,46M 21s
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27650K .......... .......... .......... .......... .......... 37% 2,34M 21s
27700K .......... .......... .......... .......... .......... 37% 1,85M 21s
27750K .......... .......... .......... .......... .......... 37% 6,95M 21s
27800K .......... .......... .......... .......... .......... 37% 2,50M 21s
27850K .......... .......... .......... .......... .......... 37% 1,52M 21s
27900K .......... .......... .......... .......... .......... 37% 6,42M 21s
27950K .......... .......... .......... .......... .......... 37% 1,24M 21s
28000K .......... .......... .......... .......... .......... 37% 7,28M 21s
28050K .......... .......... .......... .......... .......... 37% 2,42M 21s
28100K .......... .......... .......... .......... .......... 37% 1,81M 21s
28150K .......... .......... .......... .......... .......... 37% 7,88M 21s
28200K .......... .......... .......... .......... .......... 38% 1,05M 21s
28250K .......... .......... .......... .......... .......... 38% 9,49M 21s
28300K .......... .......... .......... .......... .......... 38% 6,56M 21s
28350K .......... .......... .......... .......... .......... 38% 1,21M 21s
28400K .......... .......... .......... .......... .......... 38% 8,93M 21s
28450K .......... .......... .......... .......... .......... 38% 2,39M 21s
28500K .......... .......... .......... .......... .......... 38% 1,69M 21s
28550K .......... .......... .......... .......... .......... 38% 8,35M 20s
28600K .......... .......... .......... .......... .......... 38% 2,68M 20s
28650K .......... .......... .......... .......... .......... 38% 1,52M 20s
28700K .......... .......... .......... .......... .......... 38% 6,53M 20s
28750K .......... .......... .......... .......... .......... 38% 1,16M 20s
28800K .......... .......... .......... .......... .......... 38% 8,95M 20s
28850K .......... .......... .......... .......... .......... 38% 2,63M 20s
28900K .......... .......... .......... .......... .......... 38% 1,66M 20s
28950K .......... .......... .......... .......... .......... 39% 10,5M 20s
29000K .......... .......... .......... .......... .......... 39% 2,57M 20s
29050K .......... .......... .......... .......... .......... 39% 1,67M 20s
29100K .......... .......... .......... .......... .......... 39% 9,64M 20s
29150K .......... .......... .......... .......... .......... 39% 2,57M 20s
29200K .......... .......... .......... .......... .......... 39% 1,50M 20s
29250K .......... .......... .......... .......... .......... 39% 7,51M 20s
29300K .......... .......... .......... .......... .......... 39% 1,15M 20s
29350K .......... .......... .......... .......... .......... 39% 12,6M 20s
29400K .......... .......... .......... .......... .......... 39% 4,77M 20s
29450K .......... .......... .......... .......... .......... 39% 1,26M 20s
29500K .......... .......... .......... .......... .......... 39% 11,6M 20s
29550K .......... .......... .......... .......... .......... 39% 2,47M 20s
29600K .......... .......... .......... .......... .......... 39% 1,65M 20s
29650K .......... .......... .......... .......... .......... 39% 13,0M 20s
29700K .......... .......... .......... .......... .......... 40% 2,43M 20s
29750K .......... .......... .......... .......... .......... 40% 1,51M 20s
29800K .......... .......... .......... .......... .......... 40% 48,3M 20s
29850K .......... .......... .......... .......... .......... 40% 2,29M 20s
29900K .......... .......... .......... .......... .......... 40% 1,60M 20s
29950K .......... .......... .......... .......... .......... 40% 47,3M 20s
30000K .......... .......... .......... .......... .......... 40% 1,01M 20s
30050K .......... .......... .......... .......... .......... 40% 10,5M 20s
30100K .......... .......... .......... .......... .......... 40% 5,27M 20s
30150K .......... .......... .......... .......... .......... 40% 1,11M 20s
30200K .......... .......... .......... .......... .......... 40% 23,6M 20s
30250K .......... .......... .......... .......... .......... 40%  835K 20s
30300K .......... .......... .......... .......... .......... 40%  718M 20s
30350K .......... .......... .......... .......... .......... 40%  800M 19s
30400K .......... .......... .......... .......... .......... 40% 1,09M 20s
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30450K .......... .......... .......... .......... .......... 41% 2,40M 19s
30500K .......... .......... .......... .......... .......... 41% 1,54M 19s
30550K .......... .......... .......... .......... .......... 41% 9,60M 19s
30600K .......... .......... .......... .......... .......... 41% 1,02M 19s
30650K .......... .......... .......... .......... .......... 41%  852K 19s
30700K .......... .......... .......... .......... .......... 41%  451M 19s
30750K .......... .......... .......... .......... .......... 41%  948K 19s
30800K .......... .......... .......... .......... .......... 41%  420M 19s
30850K .......... .......... .......... .......... .......... 41%  943K 19s
30900K .......... .......... .......... .......... .......... 41% 1,02M 19s
30950K .......... .......... .......... .......... .......... 41% 8,59M 19s
31000K .......... .......... .......... .......... .......... 41%  956K 19s
31050K .......... .......... .......... .......... .......... 41% 1,03M 19s
31100K .......... .......... .......... .......... .......... 41% 8,27M 19s
31150K .......... .......... .......... .......... .......... 42% 1,03M 19s
31200K .......... .......... .......... .......... .......... 42%  952K 19s
31250K .......... .......... .......... .......... .......... 42% 8,44M 19s
31300K .......... .......... .......... .......... .......... 42%  952K 19s
31350K .......... .......... .......... .......... .......... 42% 1,07M 19s
31400K .......... .......... .......... .......... .......... 42% 6,51M 19s
31450K .......... .......... .......... .......... .......... 42% 1,04M 19s
31500K .......... .......... .......... .......... .......... 42% 6,85M 19s
31550K .......... .......... .......... .......... .......... 42%  966K 19s
31600K .......... .......... .......... .......... .......... 42% 1,04M 19s
31650K .......... .......... .......... .......... .......... 42% 6,98M 19s
31700K .......... .......... .......... .......... .......... 42%  955K 19s
31750K .......... .......... .......... .......... .......... 42% 1,06M 19s
31800K .......... .......... .......... .......... .......... 42% 6,84M 19s
31850K .......... .......... .......... .......... .......... 42%  501K 19s
31900K .......... .......... .......... .......... .......... 43%  204M 19s
31950K .......... .......... .......... .......... .......... 43% 7,06M 19s
32000K .......... .......... .......... .......... .......... 43%  503K 19s
32050K .......... .......... .......... .......... .......... 43% 7,40M 19s
32100K .......... .......... .......... .......... .......... 43%  927K 19s
32150K .......... .......... .......... .......... .......... 43%  942K 19s
32200K .......... .......... .......... .......... .......... 43% 1,03M 19s
32250K .......... .......... .......... .......... .......... 43%  934K 19s
32300K .......... .......... .......... .......... .......... 43% 8,02M 19s
32350K .......... .......... .......... .......... .......... 43%  944K 19s
32400K .......... .......... .......... .......... .......... 43%  926K 19s
32450K .......... .......... .......... .......... .......... 43%  942K 19s
32500K .......... .......... .......... .......... .......... 43% 1,06M 19s
32550K .......... .......... .......... .......... .......... 43% 5,78M 19s
32600K .......... .......... .......... .......... .......... 43%  961K 19s
32650K .......... .......... .......... .......... .......... 44%  942K 19s
32700K .......... .......... .......... .......... .......... 44% 1,07M 19s
32750K .......... .......... .......... .......... .......... 44% 6,29M 19s
32800K .......... .......... .......... .......... .......... 44%  943K 19s
32850K .......... .......... .......... .......... .......... 44%  953K 19s
32900K .......... .......... .......... .......... .......... 44% 1,07M 19s
32950K .......... .......... .......... .......... .......... 44%  945K 19s
33000K .......... .......... .......... .......... .......... 44% 6,12M 19s
33050K .......... .......... .......... .......... .......... 44%  942K 19s
33100K .......... .......... .......... .......... .......... 44% 1,08M 19s
33150K .......... .......... .......... .......... .......... 44% 5,84M 19s
33200K .......... .......... .......... .......... .......... 44%  954K 19s
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33250K .......... .......... .......... .......... .......... 44%  956K 19s
33300K .......... .......... .......... .......... .......... 44% 1,06M 19s
33350K .......... .......... .......... .......... .......... 44% 5,55M 19s
33400K .......... .......... .......... .......... .......... 45%  955K 19s
33450K .......... .......... .......... .......... .......... 45% 1,08M 19s
33500K .......... .......... .......... .......... .......... 45% 5,95M 19s
33550K .......... .......... .......... .......... .......... 45%  954K 19s
33600K .......... .......... .......... .......... .......... 45% 1,07M 19s
33650K .......... .......... .......... .......... .......... 45% 6,11M 19s
33700K .......... .......... .......... .......... .......... 45%  955K 19s
33750K .......... .......... .......... .......... .......... 45% 1,07M 19s
33800K .......... .......... .......... .......... .......... 45% 6,06M 19s
33850K .......... .......... .......... .......... .......... 45%  502K 19s
33900K .......... .......... .......... .......... .......... 45% 8,62M 19s
33950K .......... .......... .......... .......... .......... 45%  940K 19s
34000K .......... .......... .......... .......... .......... 45%  944K 19s
34050K .......... .......... .......... .......... .......... 45%  944K 19s
34100K .......... .......... .......... .......... .......... 45% 1,04M 19s
34150K .......... .......... .......... .......... .......... 46%  944K 19s
34200K .......... .......... .......... .......... .......... 46% 7,46M 19s
34250K .......... .......... .......... .......... .......... 46%  962K 19s
34300K .......... .......... .......... .......... .......... 46%  943K 19s
34350K .......... .......... .......... .......... .......... 46% 1,03M 19s
34400K .......... .......... .......... .......... .......... 46%  854K 19s
34450K .......... .......... .......... .......... .......... 46% 1,03M 19s
34500K .......... .......... .......... .......... .......... 46% 7,78M 19s
34550K .......... .......... .......... .......... .......... 46%  953K 19s
34600K .......... .......... .......... .......... .......... 46%  944K 19s
34650K .......... .......... .......... .......... .......... 46% 1,03M 19s
34700K .......... .......... .......... .......... .......... 46% 8,32M 19s
34750K .......... .......... .......... .......... .......... 46%  964K 19s
34800K .......... .......... .......... .......... .......... 46%  936K 19s
34850K .......... .......... .......... .......... .......... 46% 1,01M 19s
34900K .......... .......... .......... .......... .......... 47%  943K 19s
34950K .......... .......... .......... .......... .......... 47% 9,74M 19s
35000K .......... .......... .......... .......... .......... 47%  954K 19s
35050K .......... .......... .......... .......... .......... 47% 1,00M 19s
35100K .......... .......... .......... .......... .......... 47% 9,07M 19s
35150K .......... .......... .......... .......... .......... 47%  947K 19s
35200K .......... .......... .......... .......... .......... 47%  939K 19s
35250K .......... .......... .......... .......... .......... 47% 1,03M 19s
35300K .......... .......... .......... .......... .......... 47% 7,78M 19s
35350K .......... .......... .......... .......... .......... 47%  954K 19s
35400K .......... .......... .......... .......... .......... 47% 1,04M 19s
35450K .......... .......... .......... .......... .......... 47% 7,29M 19s
35500K .......... .......... .......... .......... .......... 47%  953K 19s
35550K .......... .......... .......... .......... .......... 47% 1,04M 19s
35600K .......... .......... .......... .......... .......... 47%  939K 19s
35650K .......... .......... .......... .......... .......... 48% 6,92M 19s
35700K .......... .......... .......... .......... .......... 48%  953K 19s
35750K .......... .......... .......... .......... .......... 48% 1,05M 19s
35800K .......... .......... .......... .......... .......... 48% 6,79M 19s
35850K .......... .......... .......... .......... .......... 48%  474K 19s
35900K .......... .......... .......... .......... .......... 48%  364M 19s
35950K .......... .......... .......... .......... .......... 48%  948K 19s
36000K .......... .......... .......... .......... .......... 48%  930K 19s
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36050K .......... .......... .......... .......... .......... 48%  958K 19s
36100K .......... .......... .......... .......... .......... 48%  941K 19s
36150K .......... .......... .......... .......... .......... 48%  942K 19s
36200K .......... .......... .......... .......... .......... 48% 1,07M 19s
36250K .......... .......... .......... .......... .......... 48% 6,14M 19s
36300K .......... .......... .......... .......... .......... 48%  944K 19s
36350K .......... .......... .......... .......... .......... 49%  943K 19s
36400K .......... .......... .......... .......... .......... 49%  922K 19s
36450K .......... .......... .......... .......... .......... 49%  944K 19s
36500K .......... .......... .......... .......... .......... 49%  849K 19s
36550K .......... .......... .......... .......... .......... 49%  989K 19s
36600K .......... .......... .......... .......... .......... 49% 1,16M 19s
36650K .......... .......... .......... .......... .......... 49% 3,08M 19s
36700K .......... .......... .......... .......... .......... 49%  990K 19s
36750K .......... .......... .......... .......... .......... 49%  952K 19s
36800K .......... .......... .......... .......... .......... 49%  947K 19s
36850K .......... .......... .......... .......... .......... 49%  988K 19s
36900K .......... .......... .......... .......... .......... 49% 1,13M 19s
36950K .......... .......... .......... .......... .......... 49% 3,17M 19s
37000K .......... .......... .......... .......... .......... 49% 1012K 19s
37050K .......... .......... .......... .......... .......... 49%  968K 19s
37100K .......... .......... .......... .......... .......... 50% 1,16M 19s
37150K .......... .......... .......... .......... .......... 50% 2,48M 19s
37200K .......... .......... .......... .......... .......... 50% 1,01M 19s
37250K .......... .......... .......... .......... .......... 50% 1005K 19s
37300K .......... .......... .......... .......... .......... 50%  976K 19s
37350K .......... .......... .......... .......... .......... 50% 3,58M 19s
37400K .......... .......... .......... .......... .......... 50% 1,12M 19s
37450K .......... .......... .......... .......... .......... 50% 1,01M 19s
37500K .......... .......... .......... .......... .......... 50% 1,14M 19s
37550K .......... .......... .......... .......... .......... 50% 3,35M 19s
37600K .......... .......... .......... .......... .......... 50%  948K 19s
37650K .......... .......... .......... .......... .......... 50% 1010K 19s
37700K .......... .......... .......... .......... .......... 50% 3,82M 19s
37750K .......... .......... .......... .......... .......... 50% 1,12M 19s
37800K .......... .......... .......... .......... .......... 50% 1007K 19s
37850K .......... .......... .......... .......... .......... 51% 3,14M 19s
37900K .......... .......... .......... .......... .......... 51% 1,18M 19s
37950K .......... .......... .......... .......... .......... 51% 1,00M 19s
38000K .......... .......... .......... .......... .......... 51% 1,04M 19s
38050K .......... .......... .......... .......... .......... 51% 3,13M 19s
38100K .......... .......... .......... .......... .......... 51% 1,06M 19s
38150K .......... .......... .......... .......... .......... 51% 1,17M 19s
38200K .......... .......... .......... .......... .......... 51% 2,69M 19s
38250K .......... .......... .......... .......... .......... 51% 1,12M 19s
38300K .......... .......... .......... .......... .......... 51% 2,44M 19s
38350K .......... .......... .......... .......... .......... 51% 1,24M 19s
38400K .......... .......... .......... .......... .......... 51%  955K 19s
38450K .......... .......... .......... .......... .......... 51% 1,26M 19s
38500K .......... .......... .......... .......... .......... 51% 2,67M 18s
38550K .......... .......... .......... .......... .......... 51% 1,11M 18s
38600K .......... .......... .......... .......... .......... 52% 4,48M 18s
38650K .......... .......... .......... .......... .......... 52% 1012K 18s
38700K .......... .......... .......... .......... .......... 52% 1,35M 18s
38750K .......... .......... .......... .......... .......... 52% 2,98M 18s
38800K .......... .......... .......... .......... .......... 52%  966K 18s
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38850K .......... .......... .......... .......... .......... 52% 3,80M 18s
38900K .......... .......... .......... .......... .......... 52% 1,22M 18s
38950K .......... .......... .......... .......... .......... 52% 1,27M 18s
39000K .......... .......... .......... .......... .......... 52% 2,45M 18s
39050K .......... .......... .......... .......... .......... 52% 1,16M 18s
39100K .......... .......... .......... .......... .......... 52% 4,55M 18s
39150K .......... .......... .......... .......... .......... 52% 1,03M 18s
39200K .......... .......... .......... .......... .......... 52% 1,25M 18s
39250K .......... .......... .......... .......... .......... 52% 2,53M 18s
39300K .......... .......... .......... .......... .......... 52% 1,16M 18s
39350K .......... .......... .......... .......... .......... 53% 4,44M 18s
39400K .......... .......... .......... .......... .......... 53% 1,13M 18s
39450K .......... .......... .......... .......... .......... 53% 2,92M 18s
39500K .......... .......... .......... .......... .......... 53% 1,20M 18s
39550K .......... .......... .......... .......... .......... 53% 1,17M 18s
39600K .......... .......... .......... .......... .......... 53% 2,62M 18s
39650K .......... .......... .......... .......... .......... 53% 1,28M 18s
39700K .......... .......... .......... .......... .......... 53% 2,67M 18s
39750K .......... .......... .......... .......... .......... 53% 1,32M 18s
39800K .......... .......... .......... .......... .......... 53% 3,11M 18s
39850K .......... .......... .......... .......... .......... 53% 1,23M 18s
39900K .......... .......... .......... .......... .......... 53% 2,61M 18s
39950K .......... .......... .......... .......... .......... 53% 1,21M 18s
40000K .......... .......... .......... .......... .......... 53% 1,20M 18s
40050K .......... .......... .......... .......... .......... 53% 2,72M 18s
40100K .......... .......... .......... .......... .......... 54% 1,40M 18s
40150K .......... .......... .......... .......... .......... 54% 2,52M 18s
40200K .......... .......... .......... .......... .......... 54% 1,40M 18s
40250K .......... .......... .......... .......... .......... 54% 2,44M 18s
40300K .......... .......... .......... .......... .......... 54% 1,48M 18s
40350K .......... .......... .......... .......... .......... 54% 2,50M 18s
40400K .......... .......... .......... .......... .......... 54% 1,22M 18s
40450K .......... .......... .......... .......... .......... 54% 2,30M 18s
40500K .......... .......... .......... .......... .......... 54% 1,55M 18s
40550K .......... .......... .......... .......... .......... 54% 2,85M 18s
40600K .......... .......... .......... .......... .......... 54% 1,31M 18s
40650K .......... .......... .......... .......... .......... 54% 3,11M 18s
40700K .......... .......... .......... .......... .......... 54% 1,31M 17s
40750K .......... .......... .......... .......... .......... 54% 3,16M 17s
40800K .......... .......... .......... .......... .......... 54% 1,05M 17s
40850K .......... .......... .......... .......... .......... 55% 1,38M 17s
40900K .......... .......... .......... .......... .......... 55% 3,23M 17s
40950K .......... .......... .......... .......... .......... 55% 3,92M 17s
41000K .......... .......... .......... .......... .......... 55% 1,21M 17s
41050K .......... .......... .......... .......... .......... 55% 3,92M 17s
41100K .......... .......... .......... .......... .......... 55% 1,40M 17s
41150K .......... .......... .......... .......... .......... 55% 3,44M 17s
41200K .......... .......... .......... .......... .......... 55% 1,06M 17s
41250K .......... .......... .......... .......... .......... 55% 4,62M 17s
41300K .......... .......... .......... .......... .......... 55% 1,41M 17s
41350K .......... .......... .......... .......... .......... 55% 2,70M 17s
41400K .......... .......... .......... .......... .......... 55% 1,40M 17s
41450K .......... .......... .......... .......... .......... 55% 3,48M 17s
41500K .......... .......... .......... .......... .......... 55% 1,36M 17s
41550K .......... .......... .......... .......... .......... 56% 2,88M 17s
41600K .......... .......... .......... .......... .......... 56% 1,26M 17s
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41650K .......... .......... .......... .......... .......... 56% 2,89M 17s
41700K .......... .......... .......... .......... .......... 56% 1,32M 17s
41750K .......... .......... .......... .......... .......... 56% 3,21M 17s
41800K .......... .......... .......... .......... .......... 56% 1,45M 17s
41850K .......... .......... .......... .......... .......... 56% 2,84M 17s
41900K .......... .......... .......... .......... .......... 56% 1,48M 17s
41950K .......... .......... .......... .......... .......... 56% 2,77M 17s
42000K .......... .......... .......... .......... .......... 56% 1,39M 17s
42050K .......... .......... .......... .......... .......... 56% 2,69M 17s
42100K .......... .......... .......... .......... .......... 56% 1,35M 17s
42150K .......... .......... .......... .......... .......... 56% 3,53M 17s
42200K .......... .......... .......... .......... .......... 56% 3,43M 17s
42250K .......... .......... .......... .......... .......... 56% 1,36M 17s
42300K .......... .......... .......... .......... .......... 57% 4,11M 17s
42350K .......... .......... .......... .......... .......... 57% 1,30M 17s
42400K .......... .......... .......... .......... .......... 57% 3,20M 17s
42450K .......... .......... .......... .......... .......... 57% 1,31M 17s
42500K .......... .......... .......... .......... .......... 57% 4,42M 17s
42550K .......... .......... .......... .......... .......... 57% 1,25M 17s
42600K .......... .......... .......... .......... .......... 57% 3,51M 16s
42650K .......... .......... .......... .......... .......... 57% 3,33M 16s
42700K .......... .......... .......... .......... .......... 57% 1,52M 16s
42750K .......... .......... .......... .......... .......... 57% 3,79M 16s
42800K .......... .......... .......... .......... .......... 57% 1,18M 16s
42850K .......... .......... .......... .......... .......... 57% 3,28M 16s
42900K .......... .......... .......... .......... .......... 57% 1,48M 16s
42950K .......... .......... .......... .......... .......... 57% 2,82M 16s
43000K .......... .......... .......... .......... .......... 57% 1,54M 16s
43050K .......... .......... .......... .......... .......... 58% 3,29M 16s
43100K .......... .......... .......... .......... .......... 58% 3,73M 16s
43150K .......... .......... .......... .......... .......... 58% 1,43M 16s
43200K .......... .......... .......... .......... .......... 58% 2,55M 16s
43250K .......... .......... .......... .......... .......... 58% 1,45M 16s
43300K .......... .......... .......... .......... .......... 58% 4,04M 16s
43350K .......... .......... .......... .......... .......... 58% 1,40M 16s
43400K .......... .......... .......... .......... .......... 58% 3,91M 16s
43450K .......... .......... .......... .......... .......... 58% 3,04M 16s
43500K .......... .......... .......... .......... .......... 58% 1,59M 16s
43550K .......... .......... .......... .......... .......... 58% 3,15M 16s
43600K .......... .......... .......... .......... .......... 58% 1,31M 16s
43650K .......... .......... .......... .......... .......... 58% 4,12M 16s
43700K .......... .......... .......... .......... .......... 58% 1,52M 16s
43750K .......... .......... .......... .......... .......... 58% 3,19M 16s
43800K .......... .......... .......... .......... .......... 59% 3,93M 16s
43850K .......... .......... .......... .......... .......... 59% 1,44M 16s
43900K .......... .......... .......... .......... .......... 59% 4,12M 16s
43950K .......... .......... .......... .......... .......... 59% 1,51M 16s
44000K .......... .......... .......... .......... .......... 59% 2,39M 16s
44050K .......... .......... .......... .......... .......... 59% 1,50M 16s
44100K .......... .......... .......... .......... .......... 59% 5,63M 16s
44150K .......... .......... .......... .......... .......... 59%  769K 16s
44200K .......... .......... .......... .......... .......... 59%  550M 16s
44250K .......... .......... .......... .......... .......... 59% 6,80M 16s
44300K .......... .......... .......... .......... .......... 59% 1,46M 16s
44350K .......... .......... .......... .......... .......... 59% 2,17M 16s
44400K .......... .......... .......... .......... .......... 59% 1,17M 15s
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44450K .......... .......... .......... .......... .......... 59% 1,35M 15s
44500K .......... .......... .......... .......... .......... 59% 2,31M 15s
44550K .......... .......... .......... .......... .......... 60% 1,32M 15s
44600K .......... .......... .......... .......... .......... 60% 2,56M 15s
44650K .......... .......... .......... .......... .......... 60% 1,45M 15s
44700K .......... .......... .......... .......... .......... 60% 2,12M 15s
44750K .......... .......... .......... .......... .......... 60% 1,63M 15s
44800K .......... .......... .......... .......... .......... 60% 1,03M 15s
44850K .......... .......... .......... .......... .......... 60% 3,11M 15s
44900K .......... .......... .......... .......... .......... 60% 1,31M 15s
44950K .......... .......... .......... .......... .......... 60% 3,04M 15s
45000K .......... .......... .......... .......... .......... 60% 1,29M 15s
45050K .......... .......... .......... .......... .......... 60% 3,24M 15s
45100K .......... .......... .......... .......... .......... 60% 1,30M 15s
45150K .......... .......... .......... .......... .......... 60% 3,15M 15s
45200K .......... .......... .......... .......... .......... 60% 1,29M 15s
45250K .......... .......... .......... .......... .......... 60% 2,36M 15s
45300K .......... .......... .......... .......... .......... 61% 1,51M 15s
45350K .......... .......... .......... .......... .......... 61% 3,35M 15s
45400K .......... .......... .......... .......... .......... 61% 1,27M 15s
45450K .......... .......... .......... .......... .......... 61% 3,36M 15s
45500K .......... .......... .......... .......... .......... 61% 1,29M 15s
45550K .......... .......... .......... .......... .......... 61% 3,28M 15s
45600K .......... .......... .......... .......... .......... 61% 1,26M 15s
45650K .......... .......... .......... .......... .......... 61% 3,38M 15s
45700K .......... .......... .......... .......... .......... 61% 1,27M 15s
45750K .......... .......... .......... .......... .......... 61% 3,39M 15s
45800K .......... .......... .......... .......... .......... 61% 1,28M 15s
45850K .......... .......... .......... .......... .......... 61% 4,22M 15s
45900K .......... .......... .......... .......... .......... 61% 1,17M 15s
45950K .......... .......... .......... .......... .......... 61% 5,50M 15s
46000K .......... .......... .......... .......... .......... 61% 1,09M 15s
46050K .......... .......... .......... .......... .......... 62% 4,44M 15s
46100K .......... .......... .......... .......... .......... 62% 1,17M 15s
46150K .......... .......... .......... .......... .......... 62% 9,54M 15s
46200K .......... .......... .......... .......... .......... 62% 2,29M 15s
46250K .......... .......... .......... .......... .......... 62% 1,81M 15s
46300K .......... .......... .......... .......... .......... 62% 2,33M 15s
46350K .......... .......... .......... .......... .......... 62% 1,51M 14s
46400K .......... .......... .......... .......... .......... 62% 2,18M 14s
46450K .......... .......... .......... .......... .......... 62% 1,58M 14s
46500K .......... .......... .......... .......... .......... 62% 3,88M 14s
46550K .......... .......... .......... .......... .......... 62% 1,19M 14s
46600K .......... .......... .......... .......... .......... 62% 7,59M 14s
46650K .......... .......... .......... .......... .......... 62% 1,05M 14s
46700K .......... .......... .......... .......... .......... 62% 13,7M 14s
46750K .......... .......... .......... .......... .......... 63% 2,76M 14s
46800K .......... .......... .......... .......... .......... 63% 1,39M 14s
46850K .......... .......... .......... .......... .......... 63% 2,75M 14s
46900K .......... .......... .......... .......... .......... 63% 1,49M 14s
46950K .......... .......... .......... .......... .......... 63% 4,40M 14s
47000K .......... .......... .......... .......... .......... 63% 1,16M 14s
47050K .......... .......... .......... .......... .......... 63% 8,88M 14s
47100K .......... .......... .......... .......... .......... 63% 2,73M 14s
47150K .......... .......... .......... .......... .......... 63% 1,60M 14s
47200K .......... .......... .......... .......... .......... 63% 2,27M 14s
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47250K .......... .......... .......... .......... .......... 63% 1,56M 14s
47300K .......... .......... .......... .......... .......... 63% 2,41M 14s
47350K .......... .......... .......... .......... .......... 63% 1,52M 14s
47400K .......... .......... .......... .......... .......... 63% 4,00M 14s
47450K .......... .......... .......... .......... .......... 63% 1,21M 14s
47500K .......... .......... .......... .......... .......... 64% 4,80M 14s
47550K .......... .......... .......... .......... .......... 64% 4,38M 14s
47600K .......... .......... .......... .......... .......... 64% 1,25M 14s
47650K .......... .......... .......... .......... .......... 64% 3,59M 14s
47700K .......... .......... .......... .......... .......... 64% 1,35M 14s
47750K .......... .......... .......... .......... .......... 64% 4,89M 14s
47800K .......... .......... .......... .......... .......... 64% 1,28M 14s
47850K .......... .......... .......... .......... .......... 64% 4,63M 14s
47900K .......... .......... .......... .......... .......... 64% 4,29M 14s
47950K .......... .......... .......... .......... .......... 64% 1,28M 14s
48000K .......... .......... .......... .......... .......... 64% 3,26M 14s
48050K .......... .......... .......... .......... .......... 64% 1,41M 14s
48100K .......... .......... .......... .......... .......... 64% 4,48M 14s
48150K .......... .......... .......... .......... .......... 64% 1,28M 14s
48200K .......... .......... .......... .......... .......... 64% 4,50M 13s
48250K .......... .......... .......... .......... .......... 65% 4,39M 13s
48300K .......... .......... .......... .......... .......... 65% 1,41M 13s
48350K .......... .......... .......... .......... .......... 65% 2,67M 13s
48400K .......... .......... .......... .......... .......... 65% 1,41M 13s
48450K .......... .......... .......... .......... .......... 65% 4,42M 13s
48500K .......... .......... .......... .......... .......... 65% 1,27M 13s
48550K .......... .......... .......... .......... .......... 65% 4,34M 13s
48600K .......... .......... .......... .......... .......... 65% 4,51M 13s
48650K .......... .......... .......... .......... .......... 65% 1,43M 13s
48700K .......... .......... .......... .......... .......... 65% 4,42M 13s
48750K .......... .......... .......... .......... .......... 65% 1,27M 13s
48800K .......... .......... .......... .......... .......... 65% 3,41M 13s
48850K .......... .......... .......... .......... .......... 65% 5,97M 13s
48900K .......... .......... .......... .......... .......... 65% 1,25M 13s
48950K .......... .......... .......... .......... .......... 65% 3,72M 13s
49000K .......... .......... .......... .......... .......... 66% 1,44M 13s
49050K .......... .......... .......... .......... .......... 66% 4,38M 13s
49100K .......... .......... .......... .......... .......... 66% 5,62M 13s
49150K .......... .......... .......... .......... .......... 66% 1,25M 13s
49200K .......... .......... .......... .......... .......... 66% 3,45M 13s
49250K .......... .......... .......... .......... .......... 66% 1,44M 13s
49300K .......... .......... .......... .......... .......... 66% 2,67M 13s
49350K .......... .......... .......... .......... .......... 66% 1,60M 13s
49400K .......... .......... .......... .......... .......... 66% 4,21M 13s
49450K .......... .......... .......... .......... .......... 66% 4,24M 13s
49500K .......... .......... .......... .......... .......... 66% 1,43M 13s
49550K .......... .......... .......... .......... .......... 66% 2,63M 13s
49600K .......... .......... .......... .......... .......... 66% 1,42M 13s
49650K .......... .......... .......... .......... .......... 66% 5,03M 13s
49700K .......... .......... .......... .......... .......... 66% 5,16M 13s
49750K .......... .......... .......... .......... .......... 67% 1,20M 13s
49800K .......... .......... .......... .......... .......... 67% 4,13M 13s
49850K .......... .......... .......... .......... .......... 67% 1,42M 13s
49900K .......... .......... .......... .......... .......... 67% 5,04M 13s
49950K .......... .......... .......... .......... .......... 67% 5,24M 12s
50000K .......... .......... .......... .......... .......... 67% 1,21M 12s
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50050K ..... ..... ..... ..... ..... 67% 3,92M 12s
50100K ..... ..... ..... ..... ..... 67% 1,43M 12s
50150K ..... ..... ..... ..... ..... 67% 2,63M 12s
50200K ..... ..... ..... ..... ..... 67% 1,62M 12s
50250K ..... ..... ..... ..... ..... 67% 4,53M 12s
50300K ..... ..... ..... ..... ..... 67% 4,08M 12s
50350K ..... ..... ..... ..... ..... 67% 1,40M 12s
50400K ..... ..... ..... ..... ..... 67% 2,68M 12s
50450K ..... ..... ..... ..... ..... 67% 1,62M 12s
50500K ..... ..... ..... ..... ..... 68% 4,55M 12s
50550K ..... ..... ..... ..... ..... 68% 3,81M 12s
50600K ..... ..... ..... ..... ..... 68% 1,37M 12s
50650K ..... ..... ..... ..... ..... 68% 5,87M 12s
50700K ..... ..... ..... ..... ..... 68% 5,24M 12s
50750K ..... ..... ..... ..... ..... 68%  606K 12s
50800K ..... ..... ..... ..... ..... 68%  267M 12s
50850K ..... ..... ..... ..... ..... 68% 5,02M 12s
50900K ..... ..... ..... ..... ..... 68% 1,12M 12s
50950K ..... ..... ..... ..... ..... 68% 5,27M 12s
51000K ..... ..... ..... ..... ..... 68% 1,12M 12s
51050K ..... ..... ..... ..... ..... 68% 2,18M 12s
51100K ..... ..... ..... ..... ..... 68% 1,60M 12s
51150K ..... ..... ..... ..... ..... 68% 2,17M 12s
51200K ..... ..... ..... ..... ..... 68% 1,58M 12s
51250K ..... ..... ..... ..... ..... 69%  961K 12s
51300K ..... ..... ..... ..... ..... 69% 48,4M 12s
51350K ..... ..... ..... ..... ..... 69% 2,11M 12s
51400K ..... ..... ..... ..... ..... 69% 1,64M 12s
51450K ..... ..... ..... ..... ..... 69% 2,06M 12s
51500K ..... ..... ..... ..... ..... 69% 1,67M 12s
51550K ..... ..... ..... ..... ..... 69% 2,11M 12s
51600K ..... ..... ..... ..... ..... 69%  574K 12s
51650K ..... ..... ..... ..... ..... 69%  429M 12s
51700K ..... ..... ..... ..... ..... 69% 14,9M 12s
51750K ..... ..... ..... ..... ..... 69%  999K 12s
51800K ..... ..... ..... ..... ..... 69% 1,00M 12s
51850K ..... ..... ..... ..... ..... 69% 7,05M 12s
51900K ..... ..... ..... ..... ..... 69%  993K 12s
51950K ..... ..... ..... ..... ..... 70% 1,00M 11s
52000K ..... ..... ..... ..... ..... 70% 2,08M 11s
52050K ..... ..... ..... ..... ..... 70% 1,44M 11s
52100K ..... ..... ..... ..... ..... 70% 1,00M 11s
52150K ..... ..... ..... ..... ..... 70% 7,93M 11s
52200K ..... ..... ..... ..... ..... 70%  799K 11s
52250K ..... ..... ..... ..... ..... 70% 1,22M 11s
52300K ..... ..... ..... ..... ..... 70% 3,03M 11s
52350K ..... ..... ..... ..... ..... 70% 1,14M 11s
52400K ..... ..... ..... ..... ..... 70% 1008K 11s
52450K ..... ..... ..... ..... ..... 70% 3,63M 11s
52500K ..... ..... ..... ..... ..... 70% 1,05M 11s
52550K ..... ..... ..... ..... ..... 70% 3,63M 11s
52600K ..... ..... ..... ..... ..... 70% 1,12M 11s
52650K ..... ..... ..... ..... ..... 70% 1,17M 11s
52700K ..... ..... ..... ..... ..... 71% 3,61M 11s
52750K ..... ..... ..... ..... ..... 71%  998K 11s
52800K ..... ..... ..... ..... ..... 71% 1,16M 11s
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52850K .......... .......... .......... .......... .......... 71% 3,60M 11s
52900K .......... .......... .......... .......... .......... 71%  995K 11s
52950K .......... .......... .......... .......... .......... 71% 5,01M 11s
53000K .......... .......... .......... .......... .......... 71% 1,13M 11s
53050K .......... .......... .......... .......... .......... 71% 1,24M 11s
53100K .......... .......... .......... .......... .......... 71% 3,09M 11s
53150K .......... .......... .......... .......... .......... 71% 1,06M 11s
53200K .......... .......... .......... .......... .......... 71% 1,16M 11s
53250K .......... .......... .......... .......... .......... 71% 2,78M 11s
53300K .......... .......... .......... .......... .......... 71% 1,14M 11s
53350K .......... .......... .......... .......... .......... 71% 4,91M 11s
53400K .......... .......... .......... .......... .......... 71% 1,06M 11s
53450K .......... .......... .......... .......... .......... 72% 2,93M 11s
53500K .......... .......... .......... .......... .......... 72% 1,20M 11s
53550K .......... .......... .......... .......... .......... 72% 3,94M 11s
53600K .......... .......... .......... .......... .......... 72% 1,13M 11s
53650K .......... .......... .......... .......... .......... 72% 1,07M 11s
53700K .......... .......... .......... .......... .......... 72% 2,84M 11s
53750K .......... .......... .......... .......... .......... 72% 1,20M 11s
53800K .......... .......... .......... .......... .......... 72% 4,05M 11s
53850K .......... .......... .......... .......... .......... 72% 1,17M 11s
53900K .......... .......... .......... .......... .......... 72% 3,24M 11s
53950K .......... .......... .......... .......... .......... 72% 1,29M 11s
54000K .......... .......... .......... .......... .......... 72% 1,11M 11s
54050K .......... .......... .......... .......... .......... 72% 3,35M 11s
54100K .......... .......... .......... .......... .......... 72% 1,17M 10s
54150K .......... .......... .......... .......... .......... 72% 2,93M 10s
54200K .......... .......... .......... .......... .......... 73% 1,29M 10s
54250K .......... .......... .......... .......... .......... 73% 3,21M 10s
54300K .......... .......... .......... .......... .......... 73% 1,12M 10s
54350K .......... .......... .......... .......... .......... 73% 5,18M 10s
54400K .......... .......... .......... .......... .......... 73% 1,10M 10s
54450K .......... .......... .......... .......... .......... 73% 1,30M 10s
54500K .......... .......... .......... .......... .......... 73% 3,18M 10s
54550K .......... .......... .......... .......... .......... 73% 1,24M 10s
54600K .......... .......... .......... .......... .......... 73% 3,54M 10s
54650K .......... .......... .......... .......... .......... 73% 1,26M 10s
54700K .......... .......... .......... .......... .......... 73% 3,43M 10s
54750K .......... .......... .......... .......... .......... 73% 1,27M 10s
54800K .......... .......... .......... .......... .......... 73% 2,56M 10s
54850K .......... .......... .......... .......... .......... 73% 1,30M 10s
54900K .......... .......... .......... .......... .......... 73% 3,13M 10s
54950K .......... .......... .......... .......... .......... 74% 1,31M 10s
55000K .......... .......... .......... .......... .......... 74% 3,15M 10s
55050K .......... .......... .......... .......... .......... 74% 1,29M 10s
55100K .......... .......... .......... .......... .......... 74% 3,85M 10s
55150K .......... .......... .......... .......... .......... 74% 1,35M 10s
55200K .......... .......... .......... .......... .......... 74% 2,08M 10s
55250K .......... .......... .......... .......... .......... 74% 1,31M 10s
55300K .......... .......... .......... .......... .......... 74% 3,10M 10s
55350K .......... .......... .......... .......... .......... 74% 1,43M 10s
55400K .......... .......... .......... .......... .......... 74% 3,45M 10s
55450K .......... .......... .......... .......... .......... 74% 1,26M 10s
55500K .......... .......... .......... .......... .......... 74% 4,04M 10s
55550K .......... .......... .......... .......... .......... 74% 1,33M 10s
55600K .......... .......... .......... .......... .......... 74% 2,70M 10s
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55650K .......... .......... .......... .......... .......... 74% 1,25M 10s
55700K .......... .......... .......... .......... .......... 75% 4,08M 10s
55750K .......... .......... .......... .......... .......... 75% 1,31M 10s
55800K .......... .......... .......... .......... .......... 75% 3,10M 10s
55850K .......... .......... .......... .......... .......... 75% 1,30M 10s
55900K .......... .......... .......... .......... .......... 75% 3,09M 10s
55950K .......... .......... .......... .......... .......... 75% 5,54M 10s
56000K .......... .......... .......... .......... .......... 75% 1,11M 10s
56050K .......... .......... .......... .......... .......... 75% 1,32M 9s
56100K .......... .......... .......... .......... .......... 75% 3,10M 9s
56150K .......... .......... .......... .......... .......... 75% 5,61M 9s
56200K .......... .......... .......... .......... .......... 75% 1,41M 9s
56250K .......... .......... .......... .......... .......... 75% 3,10M 9s
56300K .......... .......... .......... .......... .......... 75% 1,32M 9s
56350K .......... .......... .......... .......... .......... 75% 4,83M 9s
56400K .......... .......... .......... .......... .......... 75% 1,15M 9s
56450K .......... .......... .......... .......... .......... 76% 4,64M 9s
56500K .......... .......... .......... .......... .......... 76% 1,33M 9s
56550K .......... .......... .......... .......... .......... 76% 3,05M 9s
56600K .......... .......... .......... .......... .......... 76% 1,33M 9s
56650K .......... .......... .......... .......... .......... 76% 3,15M 9s
56700K .......... .......... .......... .......... .......... 76% 5,12M 9s
56750K .......... .......... .......... .......... .......... 76% 1,46M 9s
56800K .......... .......... .......... .......... .......... 76% 2,53M 9s
56850K .......... .......... .......... .......... .......... 76% 1,46M 9s
56900K .......... .......... .......... .......... .......... 76% 2,83M 9s
56950K .......... .......... .......... .......... .......... 76% 1,48M 9s
57000K .......... .......... .......... .......... .......... 76% 3,49M 9s
57050K .......... .......... .......... .......... .......... 76% 1,35M 9s
57100K .......... .......... .......... .......... .......... 76% 3,10M 9s
57150K .......... .......... .......... .......... .......... 77% 6,14M 9s
57200K .......... .......... .......... .......... .......... 77% 1,09M 9s
57250K .......... .......... .......... .......... .......... 77% 6,43M 9s
57300K .......... .......... .......... .......... .......... 77% 1,41M 9s
57350K .......... .......... .......... .......... .......... 77% 2,84M 9s
57400K .......... .......... .......... .......... .......... 77% 1,54M 9s
57450K .......... .......... .......... .......... .......... 77% 3,25M 9s
57500K .......... .......... .......... .......... .......... 77% 7,29M 9s
57550K .......... .......... .......... .......... .......... 77% 1,05M 9s
57600K .......... .......... .......... .......... .......... 77% 7,16M 9s
57650K .......... .......... .......... .......... .......... 77% 1,41M 9s
57700K .......... .......... .......... .......... .......... 77% 2,71M 9s
57750K .......... .......... .......... .......... .......... 77% 1,51M 9s
57800K .......... .......... .......... .......... .......... 77% 3,32M 9s
57850K .......... .......... .......... .......... .......... 77% 7,54M 9s
57900K .......... .......... .......... .......... .......... 78% 1,06M 8s
57950K .......... .......... .......... .......... .......... 78% 8,01M 8s
58000K .......... .......... .......... .......... .......... 78% 1,41M 8s
58050K .......... .......... .......... .......... .......... 78% 2,65M 8s
58100K .......... .......... .......... .......... .......... 78% 1,51M 8s
58150K .......... .......... .......... .......... .......... 78% 3,26M 8s
58200K .......... .......... .......... .......... .......... 78% 7,76M 8s
58250K .......... .......... .......... .......... .......... 78% 1,45M 8s
58300K .......... .......... .......... .......... .......... 78% 2,61M 8s
58350K .......... .......... .......... .......... .......... 78% 1,52M 8s
58400K .......... .......... .......... .......... .......... 78% 3,06M 8s
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58450K .......... .......... .......... .......... .......... 78% 1,32M 8s
58500K .......... .......... .......... .......... .......... 78% 3,12M 8s
58550K .......... .......... .......... .......... .......... 78% 9,39M 8s
58600K .......... .......... .......... .......... .......... 78% 1,50M 8s
58650K .......... .......... .......... .......... .......... 79% 3,04M 8s
58700K .......... .......... .......... .......... .......... 79% 10,2M 8s
58750K .......... .......... .......... .......... .......... 79% 1,02M 8s
58800K .......... .......... .......... .......... .......... 79% 9,62M 8s
58850K .......... .......... .......... .......... .......... 79% 1,51M 8s
58900K .......... .......... .......... .......... .......... 79% 3,06M 8s
58950K .......... .......... .......... .......... .......... 79% 1,34M 8s
59000K .......... .......... .......... .......... .......... 79% 3,00M 8s
59050K .......... .......... .......... .......... .......... 79% 11,3M 8s
59100K .......... .......... .......... .......... .......... 79% 1,48M 8s
59150K .......... .......... .......... .......... .......... 79% 2,98M 8s
59200K .......... .......... .......... .......... .......... 79% 1,35M 8s
59250K .......... .......... .......... .......... .......... 79% 2,96M 8s
59300K .......... .......... .......... .......... .......... 79% 10,9M 8s
59350K .......... .......... .......... .......... .......... 79% 1,50M 8s
59400K .......... .......... .......... .......... .......... 80% 2,67M 8s
59450K .......... .......... .......... .......... .......... 80% 22,8M 8s
59500K .......... .......... .......... .......... .......... 80% 1,50M 8s
59550K .......... .......... .......... .......... .......... 80% 2,45M 8s
59600K .......... .......... .......... .......... .......... 80% 1,48M 8s
59650K .......... .......... .......... .......... .......... 80% 2,68M 8s
59700K .......... .......... .......... .......... .......... 80% 23,2M 8s
59750K .......... .......... .......... .......... .......... 80% 1,49M 7s
59800K .......... .......... .......... .......... .......... 80% 2,66M 7s
59850K .......... .......... .......... .......... .......... 80% 1,44M 7s
59900K .......... .......... .......... .......... .......... 80% 2,66M 7s
59950K .......... .......... .......... .......... .......... 80% 28,2M 7s
60000K .......... .......... .......... .......... .......... 80% 1,45M 7s
60050K .......... .......... .......... .......... .......... 80% 2,67M 7s
60100K .......... .......... .......... .......... .......... 80% 27,4M 7s
60150K .......... .......... .......... .......... .......... 81%  984K 7s
60200K .......... .......... .......... .......... .......... 81% 22,4M 7s
60250K .......... .......... .......... .......... .......... 81% 1,49M 7s
60300K .......... .......... .......... .......... .......... 81% 2,56M 7s
60350K .......... .......... .......... .......... .......... 81% 47,5M 7s
60400K .......... .......... .......... .......... .......... 81% 1,37M 7s
60450K .......... .......... .......... .......... .......... 81% 2,94M 7s
60500K .......... .......... .......... .......... .......... 81% 44,5M 7s
60550K .......... .......... .......... .......... .......... 81%  970K 7s
60600K .......... .......... .......... .......... .......... 81% 52,3M 7s
60650K .......... .......... .......... .......... .......... 81% 1,47M 7s
60700K .......... .......... .......... .......... .......... 81% 2,24M 7s
60750K .......... .......... .......... .......... .......... 81% 18,3M 7s
60800K .......... .......... .......... .......... .......... 81% 1,59M 7s
60850K .......... .......... .......... .......... .......... 81% 2,43M 7s
60900K .......... .......... .......... .......... .......... 82% 18,8M 7s
60950K .......... .......... .......... .......... .......... 82% 1,61M 7s
61000K .......... .......... .......... .......... .......... 82% 2,27M 7s
61050K .......... .......... .......... .......... .......... 82% 47,4M 7s
61100K .......... .......... .......... .......... .......... 82% 1,03M 7s
61150K .......... .......... .......... .......... .......... 82% 10,0M 7s
61200K .......... .......... .......... .......... .......... 82%  597K 7s
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61250K .......... .......... .......... .......... .......... 82%  646M 7s
61300K .......... .......... .......... .......... .......... 82%  959K 7s
61350K .......... .......... .......... .......... .......... 82%  312M 7s
61400K .......... .......... .......... .......... .......... 82% 2,15M 7s
61450K .......... .......... .......... .......... .......... 82% 1,59M 7s
61500K .......... .......... .......... .......... .......... 82% 2,19M 7s
61550K .......... .......... .......... .......... .......... 82% 1,61M 7s
61600K .......... .......... .......... .......... .......... 82% 2,12M 7s
61650K .......... .......... .......... .......... .......... 83% 1,60M 6s
61700K .......... .......... .......... .......... .......... 83% 2,23M 6s
61750K .......... .......... .......... .......... .......... 83% 1,57M 6s
61800K .......... .......... .......... .......... .......... 83% 2,15M 6s
61850K .......... .......... .......... .......... .......... 83% 1,61M 6s
61900K .......... .......... .......... .......... .......... 83% 2,23M 6s
61950K .......... .......... .......... .......... .......... 83% 1,60M 6s
62000K .......... .......... .......... .......... .......... 83% 2,16M 6s
62050K .......... .......... .......... .......... .......... 83% 1,58M 6s
62100K .......... .......... .......... .......... .......... 83%  719K 6s
62150K .......... .......... .......... .......... .......... 83%  474M 6s
62200K .......... .......... .......... .......... .......... 83% 7,65M 6s
62250K .......... .......... .......... .......... .......... 83% 1,05M 6s
62300K .......... .......... .......... .......... .......... 83% 1,37M 6s
62350K .......... .......... .......... .......... .......... 84% 2,10M 6s
62400K .......... .......... .......... .......... .......... 84% 1,03M 6s
62450K .......... .......... .......... .......... .......... 84% 1,29M 6s
62500K .......... .......... .......... .......... .......... 84% 2,37M 6s
62550K .......... .......... .......... .......... .......... 84% 1,03M 6s
62600K .......... .......... .......... .......... .......... 84% 1,39M 6s
62650K .......... .......... .......... .......... .......... 84% 2,08M 6s
62700K .......... .......... .......... .......... .......... 84% 1,51M 6s
62750K .......... .......... .......... .......... .......... 84% 2,36M 6s
62800K .......... .......... .......... .......... .......... 84%  945K 6s
62850K .......... .......... .......... .......... .......... 84% 1,52M 6s
62900K .......... .......... .......... .......... .......... 84% 2,34M 6s
62950K .......... .......... .......... .......... .......... 84% 1,03M 6s
63000K .......... .......... .......... .......... .......... 84% 1,40M 6s
63050K .......... .......... .......... .......... .......... 84% 2,07M 6s
63100K .......... .......... .......... .......... .......... 85% 1,50M 6s
63150K .......... .......... .......... .......... .......... 85% 2,40M 6s
63200K .......... .......... .......... .......... .......... 85% 1,02M 6s
63250K .......... .......... .......... .......... .......... 85% 1,34M 6s
63300K .......... .......... .......... .......... .......... 85% 2,26M 6s
63350K .......... .......... .......... .......... .......... 85% 1,02M 6s
63400K .......... .......... .......... .......... .......... 85% 9,34M 6s
63450K .......... .......... .......... .......... .......... 85% 1,04M 6s
63500K .......... .......... .......... .......... .......... 85% 1,33M 6s
63550K .......... .......... .......... .......... .......... 85% 2,15M 6s
63600K .......... .......... .......... .......... .......... 85% 1003K 6s
63650K .......... .......... .......... .......... .......... 85% 4,16M 5s
63700K .......... .......... .......... .......... .......... 85% 1,09M 5s
63750K .......... .......... .......... .......... .......... 85% 1,49M 5s
63800K .......... .......... .......... .......... .......... 85% 2,52M 5s
63850K .......... .......... .......... .......... .......... 86% 1,43M 5s
63900K .......... .......... .......... .......... .......... 86% 2,23M 5s
63950K .......... .......... .......... .......... .......... 86% 1,33M 5s
64000K .......... .......... .......... .......... .......... 86% 1,03M 5s
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64050K .......... .......... .......... .......... .......... 86% 2,74M 5s
64100K .......... .......... .......... .......... .......... 86% 1,38M 5s
64150K .......... .......... .......... .......... .......... 86% 2,78M 5s
64200K .......... .......... .......... .......... .......... 86% 1,39M 5s
64250K .......... .......... .......... .......... .......... 86% 2,74M 5s
64300K .......... .......... .......... .......... .......... 86% 1,39M 5s
64350K .......... .......... .......... .......... .......... 86% 2,30M 5s
64400K .......... .......... .......... .......... .......... 86% 1005K 5s
64450K .......... .......... .......... .......... .......... 86% 4,33M 5s
64500K .......... .......... .......... .......... .......... 86% 1,17M 5s
64550K .......... .......... .......... .......... .......... 86% 1,38M 5s
64600K .......... .......... .......... .......... .......... 87% 2,76M 5s
64650K .......... .......... .......... .......... .......... 87% 1,39M 5s
64700K .......... .......... .......... .......... .......... 87% 2,77M 5s
64750K .......... .......... .......... .......... .......... 87% 1,30M 5s
64800K .......... .......... .......... .......... .......... 87% 2,69M 5s
64850K .......... .......... .......... .......... .......... 87% 1,40M 5s
64900K .......... .......... .......... .......... .......... 87% 2,74M 5s
64950K .......... .......... .......... .......... .......... 87% 1,40M 5s
65000K .......... .......... .......... .......... .......... 87% 2,70M 5s
65050K .......... .......... .......... .......... .......... 87% 1,40M 5s
65100K .......... .......... .......... .......... .......... 87% 2,10M 5s
65150K .......... .......... .......... .......... .......... 87% 1,65M 5s
65200K .......... .......... .......... .......... .......... 87% 2,04M 5s
65250K .......... .......... .......... .......... .......... 87% 1,54M 5s
65300K .......... .......... .......... .......... .......... 87% 2,34M 5s
65350K .......... .......... .......... .......... .......... 88% 1,57M 5s
65400K .......... .......... .......... .......... .......... 88% 2,22M 5s
65450K .......... .......... .......... .......... .......... 88% 1,74M 5s
65500K .......... .......... .......... .......... .......... 88% 1,97M 5s
65550K .......... .......... .......... .......... .......... 88% 1,76M 5s
65600K .......... .......... .......... .......... .......... 88% 1,90M 4s
65650K .......... .......... .......... .......... .......... 88% 1,61M 4s
65700K .......... .......... .......... .......... .......... 88% 2,19M 4s
65750K .......... .......... .......... .......... .......... 88%  625K 4s
65800K .......... .......... .......... .......... .......... 88%  366M 4s
65850K .......... .......... .......... .......... .......... 88%  973K 4s
65900K .......... .......... .......... .......... .......... 88% 10,4M 4s
65950K .......... .......... .......... .......... .......... 88% 1,00M 4s
66000K .......... .......... .......... .......... .......... 88%  948K 4s
66050K .......... .......... .......... .......... .......... 88% 1,80M 4s
66100K .......... .......... .......... .......... .......... 89% 1,82M 4s
66150K .......... .......... .......... .......... .......... 89%  950K 4s
66200K .......... .......... .......... .......... .......... 89% 1,84M 4s
66250K .......... .......... .......... .......... .......... 89% 1,81M 4s
66300K .......... .......... .......... .......... .......... 89% 1,88M 4s
66350K .......... .......... .......... .......... .......... 89% 1,77M 4s
66400K .......... .......... .......... .......... .......... 89%  951K 4s
66450K .......... .......... .......... .......... .......... 89% 1,92M 4s
66500K .......... .......... .......... .......... .......... 89% 1,74M 4s
66550K .......... .......... .......... .......... .......... 89%  947K 4s
66600K .......... .......... .......... .......... .......... 89% 22,2M 4s
66650K .......... .......... .......... .......... .......... 89%  984K 4s
66700K .......... .......... .......... .......... .......... 89% 1,86M 4s
66750K .......... .......... .......... .......... .......... 89% 1,83M 4s
66800K .......... .......... .......... .......... .......... 89%  952K 4s
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66850K .......... .......... .......... .......... .......... 90%  910K 4s
66900K .......... .......... .......... .......... .......... 90%  418M 4s
66950K .......... .......... .......... .......... .......... 90%  980K 4s
67000K .......... .......... .......... .......... .......... 90% 1,81M 4s
67050K .......... .......... .......... .......... .......... 90% 1,80M 4s
67100K .......... .......... .......... .......... .......... 90%  948K 4s
67150K .......... .......... .......... .......... .......... 90%  959K 4s
67200K .......... .......... .......... .......... .......... 90%  933K 4s
67250K .......... .......... .......... .......... .......... 90%  955K 4s
67300K .......... .......... .......... .......... .......... 90% 52,3M 4s
67350K .......... .......... .......... .......... .......... 90%  918K 4s
67400K .......... .......... .......... .......... .......... 90%  953K 4s
67450K .......... .......... .......... .......... .......... 90% 1,94M 4s
67500K .......... .......... .......... .......... .......... 90% 1,70M 4s
67550K .......... .......... .......... .......... .......... 91%  948K 4s
67600K .......... .......... .......... .......... .......... 91%  947K 3s
67650K .......... .......... .......... .......... .......... 91%  950K 3s
67700K .......... .......... .......... .......... .......... 91% 1,93M 3s
67750K .......... .......... .......... .......... .......... 91% 1,72M 3s
67800K .......... .......... .......... .......... .......... 91%  948K 3s
67850K .......... .......... .......... .......... .......... 91% 1,98M 3s
67900K .......... .......... .......... .......... .......... 91% 1,70M 3s
67950K .......... .......... .......... .......... .......... 91%  725K 3s
68000K .......... .......... .......... .......... .......... 91%  941K 3s
68050K .......... .......... .......... .......... .......... 91% 6,19M 3s
68100K .......... .......... .......... .......... .......... 91% 1,07M 3s
68150K .......... .......... .......... .......... .......... 91%  948K 3s
68200K .......... .......... .......... .......... .......... 91% 6,03M 3s
68250K .......... .......... .......... .......... .......... 91% 1,07M 3s
68300K .......... .......... .......... .......... .......... 92%  944K 3s
68350K .......... .......... .......... .......... .......... 92% 6,70M 3s
68400K .......... .......... .......... .......... .......... 92%  945K 3s
68450K .......... .......... .......... .......... .......... 92% 1,05M 3s
68500K .......... .......... .......... .......... .......... 92%  950K 3s
68550K .......... .......... .......... .......... .......... 92% 7,54M 3s
68600K .......... .......... .......... .......... .......... 92% 1,03M 3s
68650K .......... .......... .......... .......... .......... 92% 6,12M 3s
68700K .......... .......... .......... .......... .......... 92% 1,08M 3s
68750K .......... .......... .......... .......... .......... 92%  952K 3s
68800K .......... .......... .......... .......... .......... 92% 5,81M 3s
68850K .......... .......... .......... .......... .......... 92% 1,08M 3s
68900K .......... .......... .......... .......... .......... 92%  951K 3s
68950K .......... .......... .......... .......... .......... 92% 6,28M 3s
69000K .......... .......... .......... .......... .......... 92% 1,07M 3s
69050K .......... .......... .......... .......... .......... 93% 6,36M 3s
69100K .......... .......... .......... .......... .......... 93% 1,05M 3s
69150K .......... .......... .......... .......... .......... 93%  944K 3s
69200K .......... .......... .......... .......... .......... 93% 6,45M 3s
69250K .......... .......... .......... .......... .......... 93% 1,01M 3s
69300K .......... .......... .......... .......... .......... 93% 1,01M 3s
69350K .......... .......... .......... .......... .......... 93% 4,29M 3s
69400K .......... .......... .......... .......... .......... 93% 1,11M 3s
69450K .......... .......... .......... .......... .......... 93% 5,26M 3s
69500K .......... .......... .......... .......... .......... 93% 1,09M 3s
69550K .......... .......... .......... .......... .......... 93% 5,13M 2s
69600K .......... .......... .......... .......... .......... 93%  986K 2s
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69650K .......... .......... .......... .......... .......... 93% 1,11M 2s
69700K .......... .......... .......... .......... .......... 93% 5,64M 2s
69750K .......... .......... .......... .......... .......... 93% 1,09M 2s
69800K .......... .......... .......... .......... .......... 94% 4,50M 2s
69850K .......... .......... .......... .......... .......... 94% 1,15M 2s
69900K .......... .......... .......... .......... .......... 94% 4,71M 2s
69950K .......... .......... .......... .......... .......... 94% 1,13M 2s
70000K .......... .......... .......... .......... .......... 94%  938K 2s
70050K .......... .......... .......... .......... .......... 94% 6,57M 2s
70100K .......... .......... .......... .......... .......... 94% 1,06M 2s
70150K .......... .......... .......... .......... .......... 94% 6,01M 2s
70200K .......... .......... .......... .......... .......... 94% 1,08M 2s
70250K .......... .......... .......... .......... .......... 94% 5,67M 2s
70300K .......... .......... .......... .......... .......... 94% 1,09M 2s
70350K .......... .......... .......... .......... .......... 94% 6,23M 2s
70400K .......... .......... .......... .......... .......... 94%  967K 2s
70450K .......... .......... .......... .......... .......... 94% 1,05M 2s
70500K .......... .......... .......... .......... .......... 94% 7,56M 2s
70550K .......... .......... .......... .......... .......... 95% 1,05M 2s
70600K .......... .......... .......... .......... .......... 95% 7,49M 2s
70650K .......... .......... .......... .......... .......... 95% 1,05M 2s
70700K .......... .......... .......... .......... .......... 95% 7,86M 2s
70750K .......... .......... .......... .......... .......... 95% 1,04M 2s
70800K .......... .......... .......... .......... .......... 95% 7,55M 2s
70850K .......... .......... .......... .......... .......... 95% 1,02M 2s
70900K .......... .......... .......... .......... .......... 95% 9,16M 2s
70950K .......... .......... .......... .......... .......... 95% 1,04M 2s
71000K .......... .......... .......... .......... .......... 95% 9,50M 2s
71050K .......... .......... .......... .......... .......... 95% 1,02M 2s
71100K .......... .......... .......... .......... .......... 95% 9,39M 2s
71150K .......... .......... .......... .......... .......... 95% 1,02M 2s
71200K .......... .......... .......... .......... .......... 95% 8,42M 2s
71250K .......... .......... .......... .......... .......... 95% 1,02M 2s
71300K .......... .......... .......... .......... .......... 96% 9,39M 2s
71350K .......... .......... .......... .......... .......... 96% 1,02M 2s
71400K .......... .......... .......... .......... .......... 96% 9,50M 2s
71450K .......... .......... .......... .......... .......... 96% 1,02M 1s
71500K .......... .......... .......... .......... .......... 96% 12,4M 1s
71550K .......... .......... .......... .......... .......... 96% 1018K 1s
71600K .......... .......... .......... .......... .......... 96% 11,6M 1s
71650K .......... .......... .......... .......... .......... 96% 1016K 1s
71700K .......... .......... .......... .......... .......... 96% 14,4M 1s
71750K .......... .......... .......... .......... .......... 96%  803K 1s
71800K .......... .......... .......... .......... .......... 96% 46,7M 1s
71850K .......... .......... .......... .......... .......... 96% 1006K 1s
71900K .......... .......... .......... .......... .......... 96% 14,6M 1s
71950K .......... .......... .......... .......... .......... 96%  488K 1s
72000K .......... .......... .......... .......... .......... 96%  193M 1s
72050K .......... .......... .......... .......... .......... 97% 1,09M 1s
72100K .......... .......... .......... .......... .......... 97% 4,55M 1s
72150K .......... .......... .......... .......... .......... 97%  998K 1s
72200K .......... .......... .......... .......... .......... 97% 1,01M 1s
72250K .......... .......... .......... .......... .......... 97% 5,33M 1s
72300K .......... .......... .......... .......... .......... 97% 1,06M 1s
72350K .......... .......... .......... .......... .......... 97% 6,55M 1s
72400K .......... .......... .......... .......... .......... 97%  950K 1s
```

```
 72450K .......... .......... .......... .......... .......... 97% 1,00M 1s
 72500K .......... .......... .......... .......... .......... 97% 10,2M 1s
 72550K .......... .......... .......... .......... .......... 97%  954K 1s
 72600K .......... .......... .......... .......... .......... 97% 1,09M 1s
 72650K .......... .......... .......... .......... .......... 97% 5,82M 1s
 72700K .......... .......... .......... .......... .......... 97% 1,06M 1s
 72750K .......... .......... .......... .......... .......... 98% 6,89M 1s
 72800K .......... .......... .......... .......... .......... 98%  966K 1s
 72850K .......... .......... .......... .......... .......... 98%  498K 1s
 72900K .......... .......... .......... .......... .......... 98%  776M 1s
 72950K .......... .......... .......... .......... .......... 98% 9,28M 1s
 73000K .......... .......... .......... .......... .......... 98%  944K 1s
 73050K .......... .......... .......... .......... .......... 98% 1,03M 1s
 73100K .......... .......... .......... .......... .......... 98%  946K 1s
 73150K .......... .......... .......... .......... .......... 98% 9,28M 1s
 73200K .......... .......... .......... .......... .......... 98%  956K 1s
 73250K .......... .......... .......... .......... .......... 98%  952K 1s
 73300K .......... .......... .......... .......... .......... 98%  926K 1s
 73350K .......... .......... .......... .......... .......... 98%  947K 0s
 73400K .......... .......... .......... .......... .......... 98% 21,6M 0s
 73450K .......... .......... .......... .......... .......... 98%  955K 0s
 73500K .......... .......... .......... .......... .......... 99%  980K 0s
 73550K .......... .......... .......... .......... .......... 99%  985K 0s
 73600K .......... .......... .......... .......... .......... 99% 10,9M 0s
 73650K .......... .......... .......... .......... .......... 99%  946K 0s
 73700K .......... .......... .......... .......... .......... 99%  956K 0s
 73750K .......... .......... .......... .......... .......... 99%  972K 0s
 73800K .......... .......... .......... .......... .......... 99% 28,6M 0s
 73850K .......... .......... .......... .......... .......... 99%  959K 0s
 73900K .......... .......... .......... .......... .......... 99%  952K 0s
 73950K .......... .......... .......... .......... .......... 99% 44,9M 0s
 74000K .......... .......... .......... .......... .......... 99%  317K 0s
 74050K .......... .......... .......... .......... .......... 99%  203M 0s
 74100K .......... .......... .......... .......... .......... 99%  994K 0s
 74150K .......... .......... .......... .......... .......... 99% 15,9M 0s
 74200K .......... .......... .......... .......... .......... 99%  945K 0s
 74250K .......... .......... .......... ...                  100%  642K=40s

2025-02-25 13:17:52 (1,81 MB/s) - 'mot20.mp4' saved [76066403/76066403]
```

Класс VideoTracker

In [396…

```python
class VideoTracker:
    def __init__(
        self,
        video_path: str,
        output_path: str,
        detector,
        tracker,
        cut_number_frames: int = 200,
        emb_size: int = 512,
        size_img: Tuple[int, int] = (128, 128),
        prefix: str = "Exp2",
        name_model: str = "convnext_tiny",
```

```
        path_to_weights: str = "./weights",
    ):
        """
        Инициализация трекера видео.

        :param video_path: Путь к входному видео.
        :param output_path: Путь для сохранения выходного видео.
        :param detector: Объект детектора (например, YoloDetector).
        :param tracker: Объект трекера (например, Tracker).
        :param cut_number_frames: Максимальное количество кадров для обработки.
        :param emb_size: Размер эмбеддингов.
        :param size_img: Размер изображений для модели.
        :param prefix: Префикс модели.
        :param name_model: Название модели.
        :param path_to_weights: Путь к весам модели.
        """
        self.video_path = video_path
        self.output_path = output_path
        self.detector = detector
        self.tracker = tracker
        self.cut_number_frames = cut_number_frames
        self.emb_size = emb_size
        self.size_img = size_img
        self.prefix = prefix
        self.name_model = name_model
        self.path_to_weights = path_to_weights

    def _get_embeddings(self, frame: np.ndarray, preds_bboxes: np.ndarray) -> np.nd
        """
        Получение эмбеддингов для bounding boxes.

        :param frame: Входное изображение.
        :param preds_bboxes: Bounding boxes (N, 4).
        :return: Эмбеддинги (N, emb_size).
        """
        return get_embeddings(
            img=frame,
            preds_bboxes=preds_bboxes,
            emb_size=self.emb_size,
            size_img=self.size_img,
            prefix=self.prefix,
            name_model=self.name_model,
            path_to_weights=self.path_to_weights,
        )

    def _tracking_visualization(self, frame: np.ndarray, preds: np.ndarray) -> np.n
        """
        Визуализация трекинга на кадре.

        :param frame: Входное изображение.
        :param preds: Данные для визуализации (N, 5) — [x1, y1, x2, y2, track_id].
        :return: Изображение с визуализацией.
        """
        # Пример реализации (замените на вашу функцию визуализации)
        for pred in preds:
            x1, y1, x2, y2, track_id = map(int, pred)
```

```
            cv2.rectangle(frame, (x1, y1), (x2, y2), (0, 255, 0), 2)
            cv2.putText(frame, str(track_id), (x1, y1 - 10), cv2.FONT_HERSHEY_SIMPL
        return frame

    def process_video(self):
        """
        Обработка видео: детекция, трекинг и визуализация.
        """
        try:
            cap = cv2.VideoCapture(self.video_path)
            if not cap.isOpened():
                raise ValueError("Ошибка при открытии видео")

            frame_size = (int(cap.get(cv2.CAP_PROP_FRAME_WIDTH)), int(cap.get(cv2.C
            length = min(self.cut_number_frames, int(cap.get(cv2.CAP_PROP_FRAME_COU
            fps = cap.get(cv2.CAP_PROP_FPS)
            fourcc = cv2.VideoWriter_fourcc(*'avc1')
            video_writer = cv2.VideoWriter(self.output_path, fourcc, fps, frame_siz

            frame_count = 0
            pbar = tqdm(total=length)

            while True:
                ret, frame = cap.read()
                if not ret or frame_count > self.cut_number_frames:
                    print("Стрим закончился")
                    break

                # Конвертируем изображение в RGB
                frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)

                # Детекция
                preds_masks, preds_bboxes = self.detector(frame)

                # Получение эмбеддингов
                embeds = self._get_embeddings(frame, preds_bboxes)

                # Трекинг
                track_ids = self.tracker.tracking(frame_count=frame_count, masks=pr
                preds = np.column_stack([preds_bboxes, track_ids])

                # Визуализация
                out_frame = self._tracking_visualization(frame, preds)
                video_writer.write(cv2.cvtColor(out_frame, cv2.COLOR_RGB2BGR))

                frame_count += 1
                pbar.update(1)

            for path in [self.output_path]:
                os.system(f'ffmpeg -y -i "{path}" -c:v libx264 -pix_fmt yuv420p "{p

        finally:
            video_writer.release()
            cap.release()
            pbar.close()
```

Запускаем трекер

In [397…

```
detector = YoloDetector()
tracker = Tracker(5)

video_tracker = VideoTracker(
    video_path="mot20.mp4",
    output_path="mot20_vis.mp4",
    detector=detector,
    tracker=tracker,
    cut_number_frames=200,
    emb_size=512,
    size_img=(128, 128),
    prefix="Exp2",
    name_model="convnext_tiny",
    path_to_weights="./weights",
)

video_tracker.process_video()
```

```
%|
| 0/200 [00:00<?, ?it/s]
13:32:40-902608 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 99.77image/s]

%|▌
| 1/200 [00:01<03:55,  1.18s/it]
13:32:41-623412 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 94.71image/s]

%|█
| 2/200 [00:01<02:57,  1.12it/s]
13:32:42-313891 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████|
20/20 [00:00<00:00, 100.48image/s]

%|▌
| 3/200 [00:02<02:40,  1.23it/s]
```

```
13:32:43-057391 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

t Feature:  50%|████████████████████████                                          | 10/2
0 [00:00<00:00, 99.16image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 92.19image/s]

%|▌
| 4/200 [00:03<02:35,  1.26it/s]
13:32:43-864098 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

t Feature:  50%|████████████████████████                                          | 10/2
0 [00:00<00:00, 96.51image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 89.36image/s]

%|▌
| 5/200 [00:04<02:36,  1.24it/s]
13:32:44-662823 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

t Feature:  47%|██████████████████████▌                                           | 9/
19 [00:00<00:00, 84.74image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 86.95image/s]

%|▌
| 6/200 [00:04<02:34,  1.25it/s]
13:32:45-438161 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 99.08image/s]

%|▌
| 7/200 [00:05<02:30,  1.28it/s]
13:32:46-198182 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

t Feature:  47%|████████████████████████▌                                        | 9/
19 [00:00<00:00, 84.47image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 84.48image/s]

%|▌
| 8/200 [00:06<02:30,  1.27it/s]
```
13:32:46-993507 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████|
19/19 [00:00<00:00, 101.63image/s]

%|▌
| 9/200 [00:07<02:28,  1.29it/s]
```
13:32:47-737989 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

t Feature:  50%|████████████████████████                                         | 10/2
0 [00:00<00:00, 94.95image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 96.34image/s]

%|▌                                                                             |
10/200 [00:08<02:27,  1.29it/s]
```
13:32:48-491444 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 92.31image/s]

%|▌
| 11/200 [00:08<02:24,  1.31it/s]
```
13:32:49-258363 INFO   Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

t Feature:  50%|████████████████████████                                          | 10/2
0 [00:00<00:00, 97.68image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 94.34image/s]

%|████▏
| 12/200 [00:09<02:25,  1.30it/s]
13:32:50-025601 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

t Feature:  37%|██████████████████                                                 | 7/
19 [00:00<00:00, 67.57image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 67.07image/s]

%|████▌
| 13/200 [00:10<02:27,  1.27it/s]
13:32:50-847342 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 95.93image/s]

%|████▋
| 14/200 [00:11<02:23,  1.29it/s]
13:32:51-597015 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/21 [00:00<?, ?image/s]

t Feature:  48%|███████████████████████                                           | 10/2
1 [00:00<00:00, 95.32image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 21/21 [00:00<00:00, 91.34image/s]

%|████▊                                                                           |
15/200 [00:11<02:23,  1.29it/s]
13:32:52-380135 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/21 [00:00<?, ?image/s]

t Feature:  48%|████████████████████████                                          | 10/2
1 [00:00<00:00, 98.02image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 21/21 [00:00<00:00, 96.75image/s]

%|▌
| 16/200 [00:12<02:22,  1.29it/s]
```
13:32:53-151851 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 99.32image/s]

%|▌
| 17/200 [00:13<02:19,  1.31it/s]
```
13:32:53-872350 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 98.34image/s]

%|▌
| 18/200 [00:14<02:18,  1.32it/s]
```
13:32:54-614309 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 91.39image/s]

%|▌
| 19/200 [00:14<02:17,  1.32it/s]
```
13:32:55-367446 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 99.14image/s]

%|▌                                                                            |
20/200 [00:15<02:15,  1.33it/s]
```

```
13:32:56-129375 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████|
20/20 [00:00<00:00, 104.13image/s]

%|████████                                                                    |
21/200 [00:16<02:14,  1.33it/s]
13:32:56-858245 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

t Feature:  50%|████████████████████████████                                | 10/2
0 [00:00<00:00, 92.14image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 91.54image/s]

%|████████▏
| 22/200 [00:17<02:13,  1.33it/s]
13:32:57-671397 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

t Feature:  47%|██████████████████████████▉                                  | 9/
19 [00:00<00:00, 87.88image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 84.81image/s]

%|████████▍
| 23/200 [00:17<02:16,  1.29it/s]
13:32:58-461681 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 95.02image/s]

%|████████▋
| 24/200 [00:18<02:15,  1.29it/s]
13:32:59-233442 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 89.91image/s]

%|          |
| 25/200 [00:19<02:14,  1.30it/s]
```
13:33:00-035210 INFO   Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 90.78image/s]

%|          |                                                                    |
26/200 [00:20<02:14,  1.30it/s]
```
13:33:00-788821 INFO   Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 97.28image/s]

%|          |
| 27/200 [00:21<02:13,  1.30it/s]
```
13:33:01-544206 INFO   Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 90.18image/s]

%|          |
| 28/200 [00:21<02:13,  1.29it/s]
```
13:33:02-304250 INFO   Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 99.38image/s]

%|          |
| 29/200 [00:22<02:10,  1.31it/s]
```
13:33:03-044126 INFO   Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 89.25image/s]

%|
| 30/200 [00:23<02:08,  1.32it/s]
```
13:33:03-787178 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

t Feature:  45%|                                                          | 9/
20 [00:00<00:00, 89.80image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 76.04image/s]

%|                                                                         |
31/200 [00:24<02:10,  1.29it/s]
```
13:33:04-616726 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 89.84image/s]

%|
| 32/200 [00:24<02:09,  1.30it/s]
```
13:33:05-327716 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 91.32image/s]

%|
| 33/200 [00:25<02:05,  1.33it/s]
```
13:33:06-037137 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████|
18/18 [00:00<00:00, 103.42image/s]

%|
| 34/200 [00:26<02:01,  1.37it/s]
```

```
13:33:06-697273 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 99.99image/s]

%|████████████
| 35/200 [00:26<01:57,  1.41it/s]
13:33:07-398494 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 93.34image/s]

%|████████████                                                              |
36/200 [00:27<01:56,  1.40it/s]
13:33:08-134643 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 94.32image/s]

%|████████████▍
| 37/200 [00:28<01:57,  1.38it/s]
13:33:08-927318 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 92.53image/s]

%|████████████▊
| 38/200 [00:29<02:00,  1.34it/s]
13:33:09-699963 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

t Feature:  50%|████████████████████████                                          | 10/2
0 [00:00<00:00, 98.79image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 94.73image/s]

%|████████████▉
| 39/200 [00:29<02:01,  1.32it/s]
```
13:33:10-438765 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 92.72image/s]

%|████████████▊
| 40/200 [00:30<01:59,  1.34it/s]
```
13:33:11-195987 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 96.06image/s]

%|████████████▉                                                                    |
41/200 [00:31<01:58,  1.34it/s]
```
13:33:11-923555 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 94.59image/s]

%|████████████▉                                                                    |
42/200 [00:32<01:56,  1.36it/s]
```
13:33:12-626586 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 98.89image/s]

%|█████████████▏
| 43/200 [00:32<01:54,  1.38it/s]
```

```
13:33:13-335438 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 93.90image/s]

%|████████████████▉
| 44/200 [00:33<01:53,  1.37it/s]
13:33:14-146294 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 83.94image/s]

%|█████████████████
| 45/200 [00:34<01:56,  1.33it/s]
13:33:14-872794 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 95.73image/s]

%|████████████████▉
| 46/200 [00:35<01:54,  1.35it/s]
13:33:15-601145 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 92.15image/s]

%|████████████████▉                                                          |
47/200 [00:35<01:53,  1.35it/s]
13:33:16-379410 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 91.42image/s]

%|█████████████████▏
| 48/200 [00:36<01:54,  1.33it/s]
13:33:17-142306 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 93.39image/s]

%|████████████████▉
| 49/200 [00:37<01:54,  1.32it/s]
13:33:17-883092 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 90.46image/s]

%|████████████████▉
| 50/200 [00:38<01:52,  1.33it/s]
13:33:18-689950 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 91.58image/s]

%|█████████████████▏
| 51/200 [00:38<01:54,  1.30it/s]
13:33:19-428489 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 96.98image/s]

%|█████████████████▍                                                          |
52/200 [00:39<01:52,  1.32it/s]
13:33:20-149122 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

t Feature:  50%|████████████████████████████                                | 10/2
0 [00:00<00:00, 95.33image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 95.73image/s]

%|█████████████████▌
| 53/200 [00:40<01:50,  1.33it/s]
```

```
13:33:20-919452 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

t Feature:  50%|████████████████████████████                                | 10/2
0 [00:00<00:00, 93.85image/s]

Extract Feature: 100%|████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 94.32image/s]

%|████████████████████▉
| 54/200 [00:41<01:50,  1.32it/s]
13:33:21-716488 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

t Feature:  45%|█████████████████████████▏                                  | 9/
20 [00:00<00:00, 89.54image/s]

Extract Feature: 100%|████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 90.09image/s]

%|████████████████████▏
| 55/200 [00:42<01:52,  1.29it/s]
13:33:22-541209 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 99.56image/s]

%|████████████████████▍
| 56/200 [00:42<01:52,  1.28it/s]
13:33:23-282508 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 94.79image/s]

%|████████████████████▋                                                     |
57/200 [00:43<01:49,  1.30it/s]
13:33:24-042155 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/21 [00:00<?, ?image/s]

t Feature:  48%|████████████████████████                                          | 10/2
1 [00:00<00:00, 96.91image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 21/21 [00:00<00:00, 95.49image/s]

%|████████████████████▉
| 58/200 [00:44<01:49,  1.30it/s]
```
13:33:24-813865 INFO Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

t Feature:  50%|████████████████████████                                          | 10/2
0 [00:00<00:00, 92.93image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 93.97image/s]

%|█████████████████████▏
| 59/200 [00:45<01:48,  1.30it/s]
```
13:33:25-580667 INFO Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/21 [00:00<?, ?image/s]

t Feature:  48%|████████████████████████                                          | 10/2
1 [00:00<00:00, 98.48image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 21/21 [00:00<00:00, 98.94image/s]

%|████████████████████▉
| 60/200 [00:45<01:47,  1.30it/s]
```
13:33:26-310406 INFO Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 97.64image/s]

%|█████████████████████▏
| 61/200 [00:46<01:44,  1.33it/s]
```
13:33:27-052182 INFO Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

t Feature:  45%|████████████████████████                                          | 9/
20 [00:00<00:00, 88.53image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 86.84image/s]

%|████████████████████                                                            |
62/200 [00:47<01:45,  1.31it/s]
```
13:33:27-829025 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 93.87image/s]

%|████████████████████                                                            |
63/200 [00:48<01:43,  1.33it/s]
```
13:33:28-577556 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 98.74image/s]

%|████████████████████▏
| 64/200 [00:48<01:42,  1.33it/s]
```
13:33:29-288707 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 95.94image/s]

%|████████████████████▎
| 65/200 [00:49<01:39,  1.35it/s]
```
13:33:30-037601 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████|
18/18 [00:00<00:00, 101.26image/s]

%|████████████████████▍
| 66/200 [00:50<01:38,  1.36it/s]
```

```
13:33:30-771092 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 99.85image/s]

%|████████████████████████
| 67/200 [00:51<01:37,  1.37it/s]
13:33:31-516577 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

t Feature:  44%|████████████████████████                                  | 7/1
6 [00:00<00:00, 69.83image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 78.68image/s]

%|████████████████████████                                                 |
68/200 [00:51<01:38,  1.34it/s]
13:33:32-247419 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 88.00image/s]

%|████████████████████████▌
| 69/200 [00:52<01:36,  1.35it/s]
13:33:32-970374 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 95.67image/s]

%|████████████████████████▏
| 70/200 [00:53<01:34,  1.37it/s]
13:33:33-657635 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 87.13image/s]

%|████████████████████████████
| 71/200 [00:53<01:32,  1.40it/s]
13:33:34-361820 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 91.07image/s]

%|████████████████████████████
| 72/200 [00:54<01:31,  1.40it/s]
13:33:35-063639 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 99.44image/s]

%|████████████████████████████                                                  |
73/200 [00:55<01:30,  1.40it/s]
13:33:35-795998 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/21 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 21/21 [00:00<00:00, 95.18image/s]

%|████████████████████████████
| 74/200 [00:56<01:32,  1.36it/s]
13:33:36-597155 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/21 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████|
21/21 [00:00<00:00, 102.01image/s]

%|████████████████████████████
| 75/200 [00:56<01:33,  1.34it/s]
13:33:37-337219 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 94.67image/s]

%|████████████████████████████▌
| 76/200 [00:57<01:31,  1.35it/s]
13:33:38-083758 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

t Feature:  42%|████████████████████████                                  | 8/1
9 [00:00<00:00, 75.84image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 78.00image/s]

%|████████████████████████████▋
| 77/200 [00:58<01:33,  1.32it/s]
13:33:38-886357 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████|
17/17 [00:00<00:00, 100.34image/s]

%|████████████████████████████                                             |
78/200 [00:59<01:31,  1.34it/s]
13:33:39-559335 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 93.04image/s]

%|████████████████████████████▊
| 79/200 [00:59<01:28,  1.36it/s]
13:33:40-283373 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 94.54image/s]

%|████████████████████████████▉
| 80/200 [01:00<01:27,  1.37it/s]
```

```
13:33:40-970288 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 93.93image/s]

%|████████████████████████████▌
| 81/200 [01:01<01:24,  1.41it/s]
13:33:41-685432 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/14 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 14/14 [00:00<00:00, 96.45image/s]

%|████████████████████████████▊
| 82/200 [01:01<01:22,  1.42it/s]
13:33:42-384396 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 95.14image/s]

%|████████████████████████████▉                                                  |
83/200 [01:02<01:23,  1.40it/s]
13:33:43-120831 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 85.00image/s]

%|█████████████████████████████                                                  |
84/200 [01:03<01:23,  1.39it/s]
13:33:43-857600 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████|
16/16 [00:00<00:00, 103.07image/s]

%|█████████████████████████████▏
| 85/200 [01:04<01:22,  1.39it/s]
13:33:44-573583 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 68.90image/s]

%|████████████████████████████████▉
| 86/200 [01:04<01:23,  1.36it/s]
```
13:33:45-335180 INFO   Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 99.46image/s]

%|█████████████████████████████████▏
| 87/200 [01:05<01:21,  1.38it/s]
```
13:33:46-008218 INFO   Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 98.87image/s]

%|█████████████████████████████████▍
| 88/200 [01:06<01:19,  1.41it/s]
```
13:33:46-687585 INFO   Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 97.42image/s]

%|█████████████████████████████████▋                                            |
89/200 [01:06<01:17,  1.43it/s]
```
13:33:47-370246 INFO   Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████|
15/15 [00:00<00:00, 101.00image/s]

%|█████████████████████████████████▉
| 90/200 [01:07<01:16,  1.44it/s]
```
13:33:48-073937 INFO   Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 84.45image/s]

%|████████████████████████████████▌
| 91/200 [01:08<01:16,  1.42it/s]
13:33:48-854781 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 86.89image/s]

%|████████████████████████████████▊
| 92/200 [01:09<01:18,  1.38it/s]
13:33:49-598773 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 88.66image/s]

%|████████████████████████████████▉
| 93/200 [01:09<01:18,  1.36it/s]
13:33:50-343991 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 89.30image/s]

%|█████████████████████████████████                                            |
94/200 [01:10<01:18,  1.36it/s]
13:33:51-077034 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 85.35image/s]

%|█████████████████████████████████▏
| 95/200 [01:11<01:18,  1.35it/s]
13:33:51-869421 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

t Feature:  44%|████████████████████████                                          | 8/1
8 [00:00<00:00, 70.52image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 73.15image/s]

%|████████████████████████████████████▉
| 96/200 [01:12<01:20,  1.30it/s]
```
13:33:52-613303 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 90.53image/s]

%|████████████████████████████████████▏
| 97/200 [01:12<01:17,  1.33it/s]
```
13:33:53-384677 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

t Feature:  50%|████████████████████████████                                      | 9/1
8 [00:00<00:00, 87.90image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 84.15image/s]

%|████████████████████████████████████▍
| 98/200 [01:13<01:17,  1.32it/s]
```
13:33:54-142891 INFO   Модель convnext_tiny успешно загружена с весами из .
```
t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 85.28image/s]

%|████████████████████████████████████                                            |
99/200 [01:14<01:16,  1.33it/s]
```
13:33:54-890323 INFO   Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 96.65image/s]

%|████████████████████████████████████                                        |
100/200 [01:15<01:14,  1.34it/s]
13:33:55-640892 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████|
17/17 [00:00<00:00, 101.41image/s]

%|████████████████████████████████████▏                                       |
101/200 [01:15<01:13,  1.34it/s]
13:33:56-364668 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

t Feature:  50%|████████████████████████████                                | 9/1
8 [00:00<00:00, 89.51image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 88.55image/s]

%|████████████████████████████████████▎                                       |
102/200 [01:16<01:13,  1.34it/s]
13:33:57-107116 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 96.91image/s]

%|████████████████████████████████████▍                                       |
103/200 [01:17<01:11,  1.35it/s]
13:33:57-794705 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████|
17/17 [00:00<00:00, 102.22image/s]

%|████████████████████████████████████▌                                       |
104/200 [01:18<01:09,  1.39it/s]
```

```
13:33:58-515330 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

t Feature:  38%|██████████████████▌                                | 6/
16 [00:00<00:00, 56.37image/s]

Extract Feature: 100%|████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 72.77image/s]

%|████████████████████████████████████▌                              |
105/200 [01:18<01:09,  1.36it/s]
13:33:59-271518 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 96.00image/s]

%|████████████████████████████████████▊                              |
106/200 [01:19<01:08,  1.38it/s]
13:33:59-964708 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 97.80image/s]

%|█████████████████████████████████████▏                             |
107/200 [01:20<01:06,  1.39it/s]
13:34:00-667148 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 93.71image/s]

%|█████████████████████████████████████▍                             |
108/200 [01:20<01:05,  1.40it/s]
13:34:01-362574 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 93.87image/s]

%|████████████████████████████████████████                                      |
109/200 [01:21<01:03,  1.42it/s]
13:34:02-073984 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 90.52image/s]

%|████████████████████████████████████████                                      |
110/200 [01:22<01:03,  1.42it/s]
13:34:02-781974 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 94.43image/s]

%|████████████████████████████████████████▏                                     |
111/200 [01:23<01:03,  1.41it/s]
13:34:03-461907 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 88.05image/s]

%|████████████████████████████████████████▍                                     |
112/200 [01:23<01:01,  1.43it/s]
13:34:04-165227 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 94.51image/s]

%|████████████████████████████████████████▌                                     |
113/200 [01:24<01:00,  1.43it/s]
13:34:04-853265 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 94.87image/s]

%|████████████████████████████████████████                                      |
114/200 [01:25<00:59,  1.44it/s]
13:34:05-587184 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 62.58image/s]

%|████████████████████████████████████████                                      |
115/200 [01:25<01:02,  1.37it/s]
13:34:06-351476 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 92.18image/s]

%|████████████████████████████████████████▏                                     |
116/200 [01:26<01:00,  1.38it/s]
13:34:07-075039 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 94.29image/s]

%|████████████████████████████████████████▍                                     |
117/200 [01:27<00:59,  1.39it/s]
13:34:07-786388 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 85.46image/s]

%|████████████████████████████████████████▌                                     |
118/200 [01:28<00:59,  1.37it/s]
13:34:08-562105 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 95.11image/s]

%|████████████████████████████████████████████                                |
119/200 [01:28<00:59,  1.36it/s]
13:34:09-277297 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████|
19/19 [00:00<00:00, 100.62image/s]

%|████████████████████████████████████████████                                |
120/200 [01:29<00:58,  1.37it/s]
13:34:10-005187 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████|
20/20 [00:00<00:00, 101.38image/s]

%|████████████████████████████████████████████▏                               |
121/200 [01:30<00:57,  1.36it/s]
13:34:10-752341 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 98.47image/s]

%|████████████████████████████████████████████▍                               |
122/200 [01:31<00:57,  1.36it/s]
13:34:11-507053 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 90.58image/s]

%|████████████████████████████████████████████▌                               |
123/200 [01:31<00:57,  1.35it/s]
13:34:12-255392 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

t Feature:  39%|████████████████████                                          | 7/1
8 [00:00<00:00, 66.40image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 75.51image/s]

%|████████████████████████████████████████████                                |
124/200 [01:32<00:57,  1.32it/s]
13:34:13-050319 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████|
16/16 [00:00<00:00, 100.28image/s]

%|████████████████████████████████████████████                                |
125/200 [01:33<00:55,  1.35it/s]
13:34:13-743847 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 96.13image/s]

%|████████████████████████████████████████████▏                               |
126/200 [01:34<00:54,  1.35it/s]
13:34:14-454442 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 98.47image/s]

%|████████████████████████████████████████████▍                               |
127/200 [01:34<00:52,  1.38it/s]
13:34:15-158866 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 95.71image/s]

%|████████████████████████████████████████████▎                               |
128/200 [01:35<00:51,  1.39it/s]
```

```
13:34:15-886100 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 84.88image/s]

%|████████████████████████████████████████████████                            |
129/200 [01:36<00:51,  1.37it/s]
13:34:16-632648 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|███████████████████████████████████████████████████████|
15/15 [00:00<00:00, 100.53image/s]

%|████████████████████████████████████████████████                            |
130/200 [01:36<00:50,  1.39it/s]
13:34:17-333660 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 96.28image/s]

%|████████████████████████████████████████████████▏                           |
131/200 [01:37<00:49,  1.40it/s]
13:34:18-059459 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 95.98image/s]

%|████████████████████████████████████████████████▍                           |
132/200 [01:38<00:49,  1.39it/s]
13:34:18-765346 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 79.95image/s]

%|████████████████████████████████████████████████▌                           |
133/200 [01:39<00:48,  1.37it/s]
13:34:19-515930 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 95.19image/s]

%|████████████████████████████████████████████████                          |
134/200 [01:39<00:47,  1.39it/s]
13:34:20-195998 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/13 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████|
13/13 [00:00<00:00, 101.96image/s]

%|████████████████████████████████████████████████                          |
135/200 [01:40<00:45,  1.43it/s]
13:34:20-825713 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/14 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 14/14 [00:00<00:00, 96.25image/s]

%|████████████████████████████████████████████████▌                         |
136/200 [01:41<00:43,  1.46it/s]
13:34:21-481316 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/12 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 12/12 [00:00<00:00, 90.93image/s]

%|████████████████████████████████████████████████▊                         |
137/200 [01:41<00:42,  1.49it/s]
13:34:22-145213 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/14 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 14/14 [00:00<00:00, 94.21image/s]

%|████████████████████████████████████████████████▉                         |
138/200 [01:42<00:41,  1.48it/s]
13:34:22-827553 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/11 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 11/11 [00:00<00:00, 96.96image/s]

%|████████████████████████████████████████████████                          |
139/200 [01:43<00:40,  1.50it/s]
13:34:23-473563 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/12 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 12/12 [00:00<00:00, 96.83image/s]

%|████████████████████████████████████████████████                          |
140/200 [01:43<00:39,  1.51it/s]
13:34:24-141049 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/13 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 13/13 [00:00<00:00, 91.28image/s]

%|████████████████████████████████████████████████▌                         |
141/200 [01:44<00:39,  1.49it/s]
13:34:24-801863 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/14 [00:00<?, ?image/s]

t Feature:  43%|████████████████████                                      | 6/1
4 [00:00<00:00, 58.66image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 14/14 [00:00<00:00, 55.61image/s]

%|████████████████████████████████████████████████▋                         |
142/200 [01:45<00:40,  1.43it/s]
13:34:25-568425 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████|
15/15 [00:00<00:00, 102.37image/s]

%|████████████████████████████████████████████████▊                         |
143/200 [01:45<00:39,  1.45it/s]
```

```
13:34:26-226655 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 93.85image/s]

%|████████████████████████████████████████████████████                    |
144/200 [01:46<00:38,  1.46it/s]
13:34:26-923315 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 99.28image/s]

%|████████████████████████████████████████████████████                    |
145/200 [01:47<00:38,  1.44it/s]
13:34:27-628204 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 98.70image/s]

%|████████████████████████████████████████████████████▏                   |
146/200 [01:47<00:37,  1.43it/s]
13:34:28-361825 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 95.01image/s]

%|████████████████████████████████████████████████████▍                   |
147/200 [01:48<00:37,  1.40it/s]
13:34:29-112184 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 96.74image/s]

%|████████████████████████████████████████████████████▋                   |
148/200 [01:49<00:37,  1.38it/s]
13:34:29-889009 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 96.10image/s]

%|████████████████████████████████████████████████████                    |
149/200 [01:50<00:37,  1.36it/s]
13:34:30-639194 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 94.74image/s]

%|████████████████████████████████████████████████████                    |
150/200 [01:50<00:36,  1.35it/s]
13:34:31-384214 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 94.50image/s]

%|████████████████████████████████████████████████████▌                   |
151/200 [01:51<00:36,  1.34it/s]
13:34:32-131081 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

t Feature:  39%|████████████████████                                    | 7/1
8 [00:00<00:00, 68.26image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 81.66image/s]

%|████████████████████████████████████████████████████▋                   |
152/200 [01:52<00:36,  1.33it/s]
13:34:32-926635 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 94.79image/s]

%|████████████████████████████████████████████████████▊                   |
153/200 [01:53<00:35,  1.32it/s]
```

```
13:34:33-642556 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 97.50image/s]

%|████████████████████████████████████████████████████                |
154/200 [01:53<00:34,  1.34it/s]
13:34:34-393292 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 89.39image/s]

%|████████████████████████████████████████████████████                |
155/200 [01:54<00:33,  1.36it/s]
13:34:35-097676 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████|
17/17 [00:00<00:00, 103.33image/s]

%|████████████████████████████████████████████████████▏               |
156/200 [01:55<00:32,  1.37it/s]
13:34:35-831587 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 86.49image/s]

%|████████████████████████████████████████████████████▍               |
157/200 [01:56<00:31,  1.35it/s]
13:34:36-568563 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 86.06image/s]

%|████████████████████████████████████████████████████▋               |
158/200 [01:56<00:30,  1.36it/s]
13:34:37-278361 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 99.90image/s]

%|████████████████████████████████████████████████████████                |
159/200 [01:57<00:29,  1.39it/s]
13:34:37-978667 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 96.45image/s]

%|████████████████████████████████████████████████████████                |
160/200 [01:58<00:28,  1.39it/s]
13:34:38-739142 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

t Feature:  44%|████████████████████████                                  | 8/1
8 [00:00<00:00, 74.34image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 78.93image/s]

%|████████████████████████████████████████████████████████▏               |
161/200 [01:59<00:29,  1.34it/s]
13:34:39-497559 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 94.70image/s]

%|████████████████████████████████████████████████████████▍               |
162/200 [01:59<00:27,  1.36it/s]
13:34:40-190882 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 98.05image/s]

%|████████████████████████████████████████████████████████▌               |
163/200 [02:00<00:26,  1.38it/s]
```

```
13:34:40-922981 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 90.84image/s]

%|████████████████████████████████████████████████████████████              |
164/200 [02:01<00:26,  1.37it/s]
13:34:41-655617 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 88.03image/s]

%|████████████████████████████████████████████████████████████              |
165/200 [02:01<00:25,  1.38it/s]
13:34:42-388301 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 97.95image/s]

%|████████████████████████████████████████████████████████████▏             |
166/200 [02:02<00:24,  1.38it/s]
13:34:43-089546 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 93.70image/s]

%|████████████████████████████████████████████████████████████▍             |
167/200 [02:03<00:24,  1.37it/s]
13:34:43-835184 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████|
19/19 [00:00<00:00, 103.04image/s]

%|████████████████████████████████████████████████████████████▏             |
168/200 [02:04<00:23,  1.37it/s]
13:34:44-571793 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

t Feature:  50%|████████████████████████████                                | 9/1
8 [00:00<00:00, 80.77image/s]

Extract Feature: 100%|████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 67.76image/s]

%|████████████████████████████████████████████████████████████            |
169/200 [02:04<00:23,  1.32it/s]
13:34:45-383797 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

t Feature:  19%|██████████                                                  | 3/
16 [00:00<00:00, 27.59image/s]

t Feature:  50%|████████████████████████████                                | 8/1
6 [00:00<00:00, 38.38image/s]

Extract Feature: 100%|████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 47.91image/s]

%|████████████████████████████████████████████████████████████            |
170/200 [02:05<00:23,  1.26it/s]
13:34:46-271231 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 92.16image/s]

%|████████████████████████████████████████████████████████████▏           |
171/200 [02:06<00:22,  1.30it/s]
13:34:46-963071 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

t Feature:  42%|████████████████████████                                    | 8/1
9 [00:00<00:00, 76.00image/s]

Extract Feature: 100%|████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 79.83image/s]

%|████████████████████████████████████████████████████████████▍           |
172/200 [02:07<00:21,  1.30it/s]
13:34:47-753800 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 87.02image/s]

%|████████████████████████████████████████████████████████████▌           |
173/200 [02:08<00:20,  1.31it/s]
13:34:48-476748 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 92.27image/s]

%|████████████████████████████████████████████████████████████          |
174/200 [02:08<00:19,  1.33it/s]
13:34:49-204814 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 96.98image/s]

%|████████████████████████████████████████████████████████████          |
175/200 [02:09<00:18,  1.35it/s]
13:34:49-939202 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 95.40image/s]

%|████████████████████████████████████████████████████████████▊         |
176/200 [02:10<00:17,  1.35it/s]
13:34:50-687199 INFO     Модель convnext_tiny успешно загружена с весами из .


t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 94.57image/s]

%|████████████████████████████████████████████████████████████▉         |
177/200 [02:10<00:17,  1.35it/s]
13:34:51-422619 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/20 [00:00<?, ?image/s]

t Feature:  50%|████████████████████████████                                | 10/2
0 [00:00<00:00, 97.77image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 20/20 [00:00<00:00, 96.69image/s]

%|████████████████████████████████████████████████████████████▌          |
178/200 [02:11<00:16,  1.34it/s]
```
13:34:52-189313 INFO   Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 93.86image/s]

%|████████████████████████████████████████████████████████████▍          |
179/200 [02:12<00:15,  1.33it/s]
```
13:34:52-968836 INFO   Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

t Feature:  44%|████████████████████████                                    | 8/1
8 [00:00<00:00, 78.18image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 65.92image/s]

%|████████████████████████████████████████████████████████████▎          |
180/200 [02:13<00:15,  1.28it/s]
```
13:34:53-793405 INFO   Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████|
18/18 [00:00<00:00, 104.90image/s]

%|████████████████████████████████████████████████████████████▏          |
181/200 [02:14<00:14,  1.31it/s]
```
13:34:54-500267 INFO   Модель convnext_tiny успешно загружена с весами из .

```
t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 89.79image/s]

%|████████████████████████████████████████████████████████████▉       |
182/200 [02:14<00:13,  1.32it/s]
13:34:55-258573 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 92.54image/s]

%|████████████████████████████████████████████████████████████▏       |
183/200 [02:15<00:12,  1.33it/s]
13:34:56-030458 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 94.18image/s]

%|████████████████████████████████████████████████████████████      |
184/200 [02:16<00:12,  1.32it/s]
13:34:56-772059 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 90.89image/s]

%|████████████████████████████████████████████████████████████      |
185/200 [02:17<00:11,  1.33it/s]
13:34:57-515843 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 87.24image/s]

%|████████████████████████████████████████████████████████████▋     |
186/200 [02:17<00:10,  1.32it/s]
13:34:58-299028 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 93.17image/s]

%|████████████████████████████████████████████████████████████████▉     |
187/200 [02:18<00:09,  1.31it/s]
13:34:59-049177 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 88.27image/s]

%|████████████████████████████████████████████████████████████████▏     |
188/200 [02:19<00:09,  1.32it/s]
13:34:59-784489 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 97.28image/s]

%|████████████████████████████████████████████████████████████████▍     |
189/200 [02:20<00:08,  1.35it/s]
13:35:00-503712 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

t Feature:  47%|████████████████████████▉                                | 9/
19 [00:00<00:00, 81.48image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 80.36image/s]

%|████████████████████████████████████████████████████████████████▋     |
190/200 [02:20<00:07,  1.32it/s]
13:35:01-287322 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 97.62image/s]

%|████████████████████████████████████████████████████████████████▊     |
191/200 [02:21<00:06,  1.33it/s]
```

```
13:35:02-074333 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/18 [00:00<?, ?image/s]

t Feature:  44%|████████████████████████                                          | 8/1
8 [00:00<00:00, 72.79image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 18/18 [00:00<00:00, 72.22image/s]

%|████████████████████████████████████████████████████████████████████▉    |
192/200 [02:22<00:06,  1.28it/s]
13:35:03-195827 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

t Feature:  47%|█████████████████████████▉                                        | 9/
19 [00:00<00:00, 88.11image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 88.59image/s]

%|████████████████████████████████████████████████████████████████████▉    |
193/200 [02:23<00:06,  1.15it/s]
13:35:03-965137 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 95.65image/s]

%|████████████████████████████████████████████████████████████████████   |
194/200 [02:24<00:05,  1.19it/s]
13:35:04-700901 INFO     Модель convnext_tiny успешно загружена с весами из .

t Feature:   0%|
| 0/19 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 19/19 [00:00<00:00, 97.31image/s]

%|████████████████████████████████████████████████████████████████████   |
195/200 [02:24<00:04,  1.24it/s]
13:35:05-421198 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 92.73image/s]

%|████████████████████████████████████████████████████████████████████▊  |
196/200 [02:25<00:03,  1.30it/s]
13:35:06-127025 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 94.71image/s]

%|████████████████████████████████████████████████████████████████████▉  |
197/200 [02:26<00:02,  1.33it/s]
13:35:06-846900 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 96.03image/s]

%|████████████████████████████████████████████████████████████████████▉  |
198/200 [02:27<00:01,  1.34it/s]
13:35:07-564729 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/16 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 16/16 [00:00<00:00, 94.33image/s]

%|████████████████████████████████████████████████████████████████████▉|
199/200 [02:27<00:00,  1.36it/s]
13:35:08-304883 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/15 [00:00<?, ?image/s]

Extract Feature: 100%|████████████████████████████████████████████████████████
| 15/15 [00:00<00:00, 87.97image/s]

%|████████████████████████████████████████████████████████████████████|
200/200 [02:28<00:00,  1.36it/s]
13:35:09-015819 INFO     Модель convnext_tiny успешно загружена с весами из .
```

```
t Feature:   0%|
| 0/17 [00:00<?, ?image/s]

Extract Feature: 100%|██████████████████████████████████████████████████████████
| 17/17 [00:00<00:00, 84.64image/s]

201it [02:29,  1.35it/s]
Стрим закончился
```

In [402…
```
os.system(f'ffmpeg -y -i "./mot20_vis.mp4" -c:v libx264 -pix_fmt yuv420p "./mot20_v
```

Out[402…
```
0
```

In [404…
```
html = f"""
<div style="display:flex; gap:20px;">
    <div>
        <h4>Исходное видео </h4>
        <video width="800" controls autoplay muted loop>
            <source src="./mot20_vis_new.mp4" type="video/mp4">
        </video>
    </div>
</div>
"""
display(HTML(html))
```

Исходное видео

Выводы

```
В целом все неплохо работает, всем советую
```